

Infrared Spectroscopy

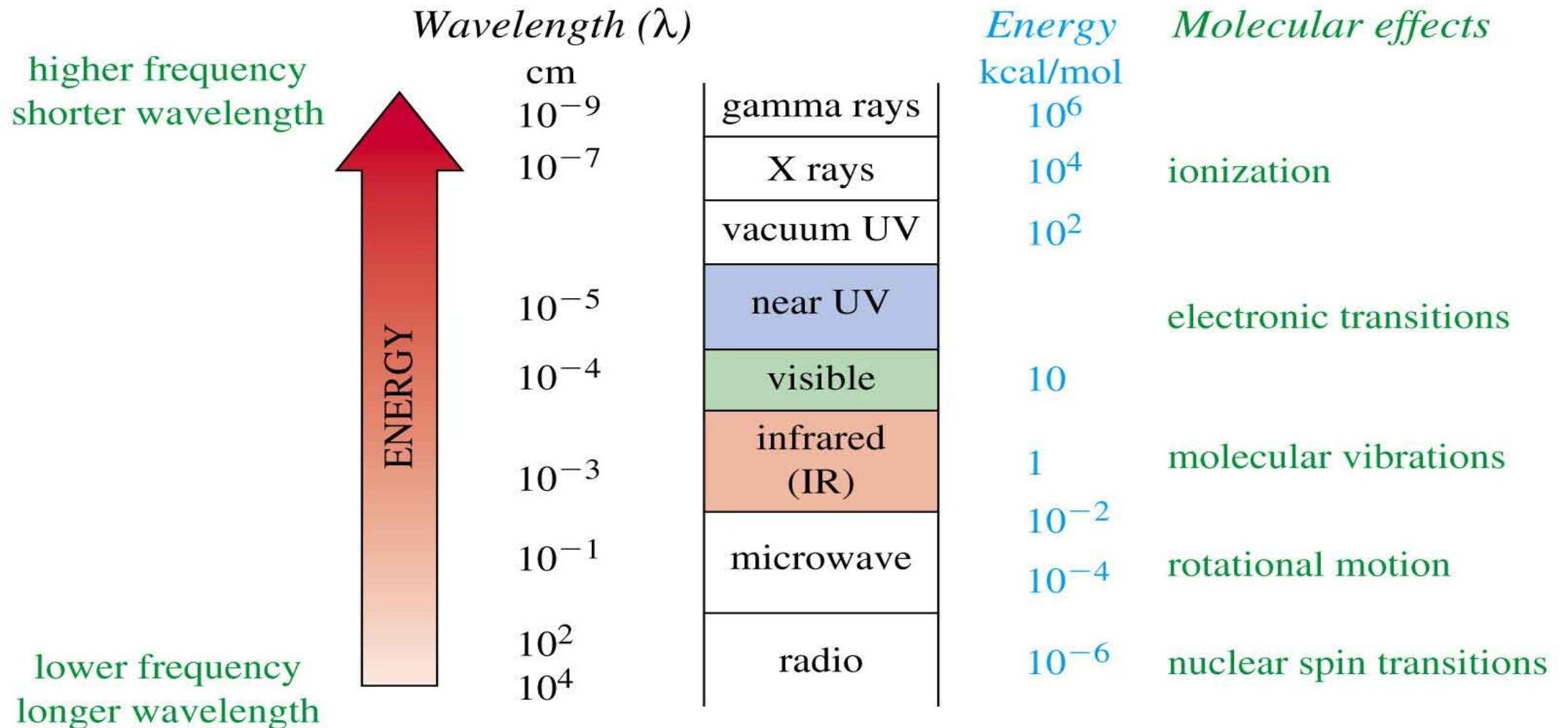
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Introduction

- All molecules of pharmaceutical interest absorb some form of electromagnetic radiation.
- Within the electromagnetic spectrum, infrared energy is a small portion which is typically divided into three regions.

Region	Wave number ($\text{v}^{-1}\text{cm}^{-1}$)	Wavelength (λ) μm
Near IR	12,800–4000	0.78–2.5
Middle IR	4000–200	2.5–50
Far IR	200–10	50–1000
Most used	4000–670	2.5–1.5

✓ **The main region for analytical purposes is the middle IR region.**



Introduction...

- IR does not have sufficient energy to cause excitation of electrons,
 - But it does cause atoms and groups of atoms of organic compounds to Vibrate faster about the **covalent bonds** that connect them.
- The energy unit wavenumber (cm^{-1}) is typically used.
- Wavenumber is the reciprocal of the IR wavelength expressed in centimeters.

Con't

- When a molecule is bombarded with radiation of a frequency that exactly matches the frequency of the vibration of one of its bonds, the molecule absorbs energy.
- Determining the wavenumbers of the energy absorbed by a particular compound gives the type of bond.
- **Therefore, the functional groups present in a molecule can be deduced from an IR spectrum**

Con't

- In their vibrations covalent bonds behave as if they were tiny springs connecting the atoms.
- Molecules can vibrate in a variety of ways, but generally bond vibration modes are divisible into two types:

(A) Stretching Vibrations - affects the bond length

(B) Bending (deformation) vibrations

- *Affect the bond angle & require lower energy than stretching vibrations*

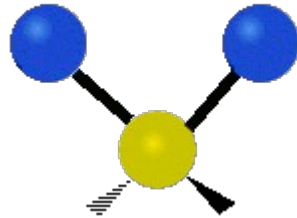
Con't

- The vibrational and rotational energy levels arise due to change in *dipole moment*.
- For absorption in the IR region, a molecule must undergo change in dipole moment during vibrations or rotations.
- The alternating electric field of the radiation interacts with the molecule and causes changes in the amplitude of one of its motion.

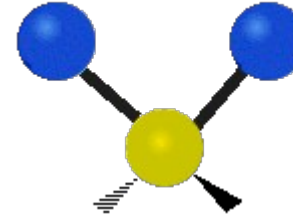
- Homonuclear species like H_2 , N_2 , O_2 or Cl_2 do not absorb IR radiations.
- ✓ no net change in their dipole moment occurs during any vibration or rotation.

Molecular vibration

Stretching

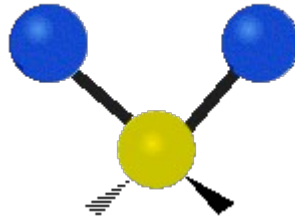


Symmetrical stretching

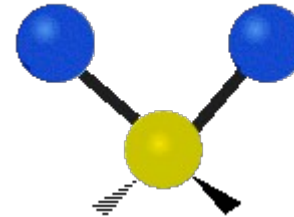


Asymmetrical stretching

In-Plane bending vibrations

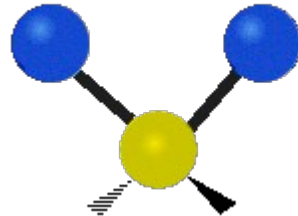


Scissoring

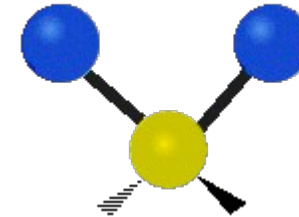


Rocking

Out-of-Plane bending vibrations



Wagging



Twisting

Vibrational Modes

- The number of possible vibrations in a poly-atomic molecule can be determined only by knowing its degree of freedom.
- Each atom in a molecule has **three degrees of freedom** in x-, y-, and z-directions
- if there are **N atoms** in a molecule there are total of **3N** degrees of freedom.
- For non-linear molecules, there are six degrees which are rotational and translational whereas there are five such degrees for linear molecules.
- Therefore, for non-linear molecules **3N-6** numbers of possible vibrations and for linear molecules **3N-5** possible vibrations are present.

E.g.

** For methane, CH_4 $3 \times 5 - 6 = 9$ possible fundamental vibrational modes.*

For benzene, C_6H_6 , $3 \times 12 - 6 = 30$

• Carbon dioxide, CO_2 , $3 \times 3 - 5 = 4$

Con't

❖ But this number may be increased or decreased.

Reasons for decrease

- Two bands might overlap (Degenerate).
- An absorption might not be in the $4000\text{--}400\text{ cm}^{-1}$ range.
- An absorption might be too weak to be observed.
- Absorptions might be too close to each other to be resolved on the instrument.
- Lack of dipole change

Reasons for increase

- Overtones –multiples of the fundamental vibrations
- Coupling-splitting of vibrational modes
- Combination tones-two vibrations .

Vibrational Frequency

- Amount of energy required to stretch a bond depends on the strength of the bond and the masses of the bonded atoms (**Hooke's Law**).
- The stronger the bond, **the greater the energy required** to stretch it.
- Frequency of the vibration is inversely related to the mass of the atoms attached to the spring, so **heavier atoms vibrate at lower frequencies**.
- Approximate wavenumber of an absorption can be calculated from the equation derived from **Hooke's law**, which describes motion of a vibrating spring:

Vibrational Frequency

Frequency $\nu_m = \frac{1}{2\pi} \sqrt{\frac{K}{\mu}} \text{ Hz}$

where K = force constant of the bond,

μ = reduced mass,

$$\mu = \frac{m_A m_B}{m_A + m_B}$$

wave number $\bar{\nu} = \frac{1}{2\pi C} \sqrt{\frac{K}{\mu}}$

C = Speed of light

Con't

□ Frequencies of normal modes of vibration of two atoms held together by a chemical bond depend on:

(a) The masses of the two vibrating atoms

(b) The force constant of the bond between them & bond order: (relative stiffness of the bond).

➤ **K** which is a measure of **bond strength**, and is inversely proportional to the **reduced mass** .

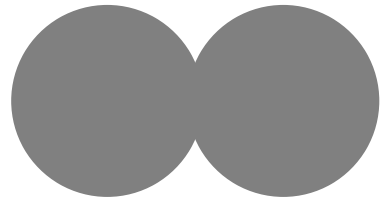
❖ Frequencies will increase with increasing bond strength (bond order) and decreasing mass.

- Triple bonds are stiffer (vibrate at higher frequencies) than double bonds, and double bonds are stiffer than single bonds.

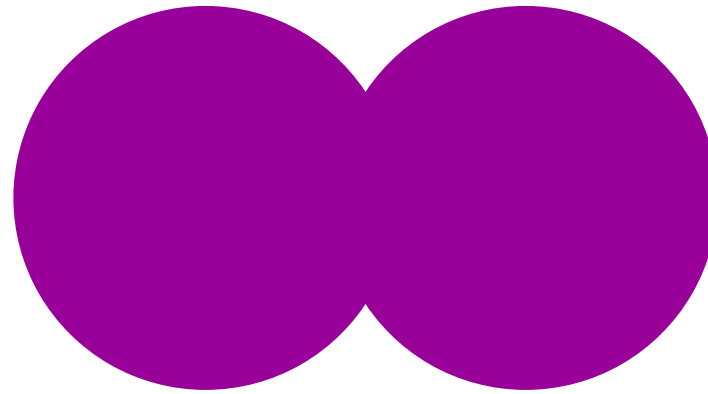
Con't

- Lighter atoms like C, N, and O vibrate at higher frequency due to their low reduced mass.
- The force constants for triple bond, double bond, and single bond are in order $C \equiv C > C = C > C - C$ and so are the frequencies $2100 > 1600 > 1200 \text{ cm}^{-1}$ respectively.

How does the mass influence the vibration?



MM = 2 g/mole



MM = 254 g/mole

The greater the mass - the lower the wavenumber

(c) Slightly depends also on the other atoms attached to the two atoms concerned (Resonance and inductive effects).

(d) The electronegativity of the two vibrating atoms (Dipole moment effects). Examples:

C – H	2850 – 2960 cm ⁻¹	N – H	3300 – 3500 cm ⁻¹
O – H	3590 – 3650 cm ⁻¹	C – C	Around 1250 cm ⁻¹
C – N	Around 1350 cm ⁻¹	C – O	Around 1400 cm ⁻¹
C = C	1620 – 1680 cm ⁻¹	C = N	1620 – 1690 cm ⁻¹
C = O	1630 – 1780 cm ⁻¹		
C ≡ C	2100 – 2260 cm ⁻¹	C ≡ N	2220 – 2260 cm ⁻¹ .

There are two general regions in the infrared spectrum:

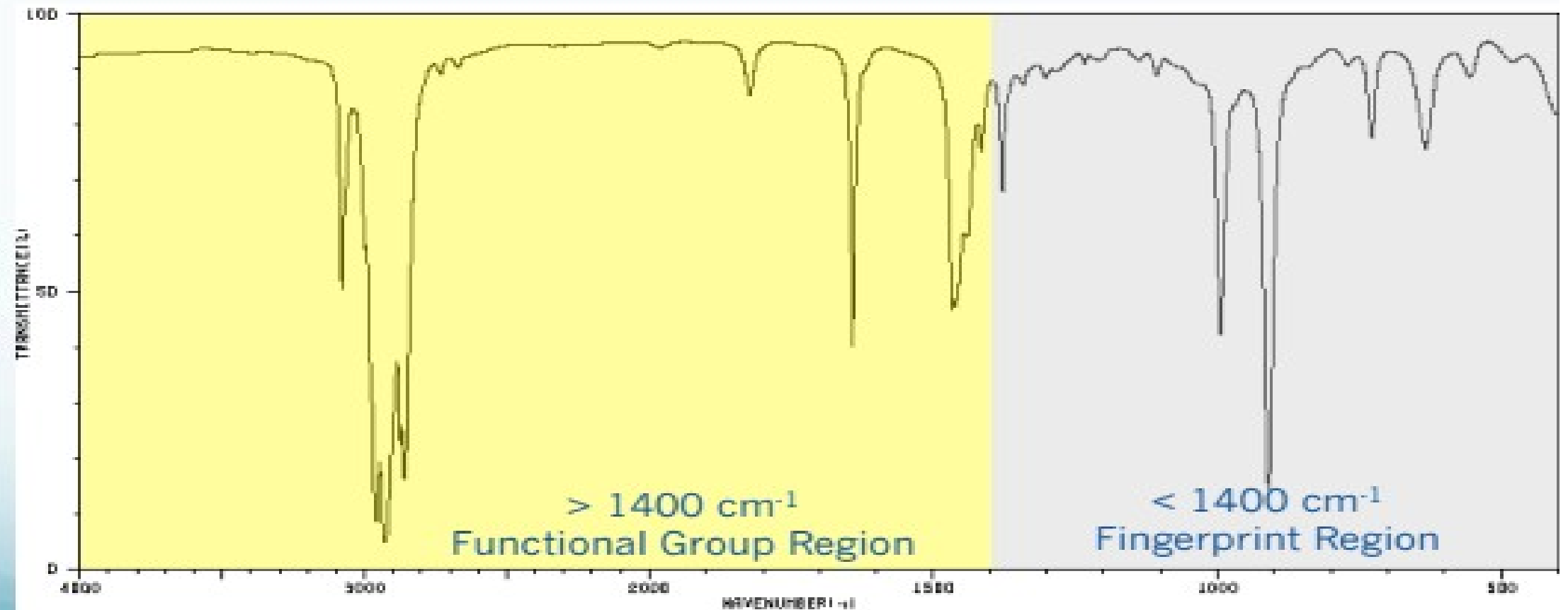
1. Functional Group frequency region :

- Having a wavelength ranging from 2.5 to 8.0 μm and a wave number from 4000-1300 cm^{-1}
- Here, the stretching & bending vibrational bands associated with specific structural or functional groups are observed frequently.

2. Fingerprint region:

- Having a wavelength ranging from 8.0-25 μ and a wave number from 1300-400 cm^{-1} .
- Here, the vibrational modes depend solely and strongly on the rest of the molecule. **proof of identity**

Two General Regions in an IR Spectrum:



The Intensity of Absorption Bands

1. It depends on the **size of the change in dipole moment** associated with the vibration:

✓ The greater the change in dipole moment, the more intense the absorption.

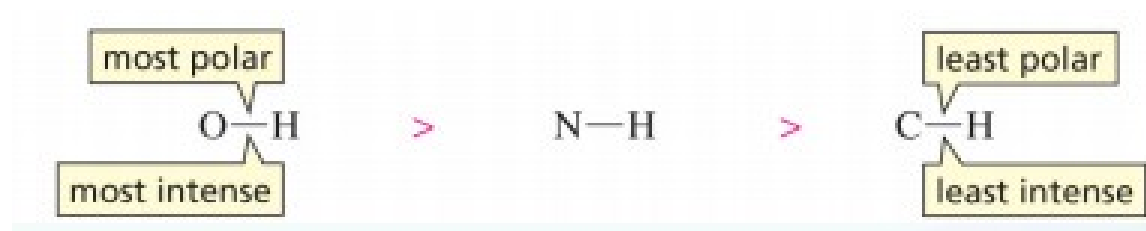
➤ The stretching vibration of an O-H bond will be associated with a greater change in dipole moment than that of an N-H bond b/c the bond is more polar.

➤ Consequently, the stretching vibration of the O-H bond will be more intense.

➤ Likewise, the stretching vibration of an N-H bond is more intense than that of a

• C-H bond because the N-H bond is more polar.

• $O-H > N-H > C-H$



Con't

2. the **number of bonds** responsible for the absorption.

- E.g., the absorption band for the C – H stretch will be more intense for a compound
 - such as octyl iodide, which has **17** C – H bonds, than for methyl iodide, which has only **3** C – H bonds.

3. The **concentration of the sample** used to obtain an IR spectrum also affects the intensity of the absorption bands.

- Concentrated samples have greater numbers of absorbing molecules and, therefore, more intense absorption bands.

The Position of Absorption Bands

Hydrogen Bonding

- Alters both Stretching and bending vibrations:

(a) Stretching vibrations:

- Cause vibrations to move to lower frequencies
 - ✓ To longer wavelength with increased intensity
 - ✓ Cause band widening

• (b) Bending vibrations:

- ✓ Cause shift to shorter wavelengths.
- ✓ Less pronounced than stretching vibrations.

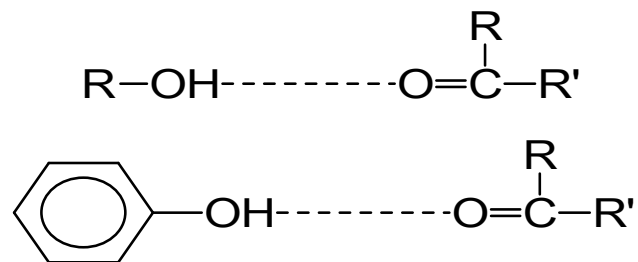
➤ Hydrogen Bonding have two types:

- I. Inter-molecular
- II. Intra-molecular

(a) Intermolecular Hydrogen Bonding:

- Involves association of two or more molecules of the same or different compounds.

<u>Type</u>	<u>Frequency reduction</u>		<u>Examples</u>
	$X - H \cdots \cdots Y = C$		
Weak bonding	300	15	Alcohols & phenols with carbonyl cpds
Strong bonding	> 500	50	Acid Dimers



Weak H.B.



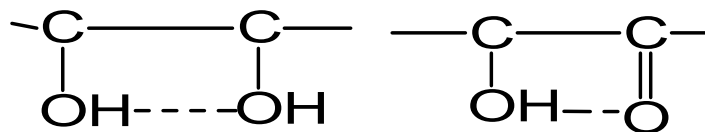
Strong H.B.

(b) Intramolecular Hydrogen Bonding:

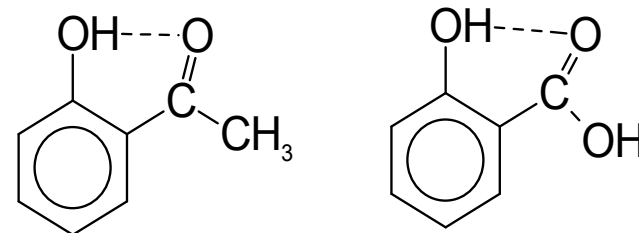
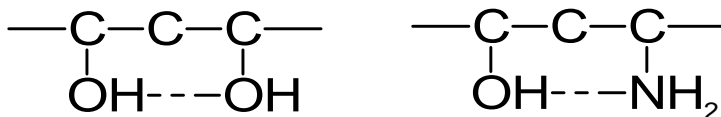
- Formed when the proton donor and acceptor are present in a single molecule.
- It is not affected by dilution, but temperature dependent.

<u>Type</u>	<u>X - H</u>	<u>Y = C</u>	<u>Examples</u>
Weak	100	10	1,2-diols, α-hydroxy ketones
Medium	100-300	50	1,3-diols, β-hydroxy amines
Strong	> 300	100	o-hydroxy aryl ketones or acids

Weak



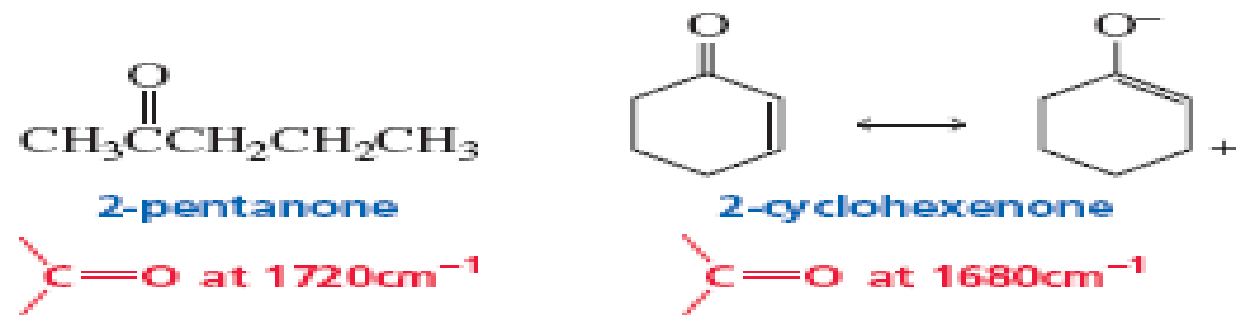
Medium



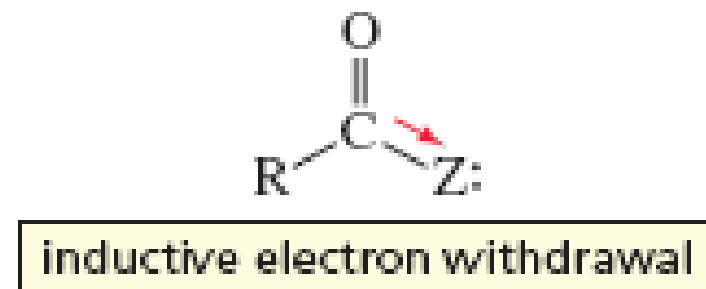
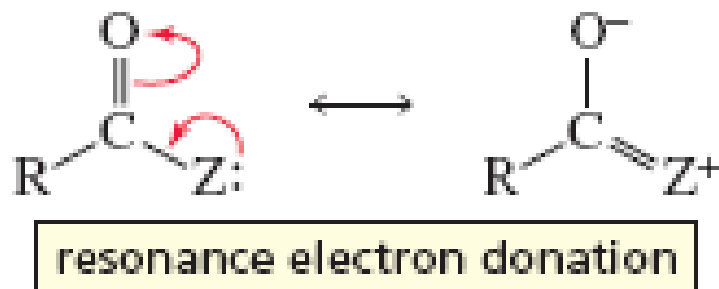
Strong

Electrons Delocalization, Resonance and Inductive Effects

- The exact position of absorption band depends on other structural features of the molecule, such as
 - ✓ electron delocalization,
 - ✓ the electronic effect of neighboring substituents, and of course
 - ✓ hydrogen bonding.
- For example, the IR of the carbonyl group of 2-pentanone absorbs at 1720 cm^{-1} whereas the IR spectrum of the carbonyl group of 2-cyclohexenone absorbs at a lower frequency (1680 cm^{-1}) ?
- ❖ 2-Cyclohexenone absorbs at a lower frequency because the carbonyl group has less
 - ❖ double-bond character due to electron delocalization.
 - ❖ Because a single bond is weaker than a double bond, a carbonyl group with significant
 - ❖ single bond character will stretch at a lower frequency than will one with little or no single-bond character.



- Putting an atom other than carbon next to the carbonyl group also causes the position of carbonyl absorption band to shift.
- Whether it shifts to a lower or to a higher frequency depends on whether the predominant effect of the atom is to donate electrons by resonance or to withdraw electrons inductively.

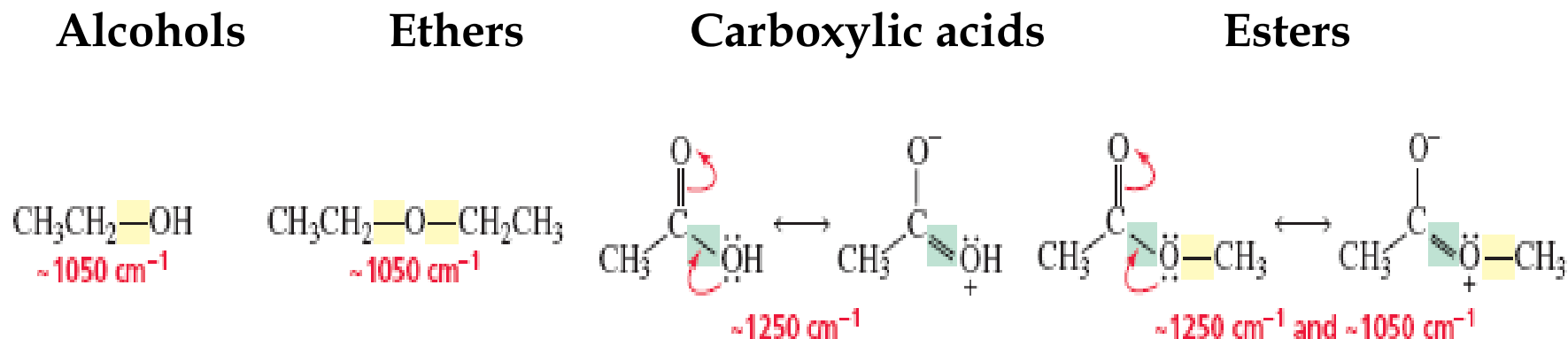


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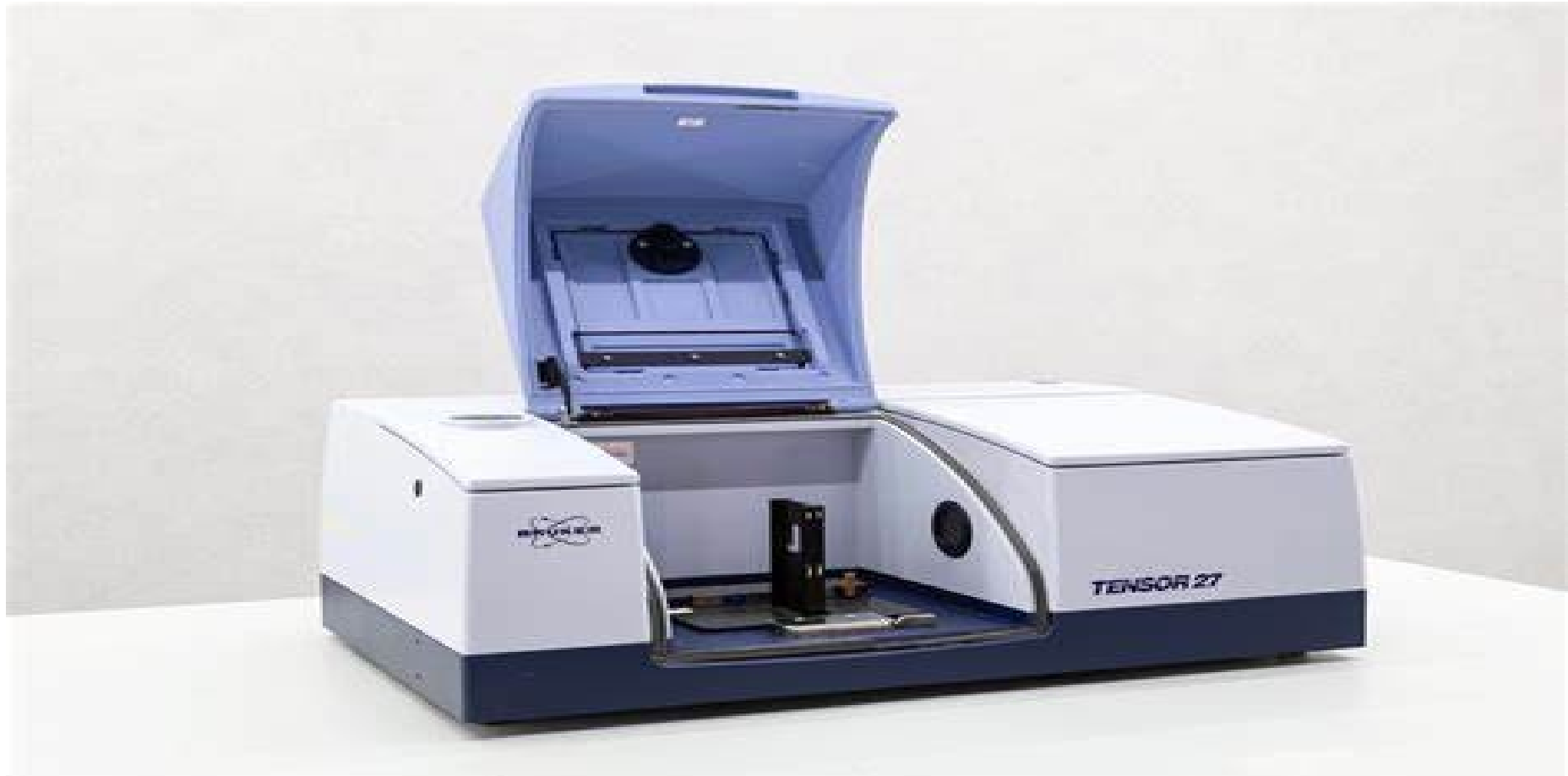
- E.g. The effect of the nitrogen of an amide (CO-NH_2) is **electron donation by resonance**.
- Therefore, the carbonyl group of an amide has **less double-bond character** than
 - does the carbonyl group of a ketone, so it is **weaker and stretches more easily** (1660 cm^{-1}).
- In contrast, the effect of the oxygen of an ester is **inductive electron withdrawal**, so the resonance contributor with the $\text{C} - \text{O}$ single bond contributes less to the hybrid.
- The carbonyl group of an ester, therefore, has **more double-bond character** than does the carbonyl group of a ketone, so the former **is stronger and harder to stretch** (1740 cm^{-1}).
- A $\text{C} - \text{O}$ bond normally shows a stretch between 1250 and 1050 cm^{-1} .

Con't

- If, however, the C – O bond is in a carboxylic acid, the stretch will occur at the higher end of the range.
- The position of the C – O absorption varies because the C – O bond in an alcohol is a pure **single bond**,
- whereas the C – O bond in a carboxylic acid has **partial double-bond** character that is due to resonance electron donation.
- Esters show C – O stretches **at both ends of the range** because esters have two
 - C– O **single bonds**—one that is a **pure single bond** and one that has **partial double-bond** character.



INSTRUMENTATION



INSTRUMENTATION

- IR spectra are obtained by detecting changes in transmittance (or absorption) intensity as a function of frequency.
- Most commercial instruments separate and measure IR radiation using dispersive spectrometers or Fourier transform spectrometers /FT-IR/.

Basic components include

Radiation sources

Monochromators/Interferometers/

Sample compartment

Detector

□ Radiation sources

- An inert solid is electrically heated to a temperature in the range 1500 to 2000 °C.
- The heated material will then emit **infra red** radiation.
- *Three popular types of sources.*
 - **Nernst glower** : is a cylinder (1-2 mm diameter, approximately 20 mm long) of rare earth oxides. The Nernst glower can reach temperatures of 2200 °C .
 - **Globar source** : is a silicon carbide rod (5mm diameter, 50mm long) which is electrically heated to about 1500 °C.
 - **Nichrome wire**,: electrically heated to 1100 °C. It produces a lower intensity of radiation than the Nernst or Globar sources, but has a **longer working life**.

□ Monochromators

- The monochromator is a device used to disperse a broad spectrum of radiation and provide a continuous series of electromagnetic energy bands of determinable wavelength or frequency range.
- Prisms or gratings are the dispersive components used in conjunction with variable-slit mechanisms and mirrors.
- Used only in **dispersive IR spectrometer**.

□ Interferometers

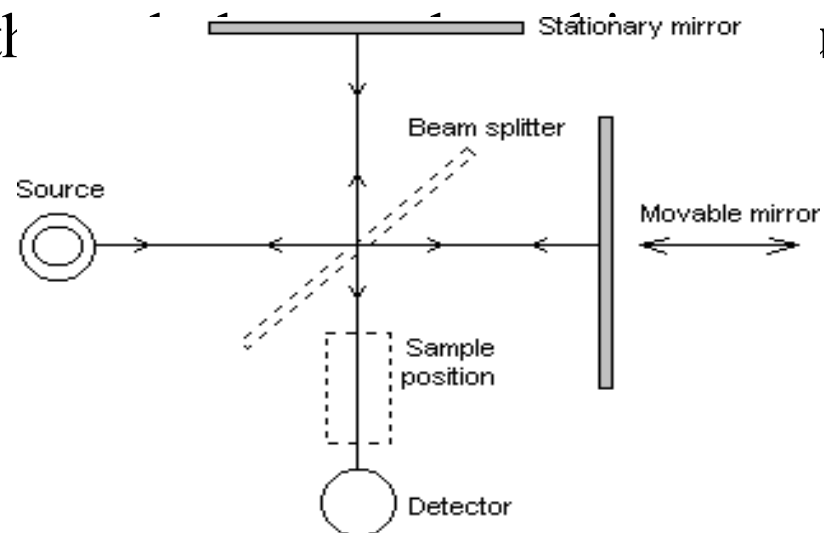
- An FT-IR instrument uses a system called an interferometer to **collect a spectrum**.
- It consists of three active components: a **moving mirror**, a **fixed mirror**, and a **beam splitter**.

□ Interferometers...

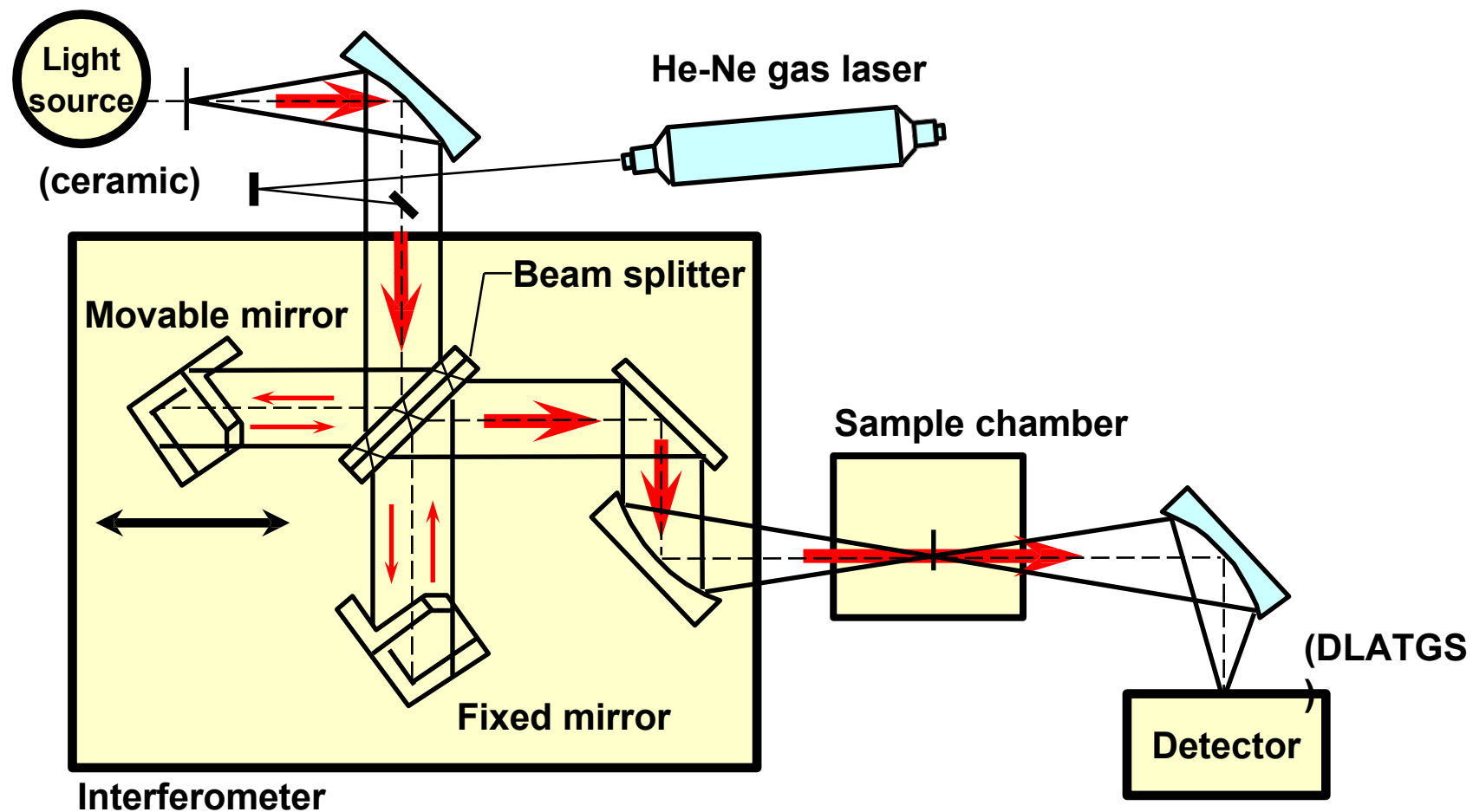
- The energy goes from the source to the beam splitter which splits the beam into two parts.
- One part is transmitted to a moving mirror; one part is reflected to a fixed mirror.
- The moving mirror moves back and forth at a constant velocity.
- The two beams are reflected from the mirrors and recombined at the beam splitter.
- The beam from the moving mirror has traveled a different distance than the beam from the fixed mirror.
- When the beams are combined an interference pattern is created, since some of the wavelengths recombine constructively and some destructively.

□ Interferometers...

- If the movable mirror moves away from the beam splitter at a constant speed, radiation reaching the detector goes through a steady sequence of maxima and minima as the interference alternates between constructive and destructive phases.
- This interference pattern is called **interferogram**.
- The resulting beam then passes through a lens, which is optically focused on the detector.



FT Optical System Diagram



□ Detectors

- Most detectors used in dispersive IR spectrometers can be categorized into two classes: **thermal detectors** and **photon detectors**.
- Thermal detectors include **thermocouples, thermistors, and pneumatic devices (Golay detectors)**.
- They measure the **heating effect** produced by infrared radiation.
- ❖ A variety of physical property changes are quantitatively determined: expansion of a non absorbing gas (Golay detector), electrical resistance (thermistor), and voltage at junction of dissimilar metals (thermocouple).

☐ Detectors...

- Photon detectors rely on the **interaction of IR radiation and a semiconductor material.**

- Non-conducting electrons are **excited to a conducting state**. Thus, a small current or voltage can be generated.

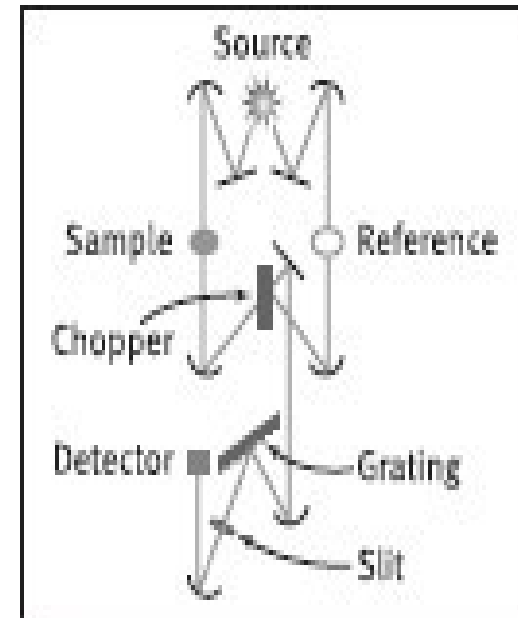
- ☐The response times of many detectors (for example, thermocouple and thermistor) used in dispersive IR instruments are too slow for the rapid scan times (1 sec or less) of the interferometer.

- The two most popular detectors for a FTIR spectrometer are **deuterated triglycine sulfate (DTGS)** and **mercury cadmium telluride (MCT)**.

Types of instruments

❑ Dispersive IR spectrometer

- The dispersive element in dispersive instruments is contained within a monochromator.
- Figure below shows the optical path of a dispersive infrared spectrometer which uses a grating monochromator.
- Dispersion occurs when energy falling on the entrance slit is collimated onto the Dispersive element.
- The dispersed radiation is then reflected back to the exit slit, beyond which lies the detector.
- The dispersed spectrum is then scanned

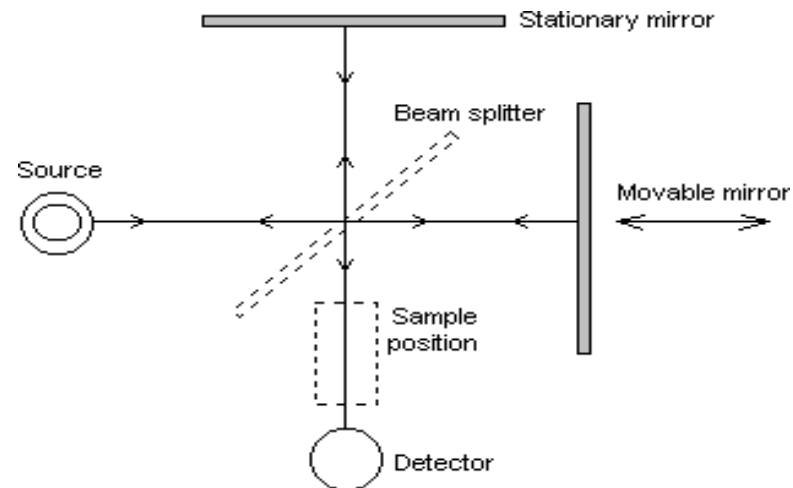


Types of instruments...

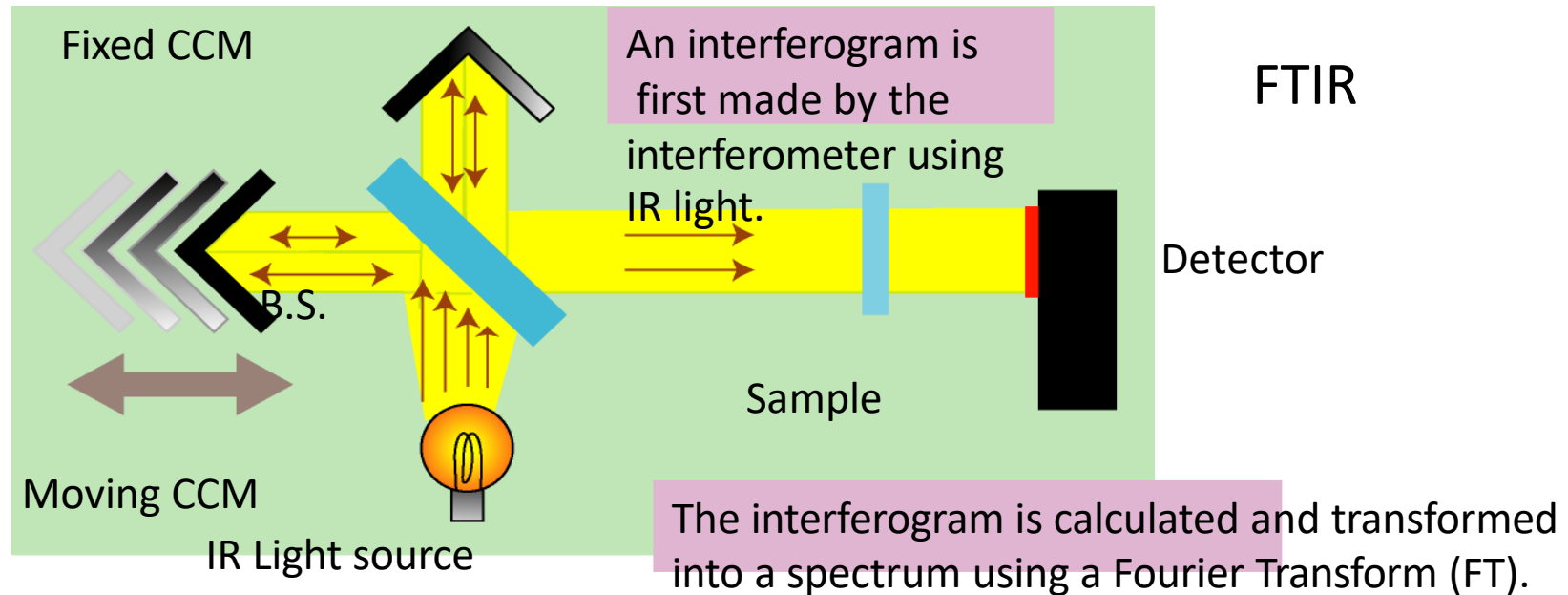
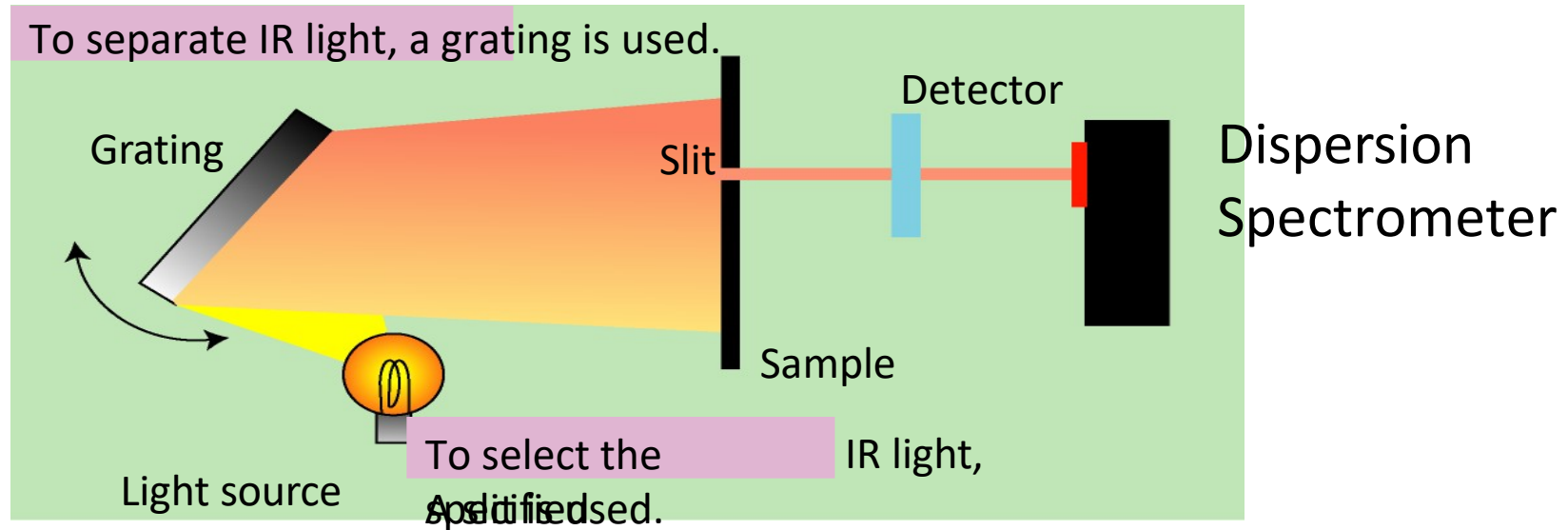
❑ Fourier transform-IR spectrometer

- The IR radiation from a broadband source is first directed into an interferometer,
 - where it is divided and then recombined after the split beams travel different optical paths to generate constructive and destructive interference.
- Next, the resulting beam passes through the sample compartment and reaches to the detector.

- Most FTIR spectrometers are single-beam instruments



Comparison Between Dispersion Spectrometer and FTIR



Advantages of FT-IR

Better speed

- A complete spectrum can be obtained during a single scan of the moving mirror, while the detector observes all frequencies simultaneously.
- An interferometer in an FT-IR instrument does not separate energy into individual frequencies for measurement of the infrared spectrum.
- In contrast, every wavelength across the spectrum must be measured individually in a dispersive spectrometer.
- This is a slow process, and typically only one measurement scan of the sample is made in a dispersive instrument.
- FT-IR advantage is that many scans can be completed and combined on an FT-IR in a shorter time than one scan on a dispersive instrument.

Advantage...

- **Better sensitivity and Resolution**

② An FT-IR instrument does not use a slit to limit the individual frequency reaching the sample and detector as a dispersive instrument does.

There are also fewer mirror surfaces in an FT-IR spectrometer, so there are less reflection losses than in a dispersive spectrometer.

Overall, more energy reaches the sample and hence the detector in an FT-IR spectrometer than in a dispersive spectrometer.

This means that the signal-to-noise ratio of an infrared spectrum measured on an FT-IR is higher than the signal-to-noise ratio attained on a dispersive instrument

Higher sensitivity and clear spectrum

Simpler mechanical design

- ☐ There is only one moving part, the moving mirror, resulting in better reliability.

Elimination of stray light and emission contributions

- ☐ The interferometer in FTIR modulates all the frequencies.
- ☐ The unmodulated stray light and sample emissions (if any) are not detected.

Powerful data station

- ☐ Modern FTIR spectrometers are usually equipped with a powerful, computerized data system.
- ☐ It can perform a wide variety of data processing tasks such as Fourier transformation, baseline correction, and library searching.

□ Sample cells

- It is possible to analyze samples in the liquid, solid or gaseous forms when using this approach.
- The material used to contain the Sample must always be transparent to IR radiation. This limits the selection of to certain material only.

Window material	Useful range (cm ⁻¹)	Properties
NaCl	40 000–600	Soluble in water; slightly soluble in alcohol; low cost; fair resistance to mechanical and thermal shock; easily polished
KBr	43 500–400	Soluble in water and alcohol; slightly soluble in ether; hygroscopic; good resistance to mechanical and thermal shock
CaF ₂	77 000–900	Insoluble in water; resists most acids and bases; does not fog; useful for high-pressure work
BaF ₂	66 666–800	Insoluble in water; soluble in acids and NH ₄ Cl; does not fog; sensitive to thermal and mechanical shock
KCl	33 000–400	Similar properties to NaCl but less soluble; hygroscopic
CsBr	42 000–250	Soluble in water and acids; hygroscopic
CsI	42 000–200	Soluble in water and alcohol; hygroscopic

Sample cells...

Sample preparation

➤ Solid samples- 3 different approaches

- First- the sample may be ground to a powder and made into a thick slurry or mull by grinding it with viscous liquid such as **Nujol (paraffin oil)**.
- The mull is placed between two sodium chloride plates and plates are subjected to IR beam.
- Second- is the **KBr pellet method**, which involves mixing a finely ground solid sample with powdered KBr . The mixture is pressed under very high pressure to form a small disc about 1cm in diameter.
 - KBr pellet is then mounted on holder and is placed in sample beam of IR spectrophotometer.
- Third- the **sample is deposited** on the surface of the KBr or NaCl cell by evaporating a **solution** of the solid to dryness.
- IR radiation is then passed through the thin layer deposited



Sample cells...

Sample preparation...

- ✓ In general, analysis of solid samples **do not yield spectra as fine as liquid samples.**
- **Liquid samples-** Liquids are studied neat or in solution.
 - In case of neat liquid, a thin film of < 0.01 mm thickness is obtained by pressing the liquid between two sodium chloride plates and plates are subjected to IR beam.
 - Spectra of solutions are obtained by taking 1-10 % solution of the sample in an appropriate solvent in cells of 0.1-1 mm thickness.
 - A compensating cell, containing pure solvent is placed in the reference beam of the instrument.
 - The choice of solvent depends on the **solubility** of the sample and its own **minimal absorption** in IR region.
 - **Carbon tetrachloride, chloroform and carbon disulfide are preferred solvents.**

Sample cells...

Sample preparation...

- Gases- the gas sample cell is similar to the cell for liquid samples.
- Gases have densities which are several orders of magnitude less than liquids, and hence pathlengths must be correspondingly greater, usually 10 cm or longer.

Con't

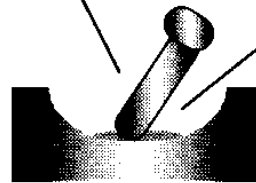
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Preparation as mull

Sandwich between NaCl discs



Grind sample in
liquid paraffin

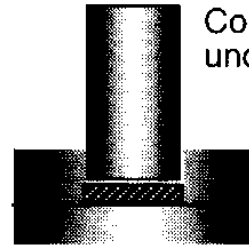


Preparation as KBr disc

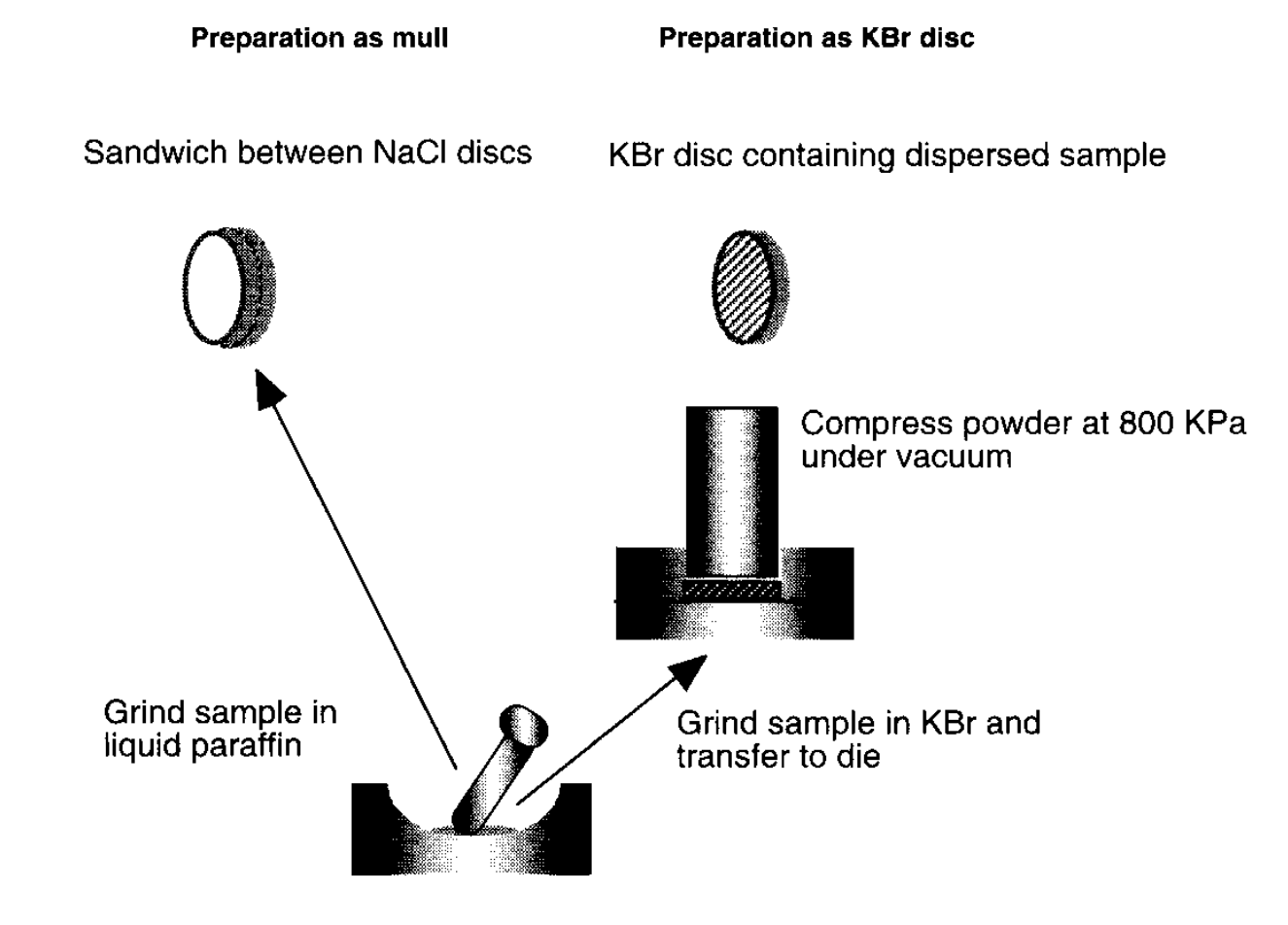
KBr disc containing dispersed sample



Compress powder at 800 KPa
under vacuum



Grind sample in KBr and
transfer to die



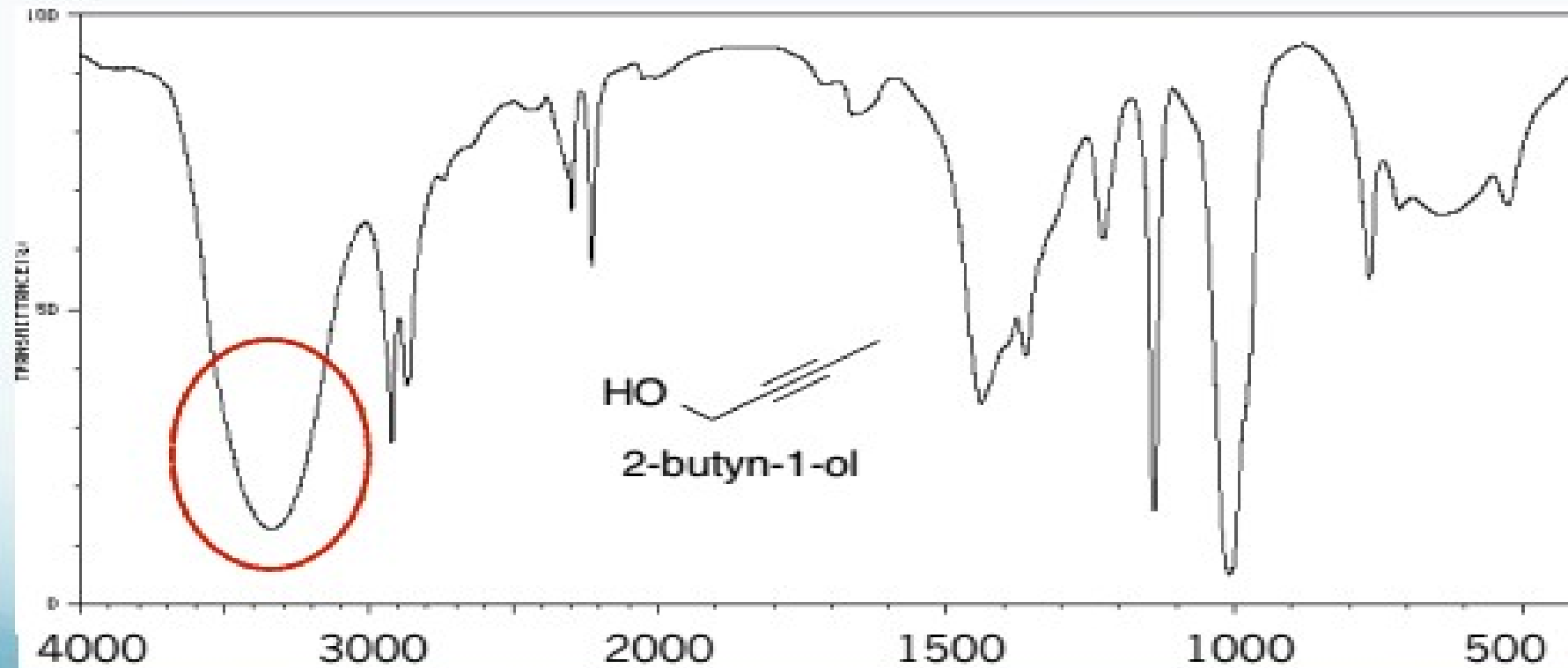
Solvents

1. Must be transparent in the region studied: no single solvent is transparent throughout the entire IR region
2. Water and alcohols are seldom employed to avoid O-H band of water .
3. Must be chemically inert (does not react with substance or cell holder).
 - CCl_4 , CS_2 , or CHCl_3 ; may be used but we should consider its IR spectrum

Location of Peaks in an IR Spectrum

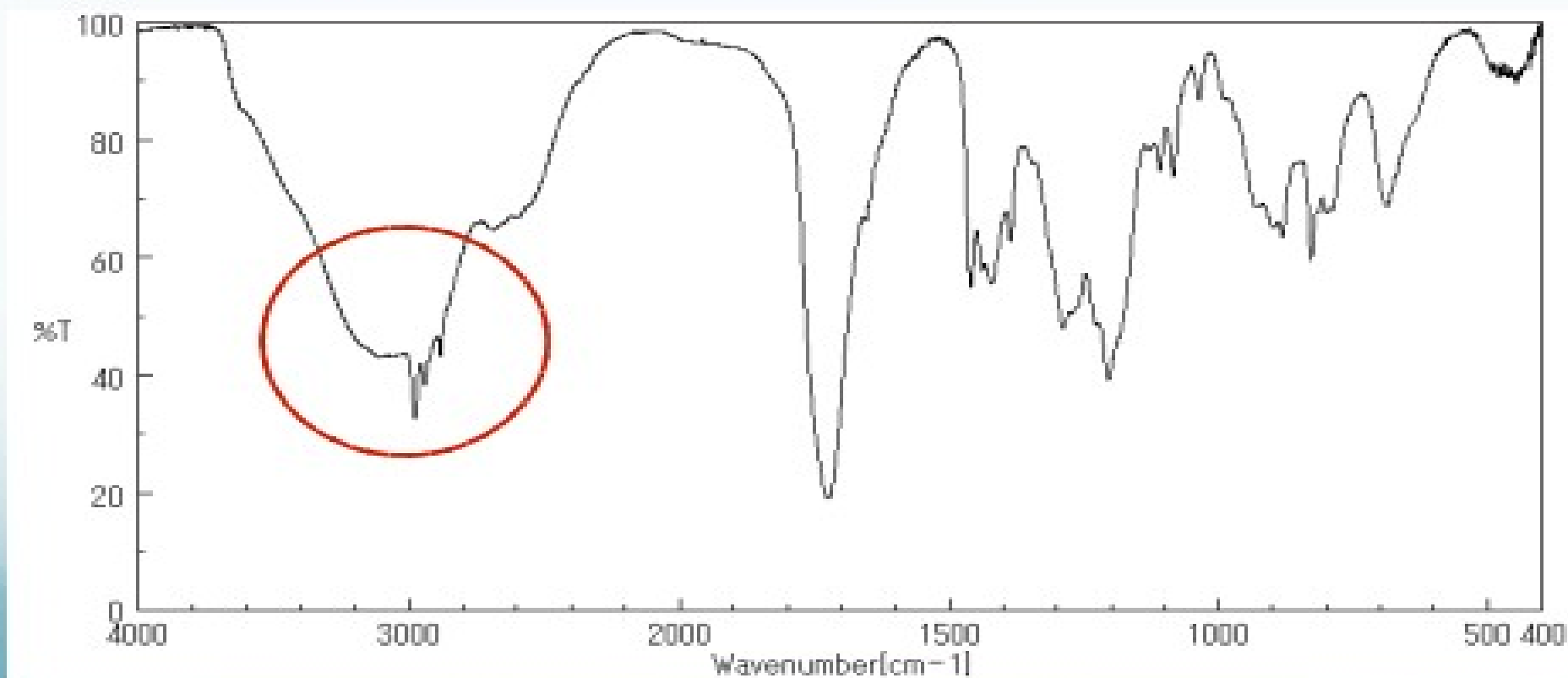
Hydroxyl Group, OH

Generally broad and intense $\sim 3300\text{ cm}^{-1}$



Carboxylic Acid OH Group

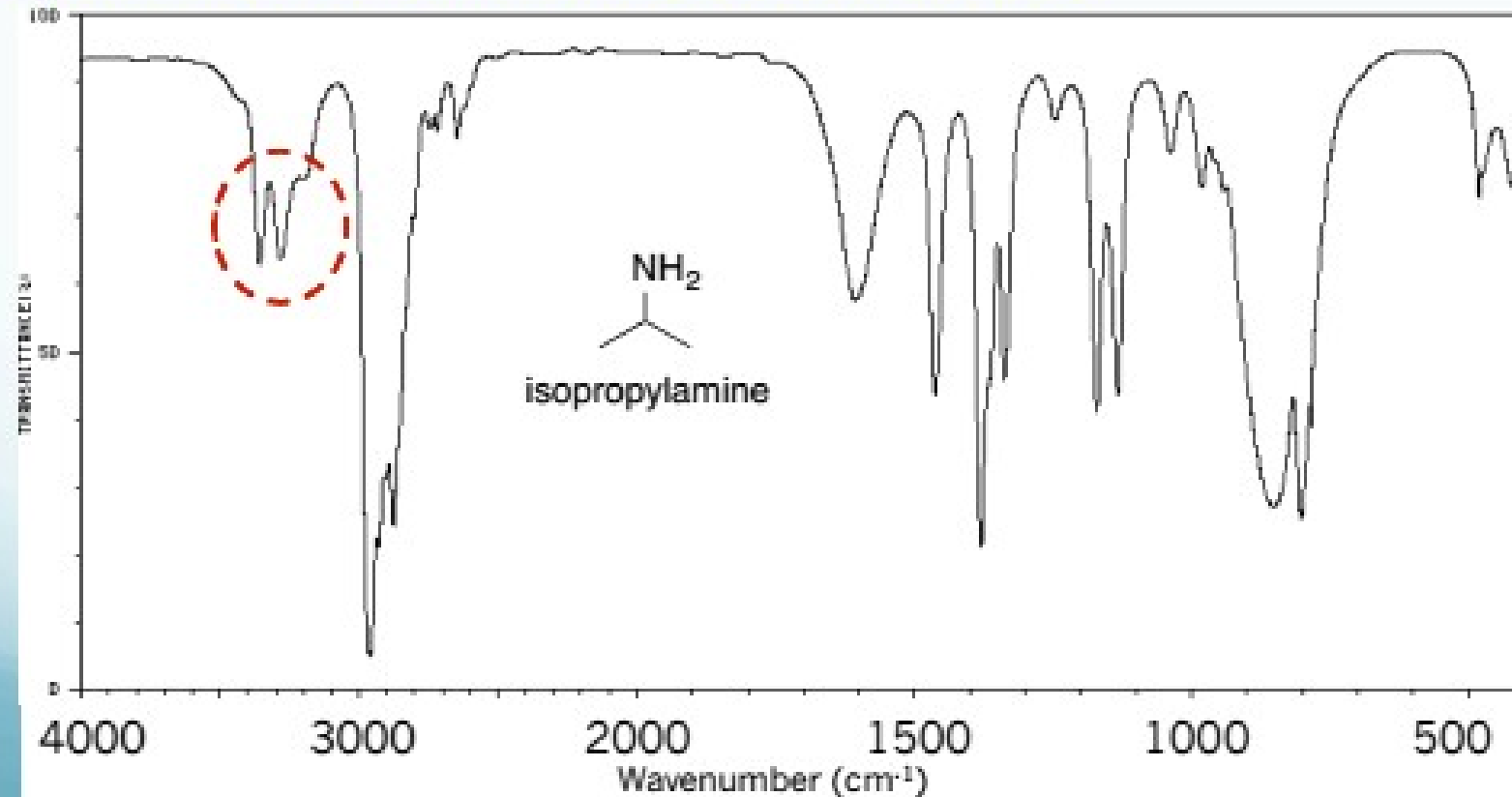
A Carboxylic Acid O–H stretch is generally lower in frequency and even more broad than an alcohol OH. Often $\sim 3000\text{ cm}^{-1}$



Amine Group, NH and NH₂

Generally less broad and less intense than the OH stretch. Generally appear between 3300 and 3500 cm⁻¹

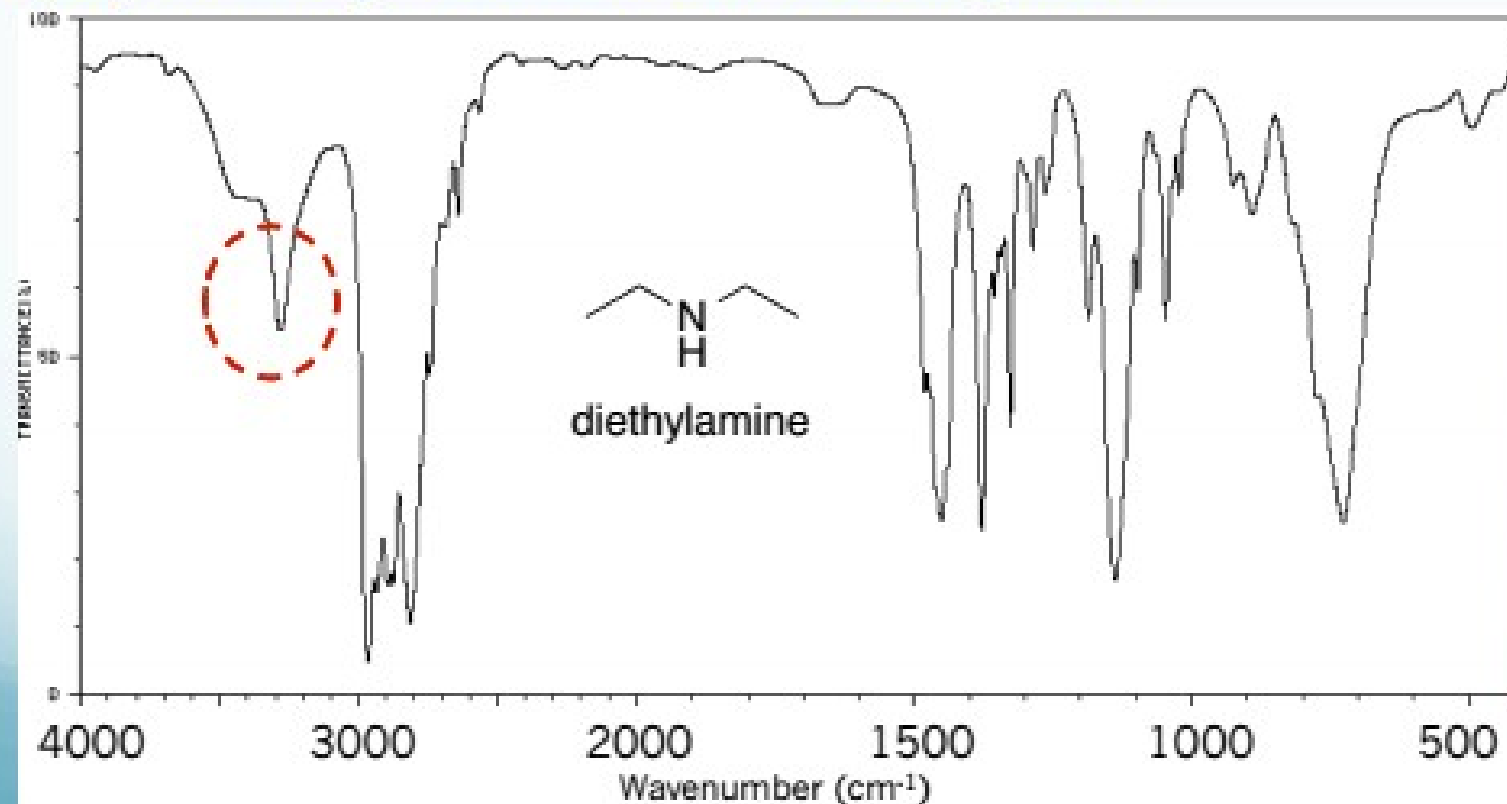
Primary amines R-NH₂ exhibit two N-H stretching bands.



Amine Group, NH and NH₂

Generally less broad and less intense than the OH stretch. Generally appear between 3300 and 3500 cm⁻¹

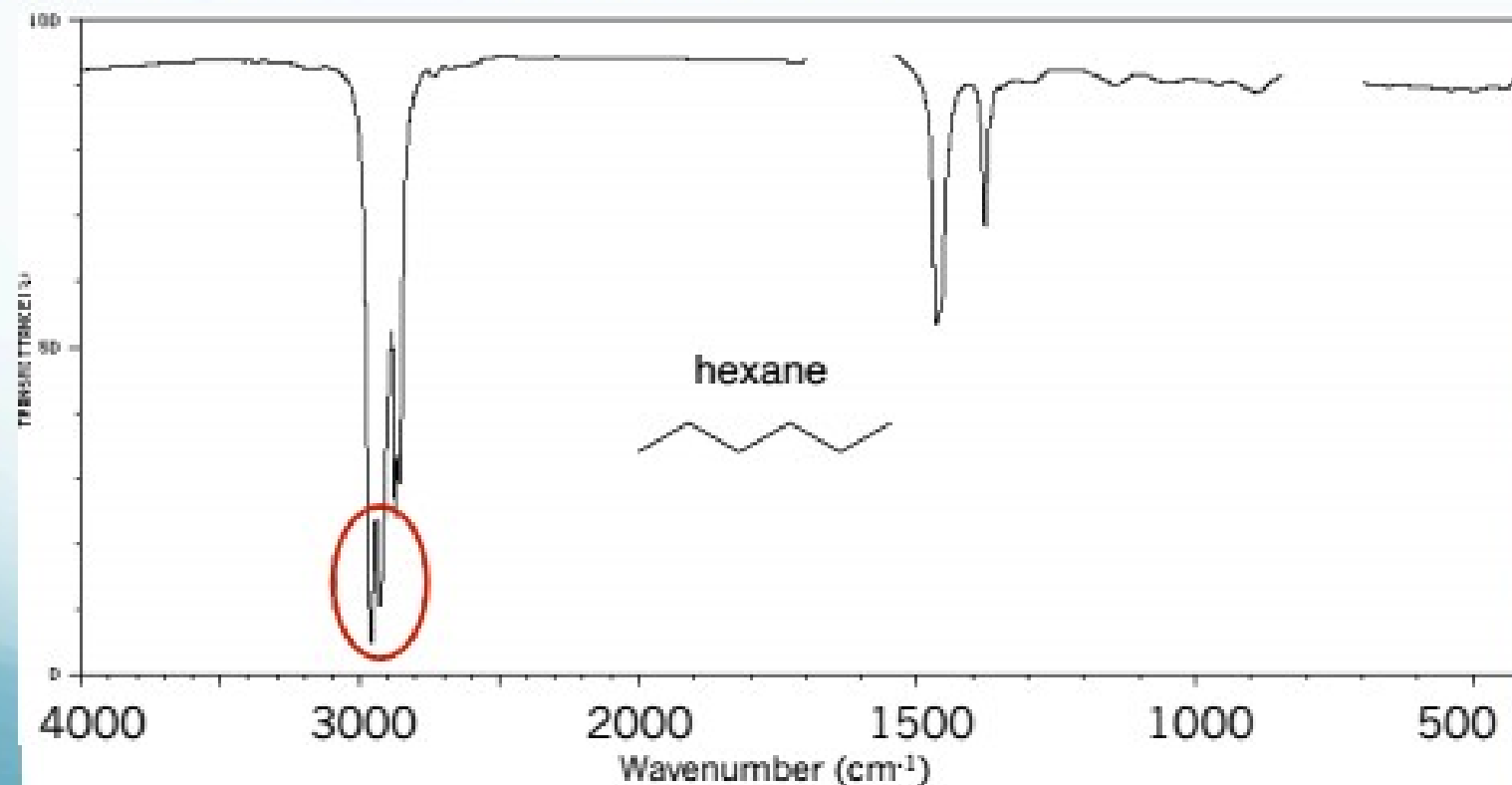
Secondary amines R₂NH exhibit one N-H stretching band.



Carbon-Hydrogen, CH

All C-H stretches occur between 2700 and 3300 cm^{-1} . They are relatively intense and typically quite narrow.

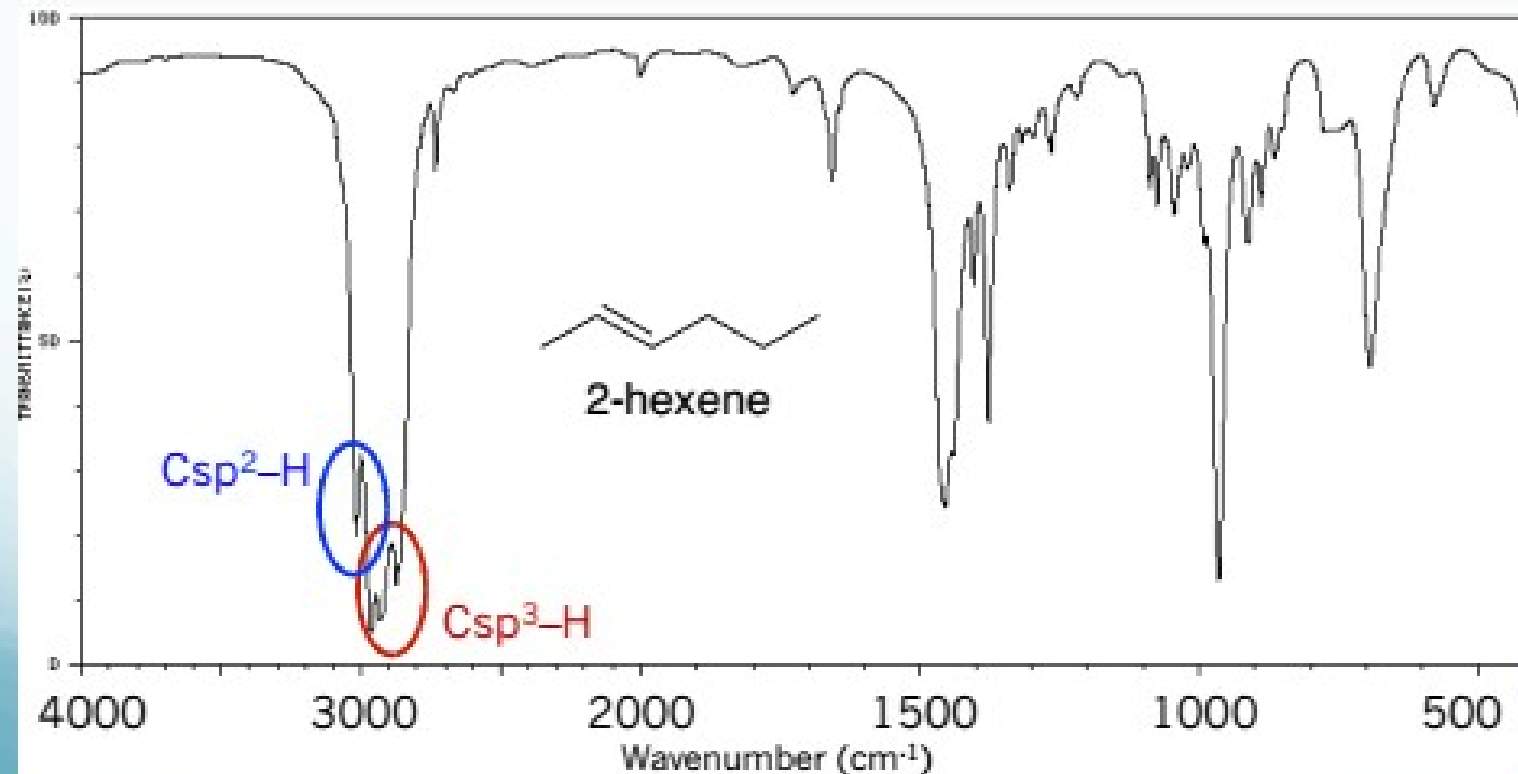
$\text{Csp}^3\text{-H}$ ($-\text{CH}_2\text{-H}$) appear just below 3000 cm^{-1}



Carbon-Hydrogen, CH

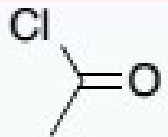
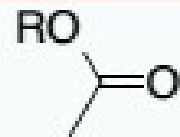
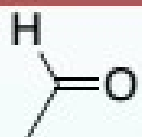
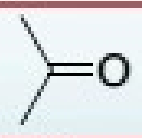
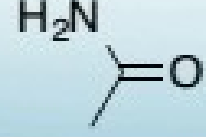
All C–H stretches occur between 2700 and 3300 cm^{-1} . They are relatively intense and typically quite narrow.

$\text{Csp}^2\text{--H}$ ($\text{C}=\text{C}\text{--H}$) appear just above 3000 cm^{-1}



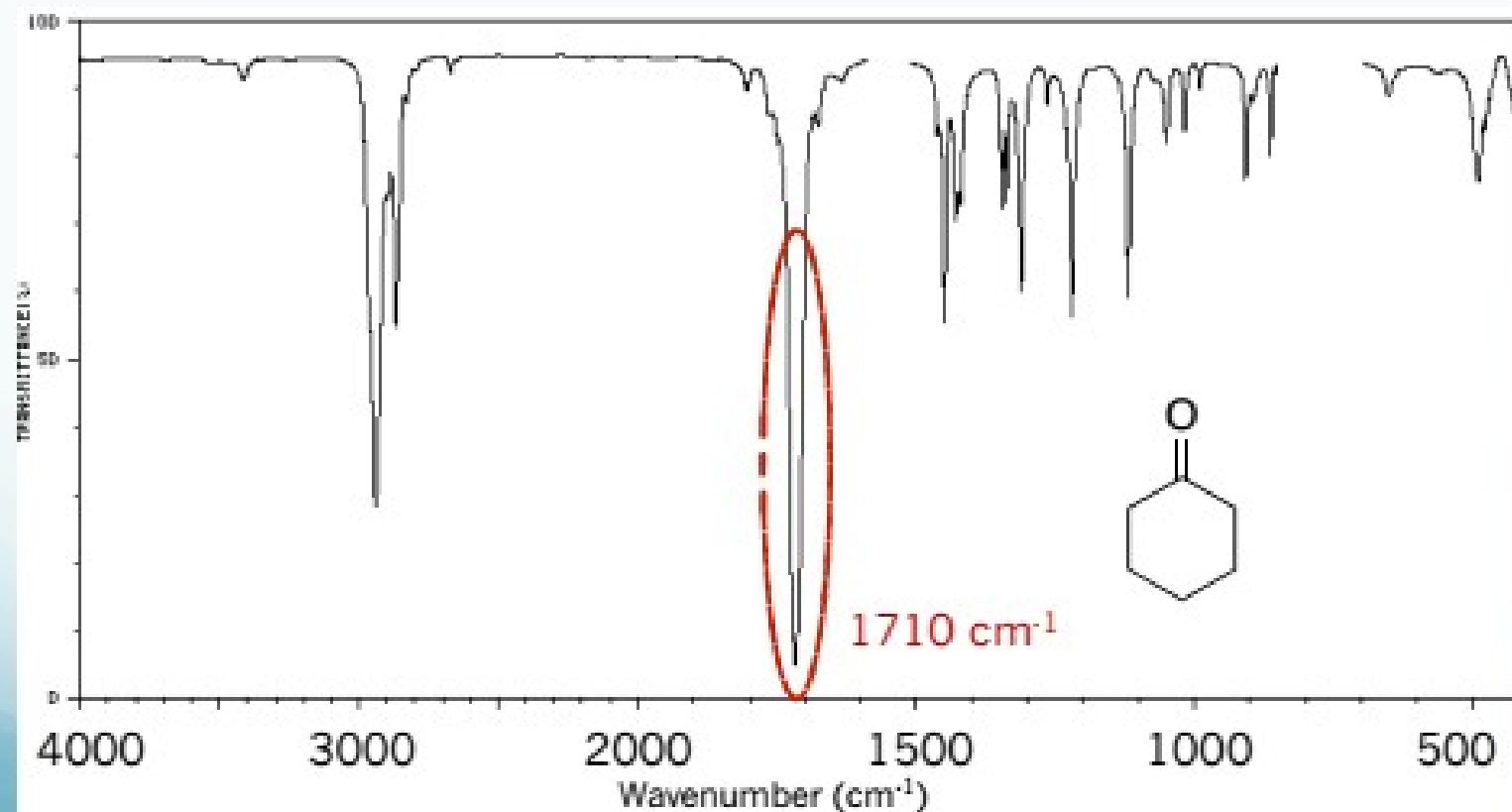
Carbonyl Group, C=O

The C=O stretch is very strong and generally appears between 1650-1850 cm^{-1}

	1750-1850
	1700-1750
	1720-1740
	1680-1750
	1650-1700

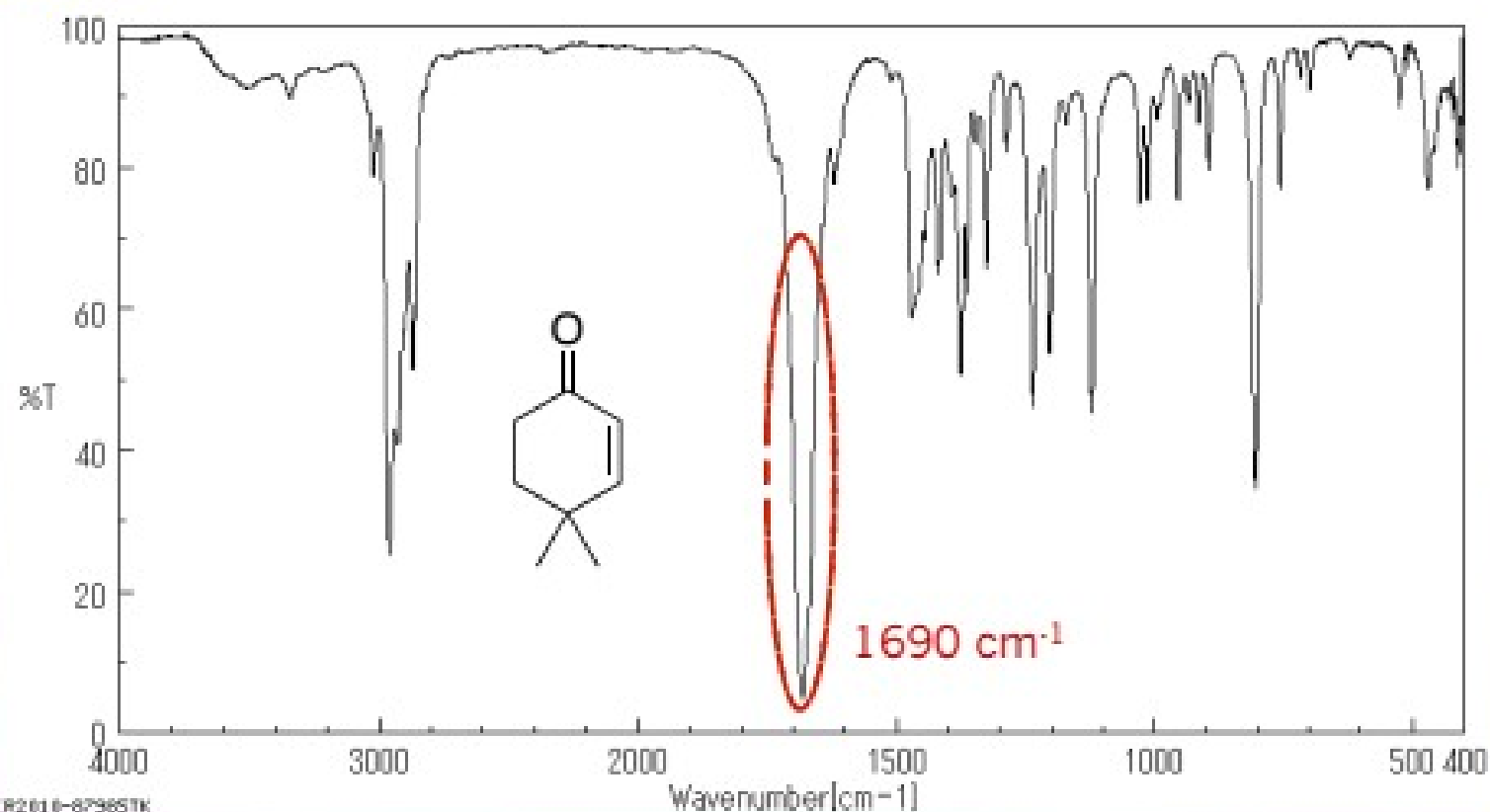
Carbonyl Group, C=O

The C=O stretch is very strong and generally appears between 1650-1850 cm^{-1}



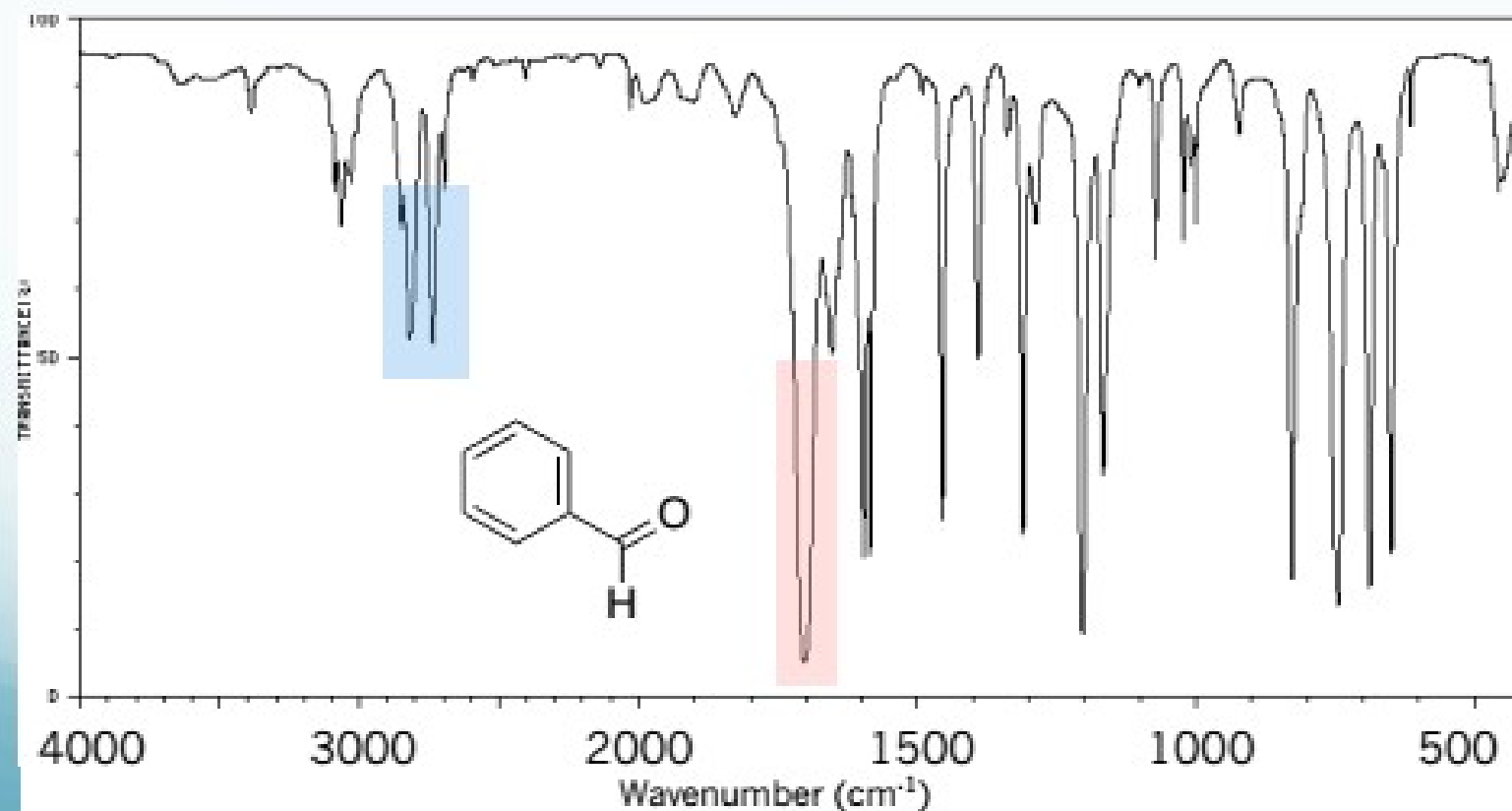
Carbonyl Group, C=O

When the C=O is in conjugation to a C=C or a C≡C, the frequency is lowered 20-40 cm^{-1}



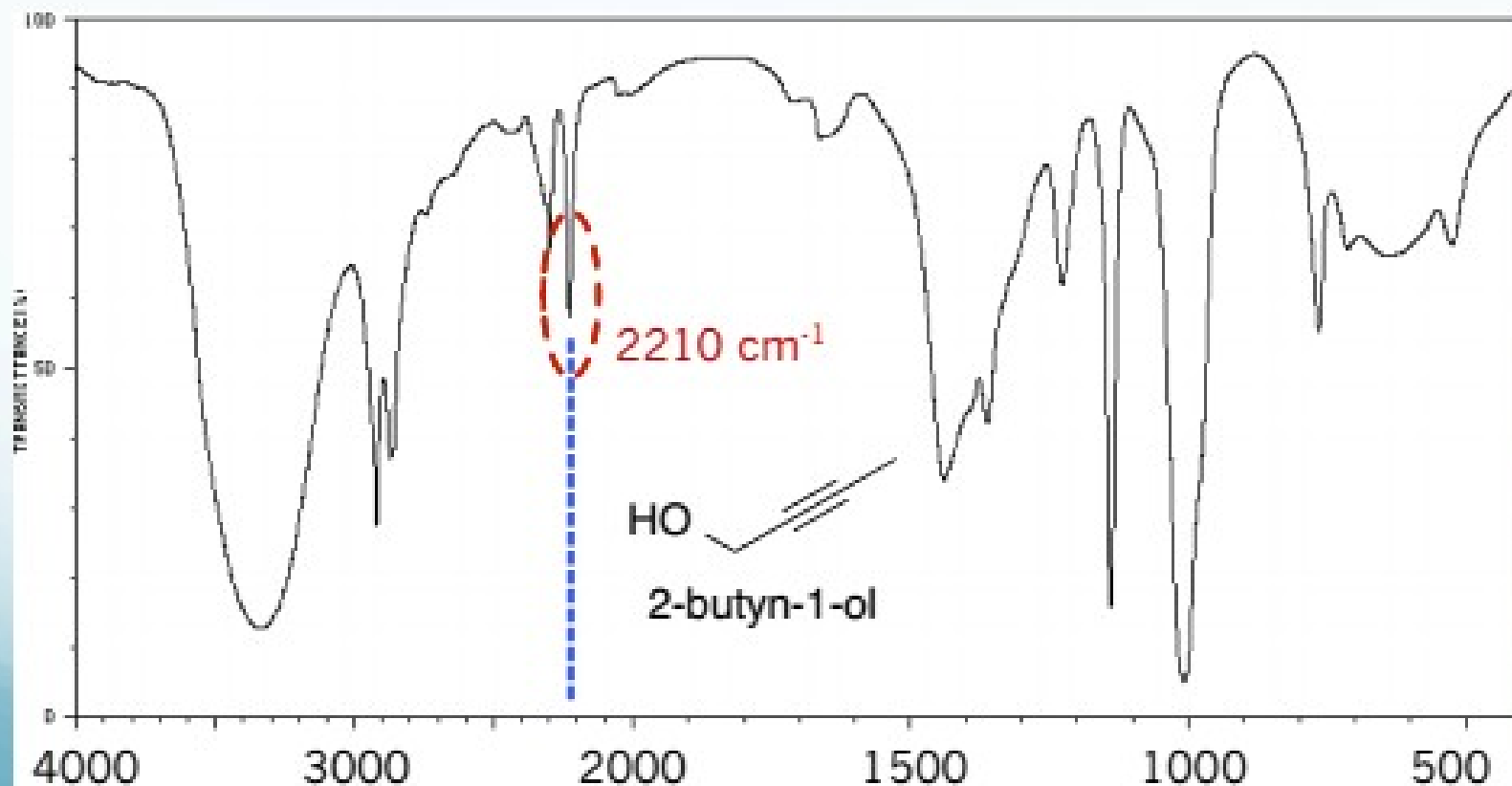
Carbonyl Group: Aldehyde

In addition to the C=O stretch, the aldehyde C-H stretch appears as two bands. One will be at ~ 2720 and the other at ~ 2820 cm^{-1} .



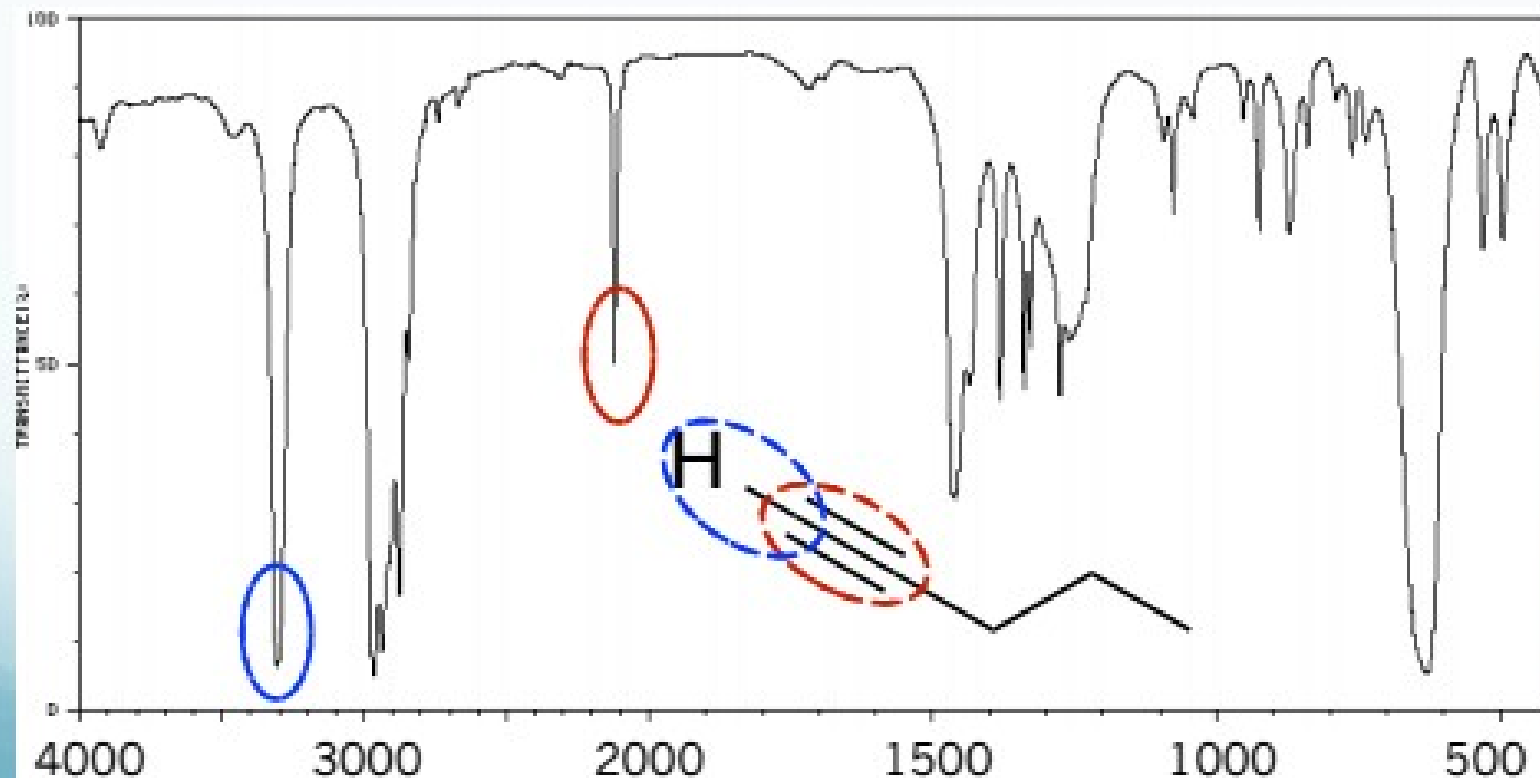
Alkyne ($\text{C}\equiv\text{C}$) and Nitrile ($\text{C}\equiv\text{N}$) Groups

Both $\text{C}\equiv\text{C}$ and $\text{C}\equiv\text{N}$ bonds appear as a sharp peak between 2100 and 2260 cm^{-1} . Internal alkynes and nitriles are generally difficult to distinguish.



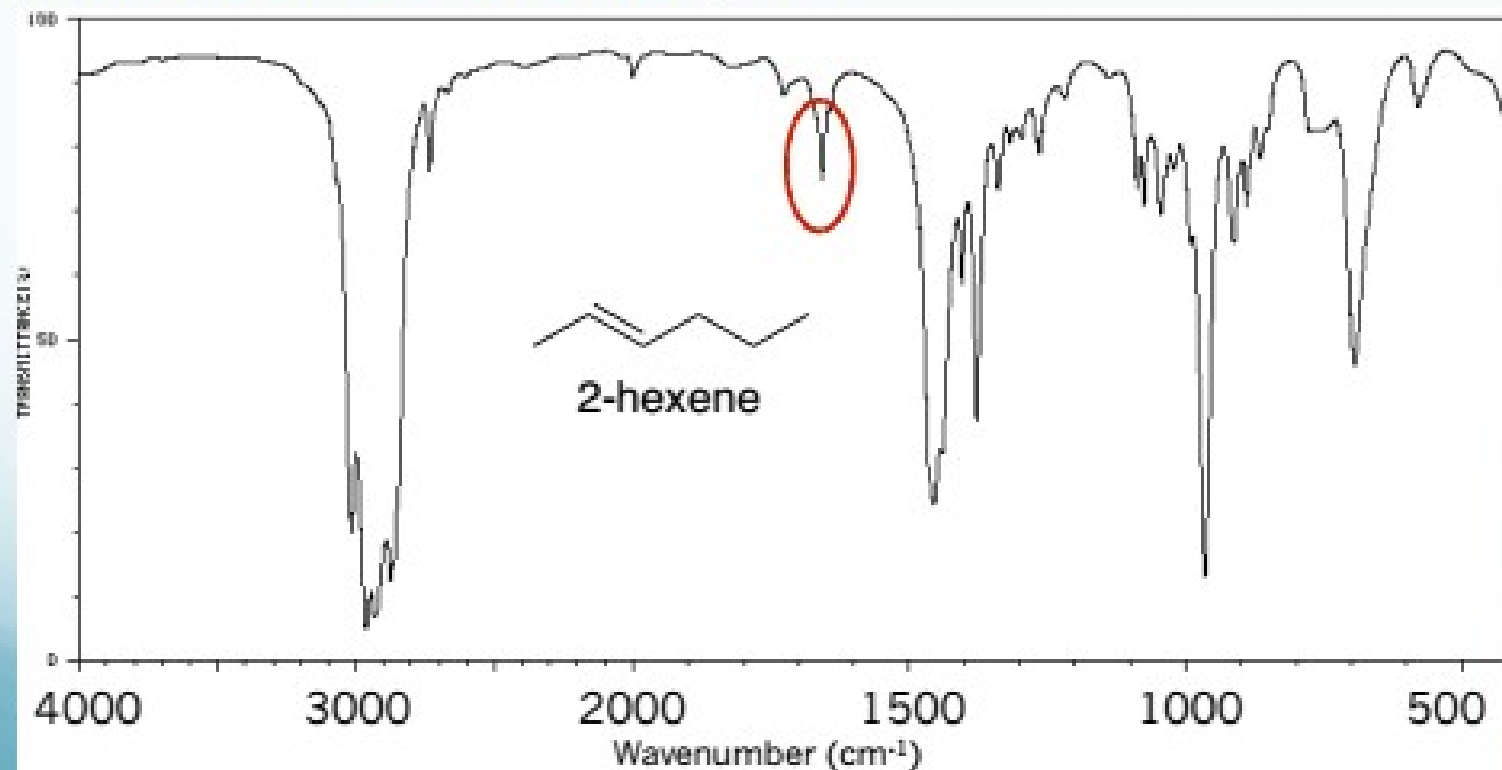
Alkyne ($\text{C}\equiv\text{C}$) and Nitrile ($\text{C}\equiv\text{N}$) Groups

Terminal alkynes are much easier to identify. In addition to the $\text{C}\equiv\text{C}$ stretch, a sharp C-H stretch at $\sim 3300\text{ cm}^{-1}$ will also be observed.



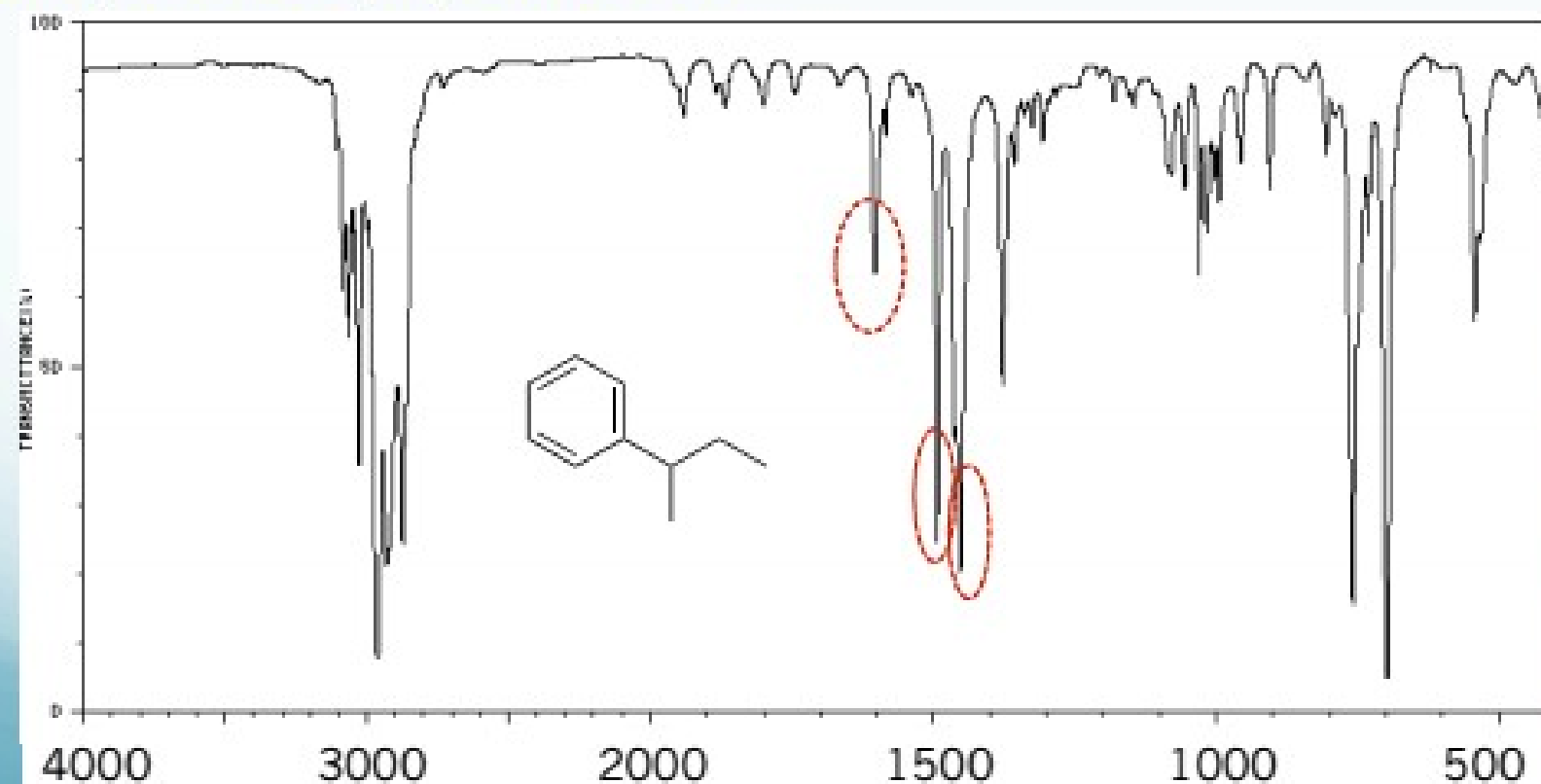
Carbon-Carbon Double Bond (C=C)

C=C stretches are one of the more difficult functional groups to identify in an IR spectrum. Alkene stretches occur between 1620 and 1680 cm^{-1} . They are sharper and less intense than C=O signals.



Aromatic Carbon-Carbon Double Bond (C=C)

C=C stretches corresponding to an aromatic ring generally occur between 1450 and 1600 cm^{-1} . There can be up to three signals: One at ~ 1600 , One at ~ 1500 , and one slightly $< 1500 \text{ cm}^{-1}$.



Carbon Oxygen Single Bond Stretch (C–O)

$\text{Csp}^3\text{--O}$ 1000-1100 cm^{-1}

$\text{Csp}^2\text{--O}$ 1200-1300 cm^{-1}



The carbon—oxygen bond in **an alcohol** is a **pure single bond**.

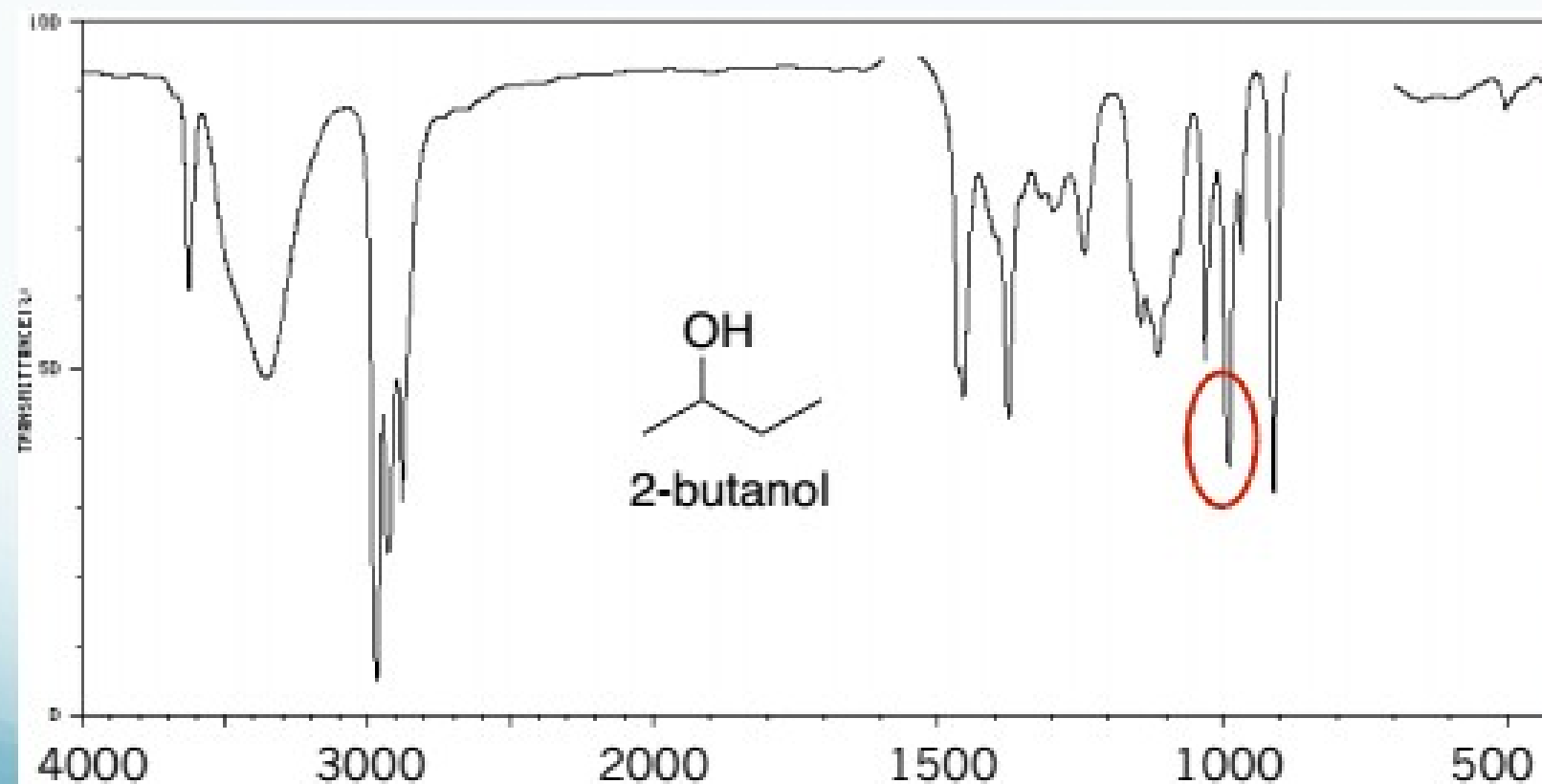
The carbon—oxygen bond in **an ether** is a **pure single bond**.

The carbon—oxygen single bond in **a carboxylic acid** has **partial double bond character**.

One carbon—oxygen single bond in **an ester** is a **pure single bond** and one has **partial double bond character**.

Carbon Oxygen Single Bond Stretch (C-O)

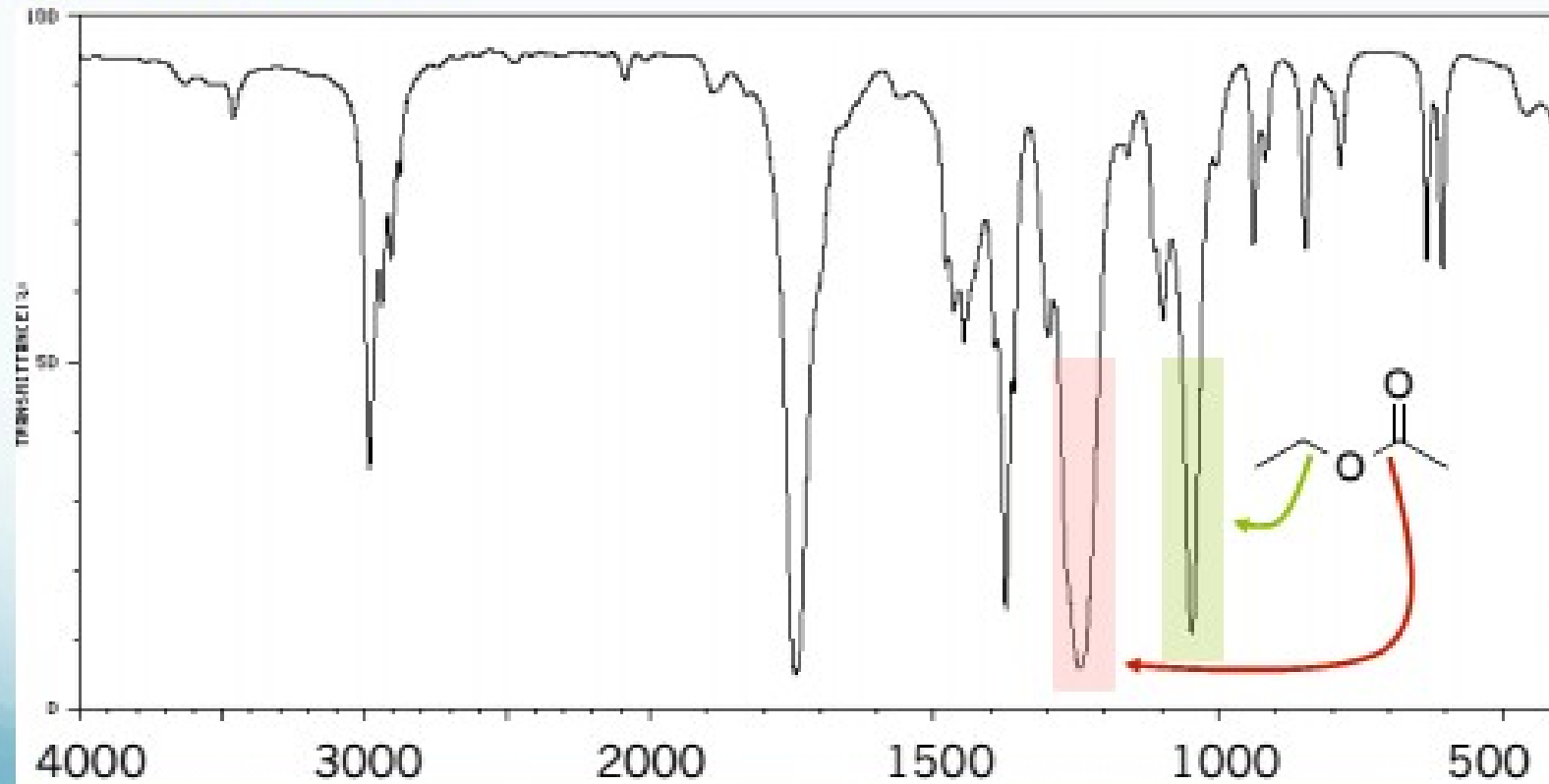
$\text{Csp}^3\text{-O}$ 1000-1100 cm^{-1}



Carbon Oxygen Single Bond Stretch (C-O)

$\text{Csp}^3\text{-O}$ 1000-1100 cm^{-1}

$\text{Csp}^2\text{-O}$ 1200-1300 cm^{-1}

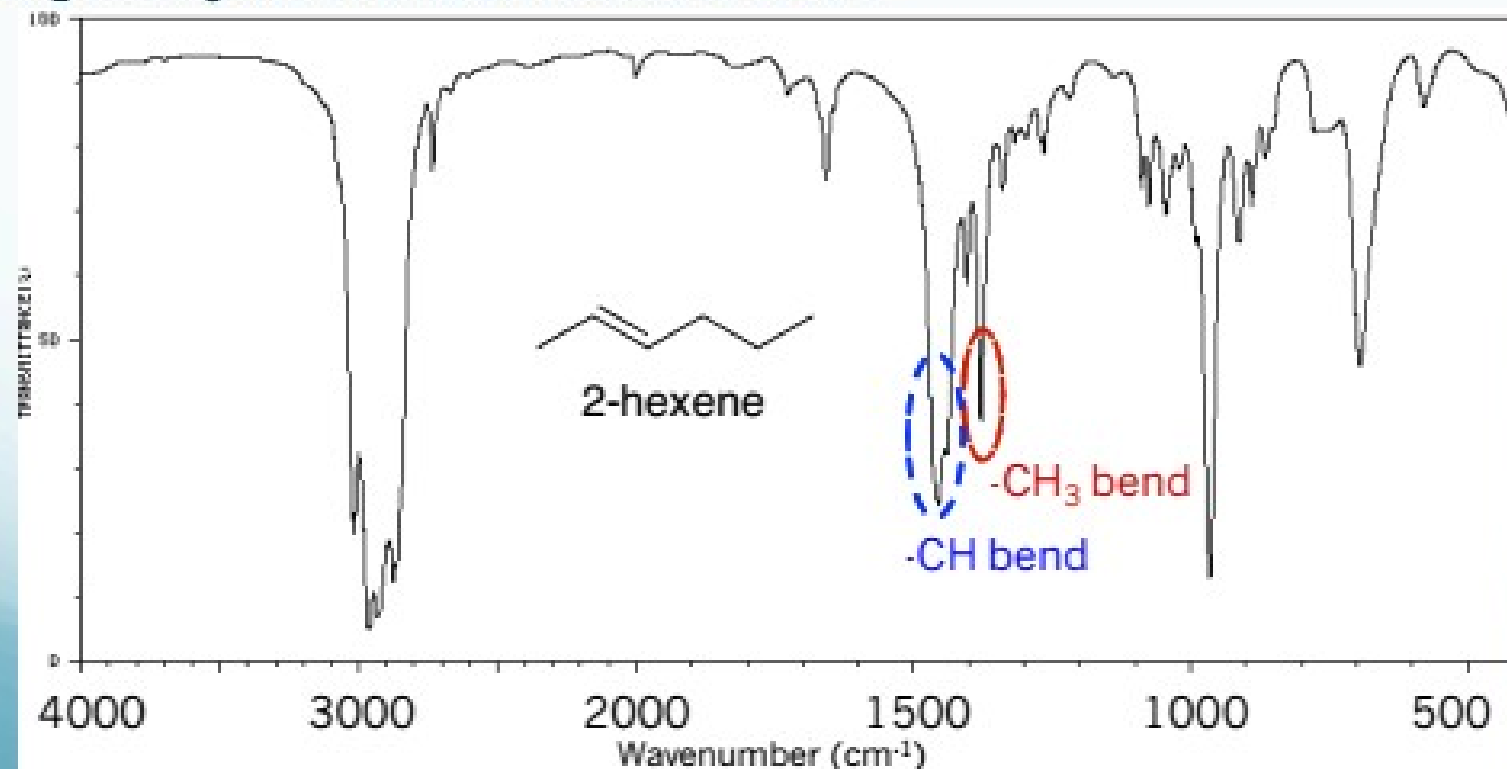


Carbon-Hydrogen (C-H) Bending Vibrations

Bending vibrations are generally less pronounced than stretching. Additionally, because the bending vibrations are generally found in the fingerprint region, they are often hard to identify.

-CH₃ bending just below 1400 cm⁻¹

R₂CH₂ and R₃CH bending just above 1400 cm⁻¹



How To Approach The Analysis Of A Spectrum

1. Is a carbonyl group present?

- The C=O group gives rise to a strong absorption in the region 1820-1660 cm^{-1} . peak is often the **strongest in the spectrum and of medium width.** (You can't miss it)

2. If C=O is present, check the following types (if absent, go to 3).

➤ ACIDS is OH also present?

- ✓ *broad* absorption near 3400-2400 cm^{-1} (usually overlaps C-H)

➤ AMIDES is NH also present?

- ✓ medium absorption near 3500 cm^{-1}

➤ ESTERS is C-O also present?

- ✓ strong intensity absorptions near 1300-1000 cm^{-1}

➤ ANHYDRIDES

- ✓ have *two* C=O bands near 1810 and 1760 cm^{-1}

How To Approach The Analysis....

➤ **ALDEHYDES** is CH present?

✓ Two weak absorptions near 2850 & 2750 cm^{-1} on the right-hand side of CH band

➤ **KETONES** The above 5 choices have been eliminated

3. If C=O is absent

➤ **ALCOHOLS** Check for OH

➤ **PHENOLS** - *broad* absorption near 3400-2400 cm^{-1}

✓ confirm this by finding C-O near 1300-1000 cm^{-1}

➤ **AMINES** Check for NH medium absorptions(s) near 3500 cm^{-1}

➤ **ETHERS** Check for C-O (and absence of OH) near 1300-1000 cm^{-1}

How To Approach The Analysis...

4. Double Bonds and/or Aromatic Rings

- C=C is a *weak* absorption near 1650 cm^{-1}
- Medium to strong absorptions in the region $1650\text{-}1450\text{ cm}^{-1}$ often imply an aromatic ring.
- Confirm the above by consulting the CH region; aromatic and vinyl CH occurs .
- The left of 3000 cm^{-1} (aliphatic CH occurs to the right of this value)

How To Approach The Analysis...

5. Triple Bonds

- $\text{C}\equiv\text{N}$ is a medium, sharp absorption near 2250 cm^{-1}
- $\text{C}\equiv\text{C}$ is a weak but sharp absorption near 2150 cm^{-1}
- Check also for acetylenic CH near 3300 cm^{-1}

6. Nitro Groups

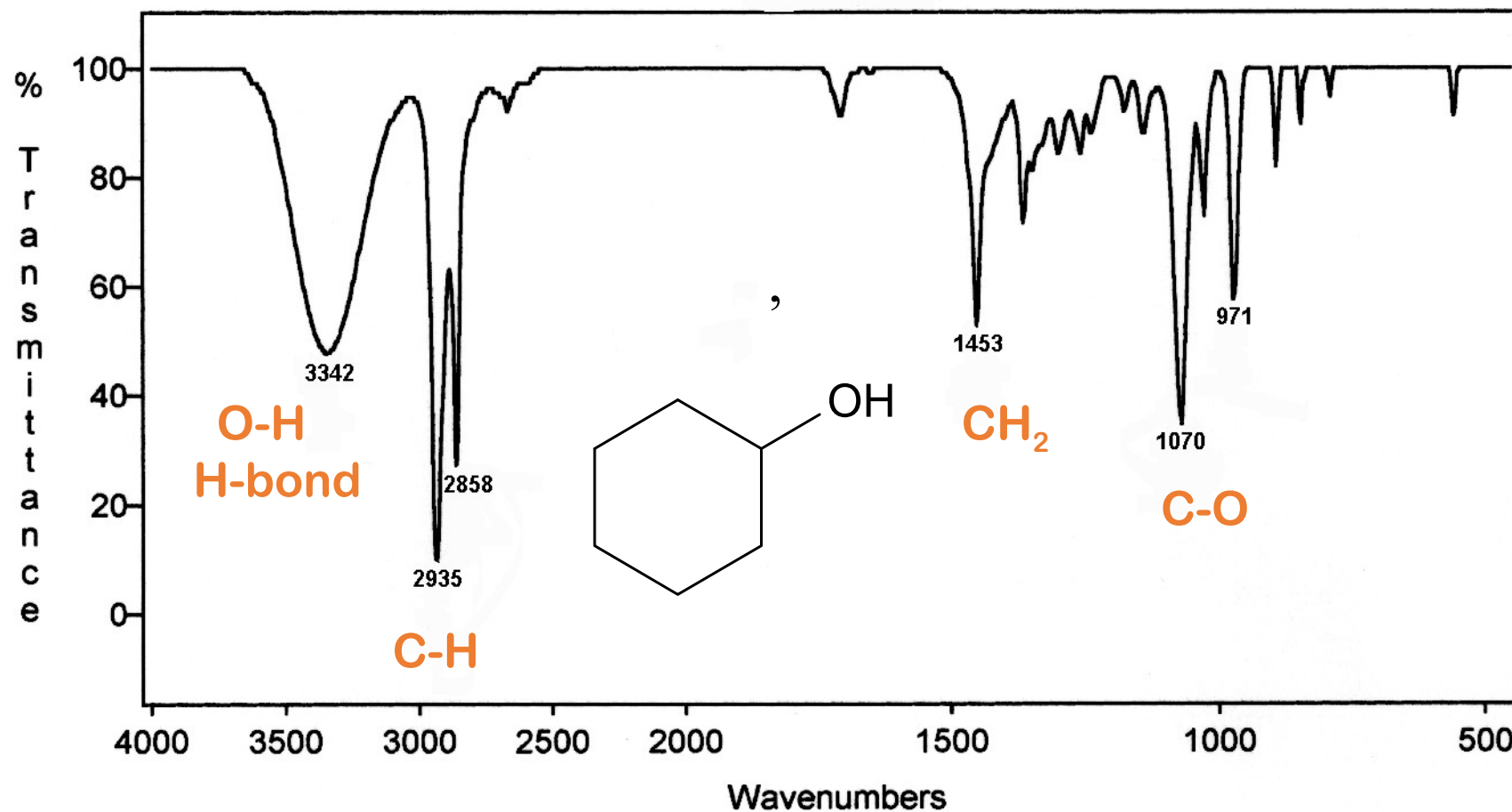
- two strong absorptions at $1600 - 1500\text{ cm}^{-1}$ and $1390-1300\text{ cm}^{-1}$

7. Hydrocarbons

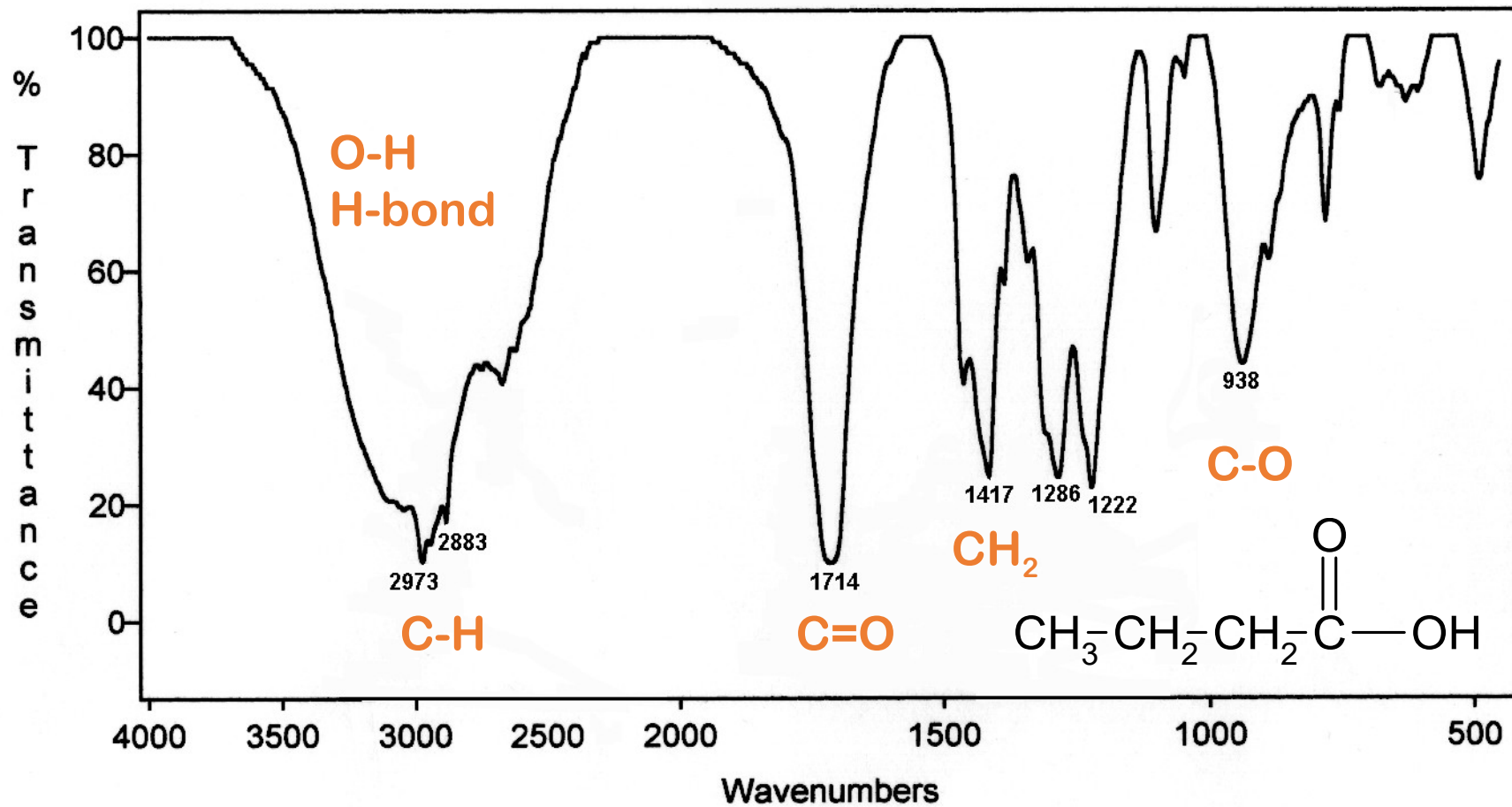
- none of the above are found
- major absorptions are in CH region near 3000 cm^{-1}
- very simple spectrum, only other absorptions near 1450 cm^{-1} and 1375 cm^{-1} .

ALCOHOL

Cyclohexanol

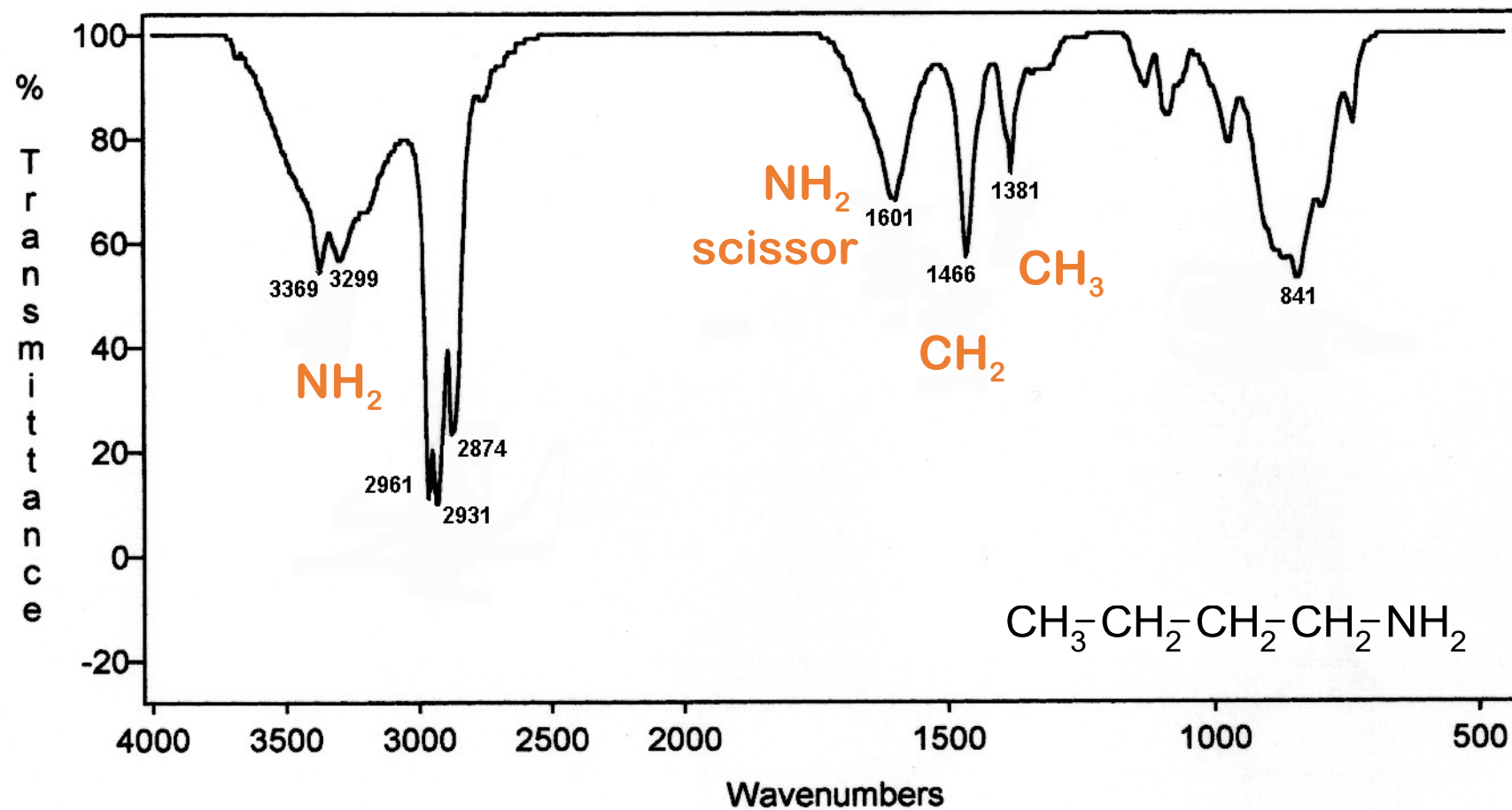


Butanoic Acid



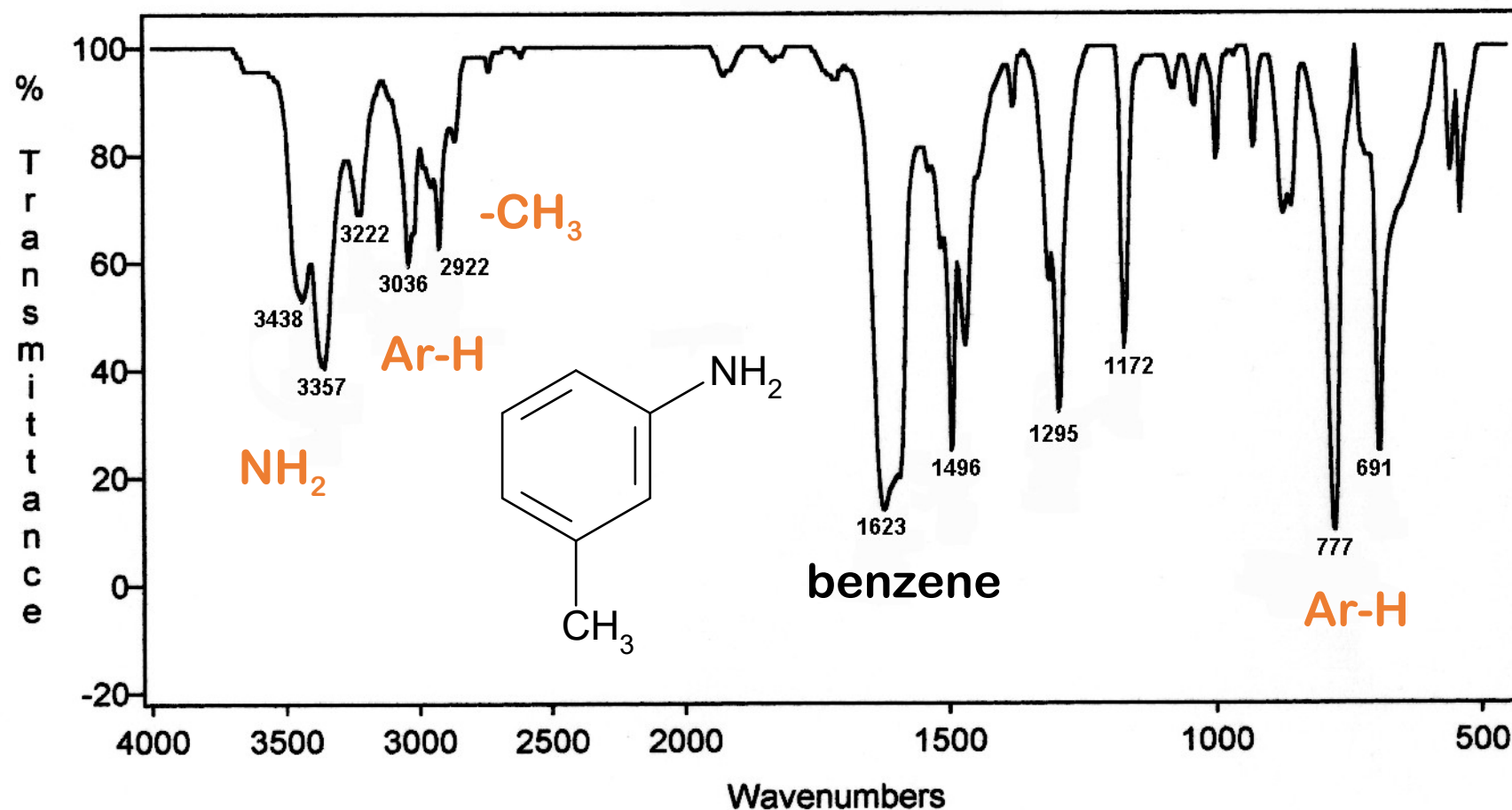
PRIMARY AMINE
aliphatic

1-Butanamine



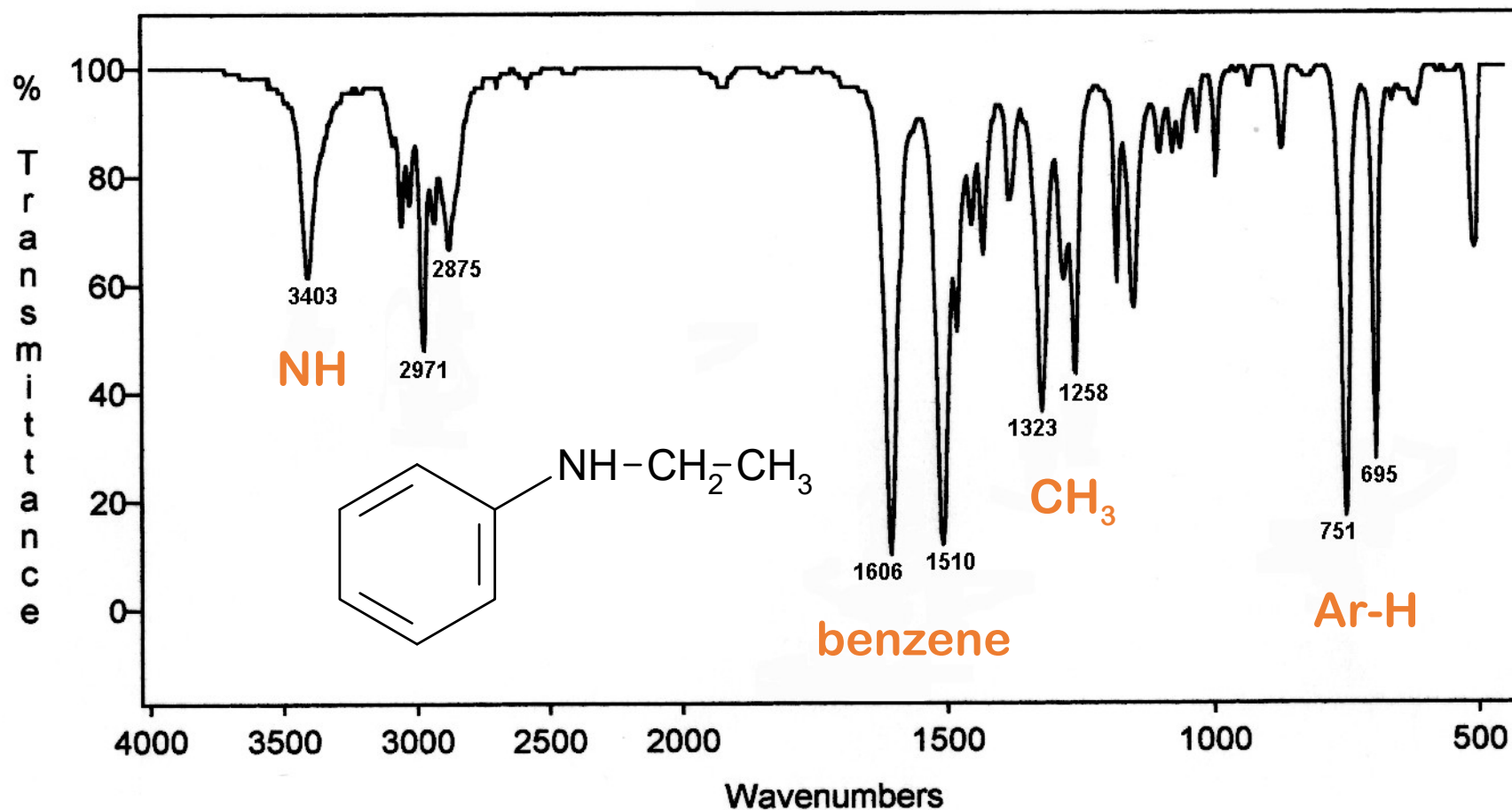
PRIMARY AMINE
aromatic

3-Methylbenzenamine



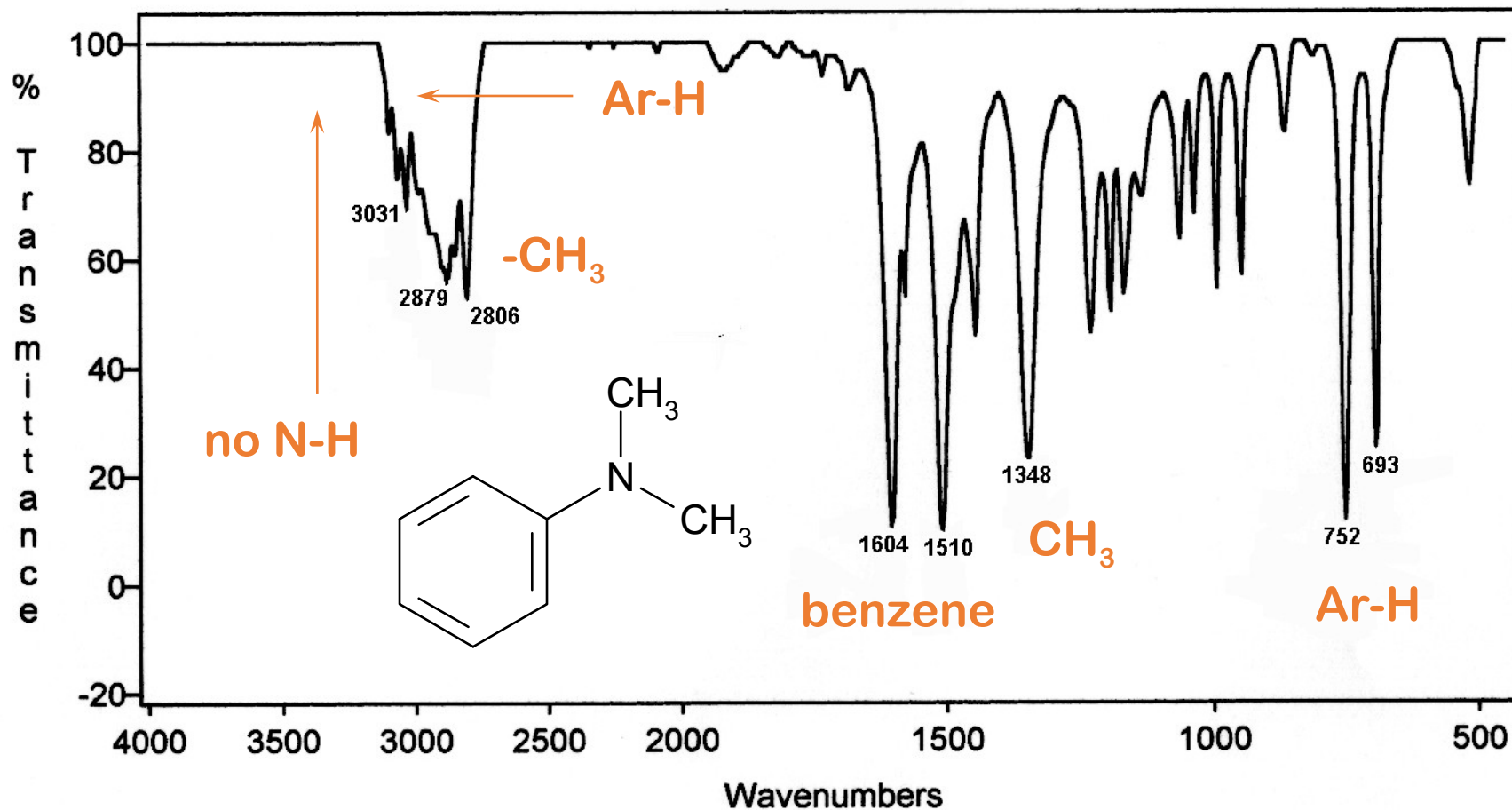
SECONDARY AMINE

N-Ethylbenzenamine



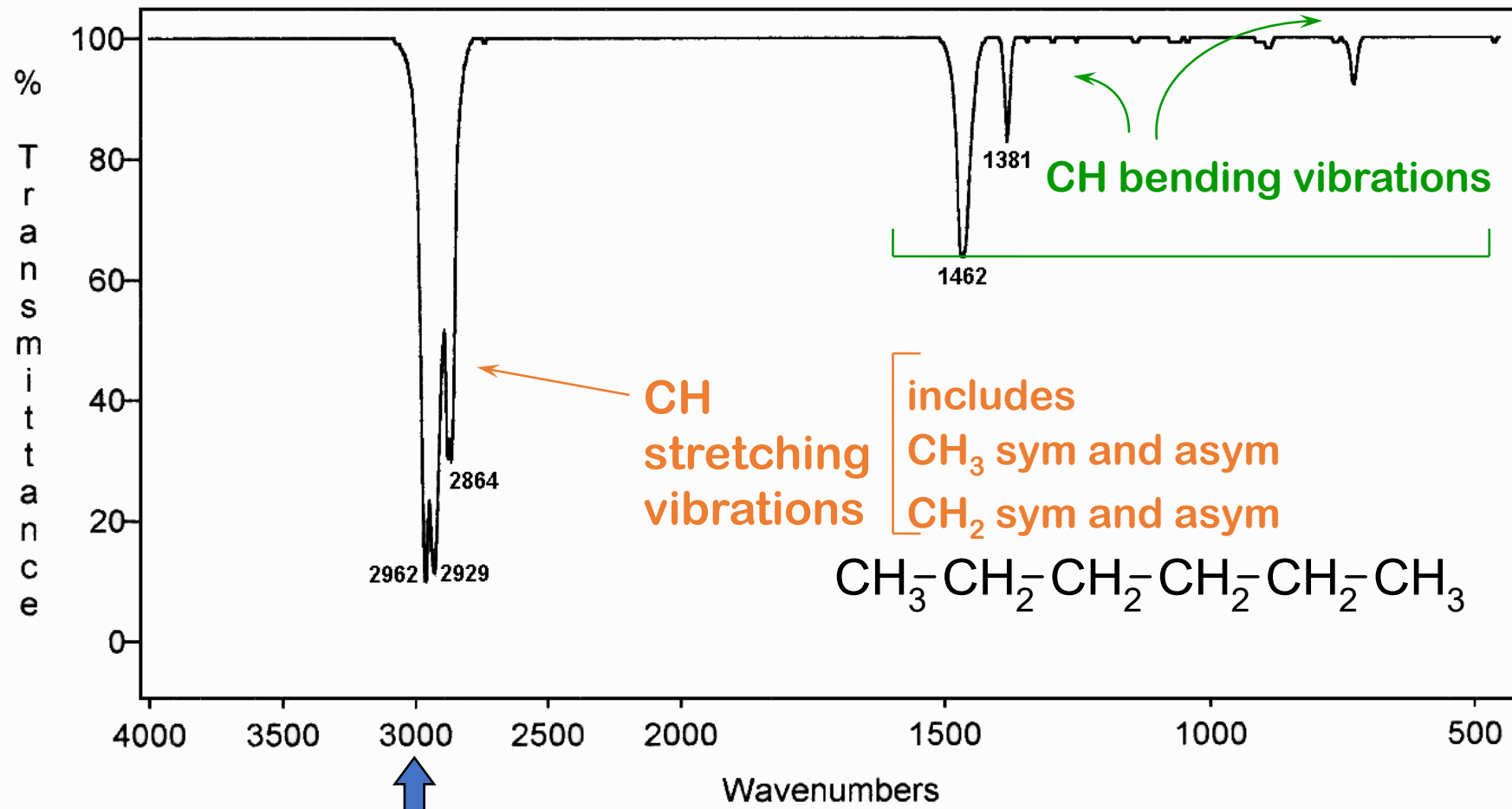
TERTIARY AMINE

N,N-Dimethylaniline



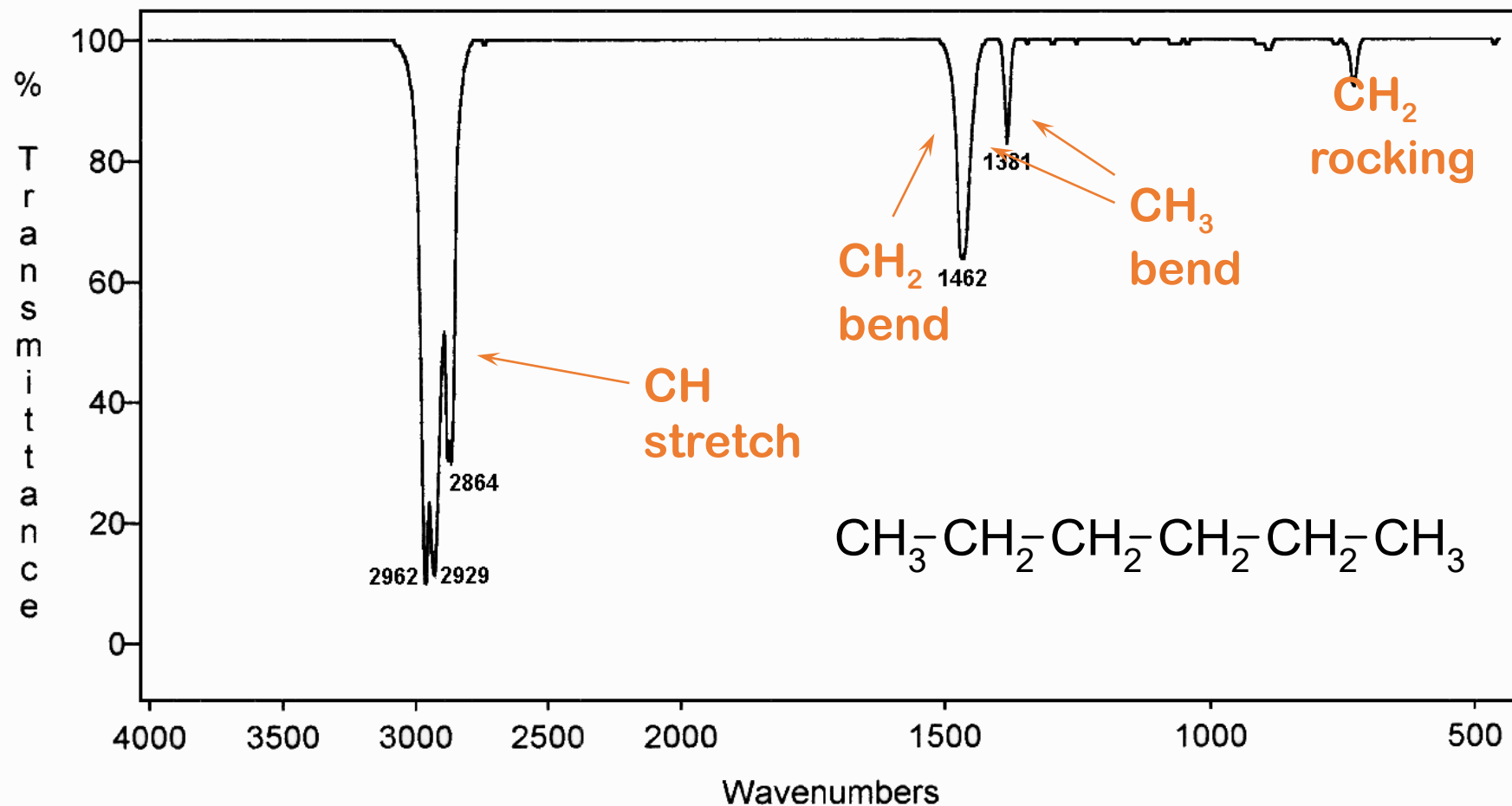
Hexane

ALKANE



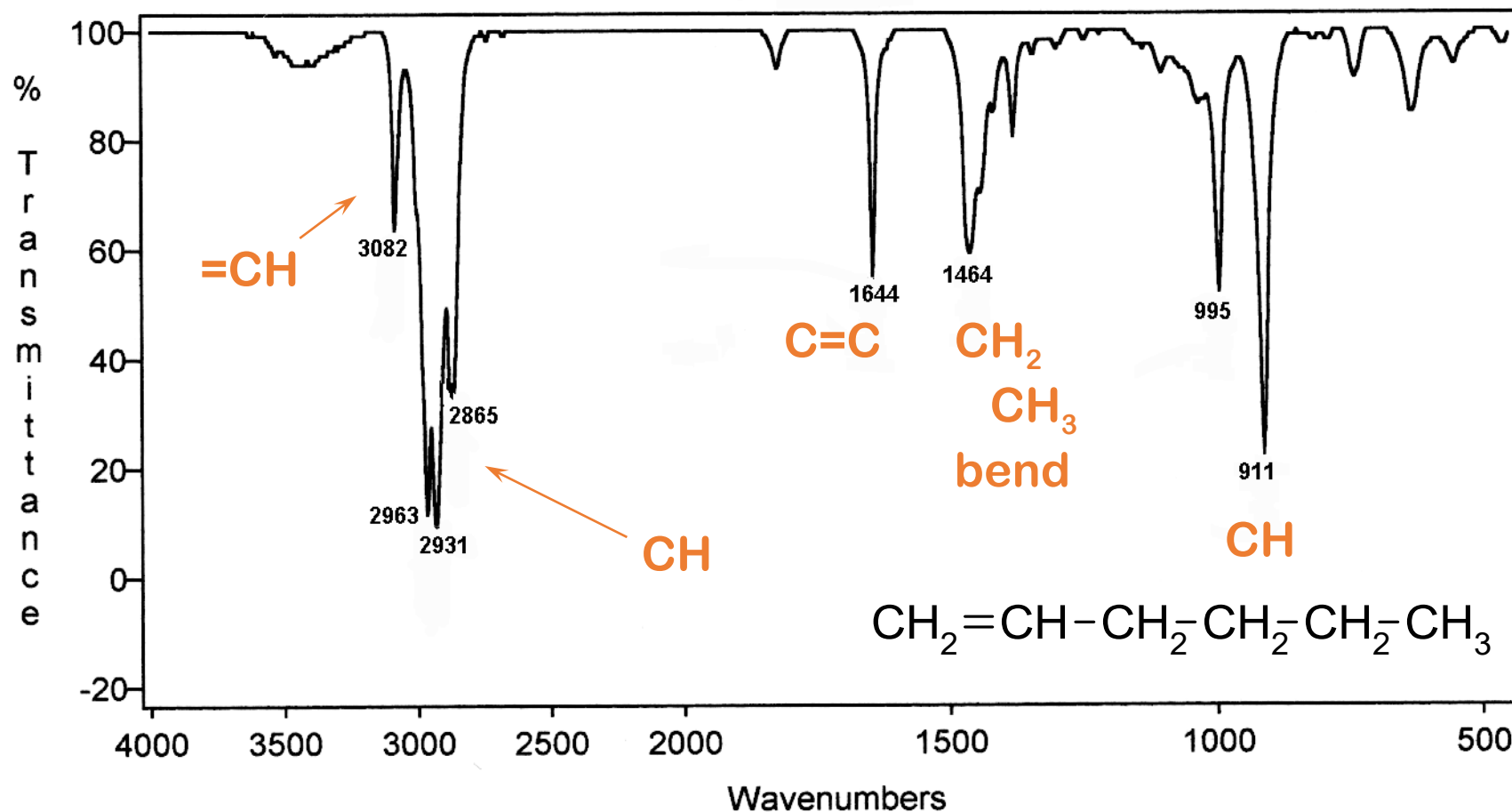
Hexane

ALKANE



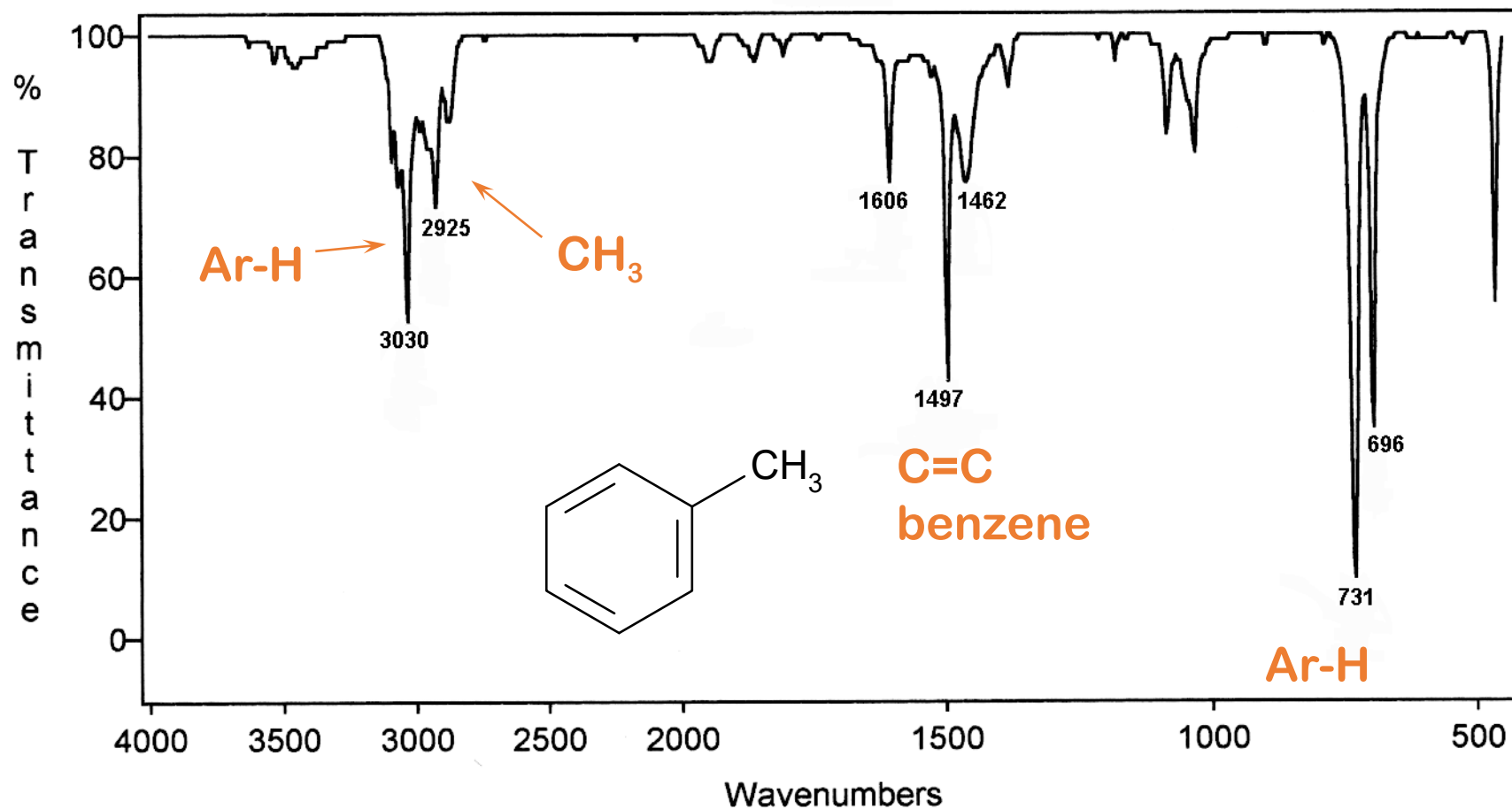
ALKENE

1-Hexene

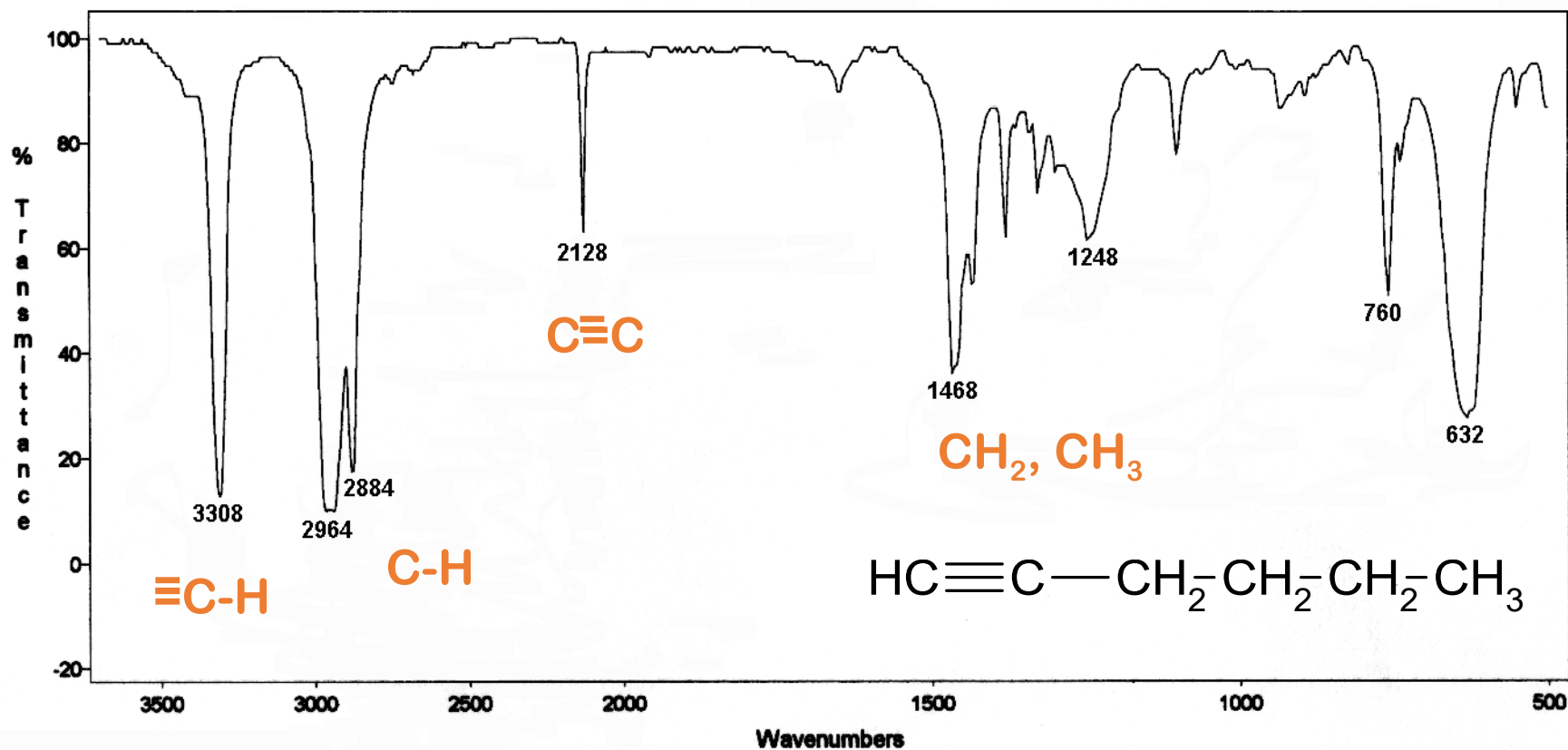


AROMATIC

Toluene



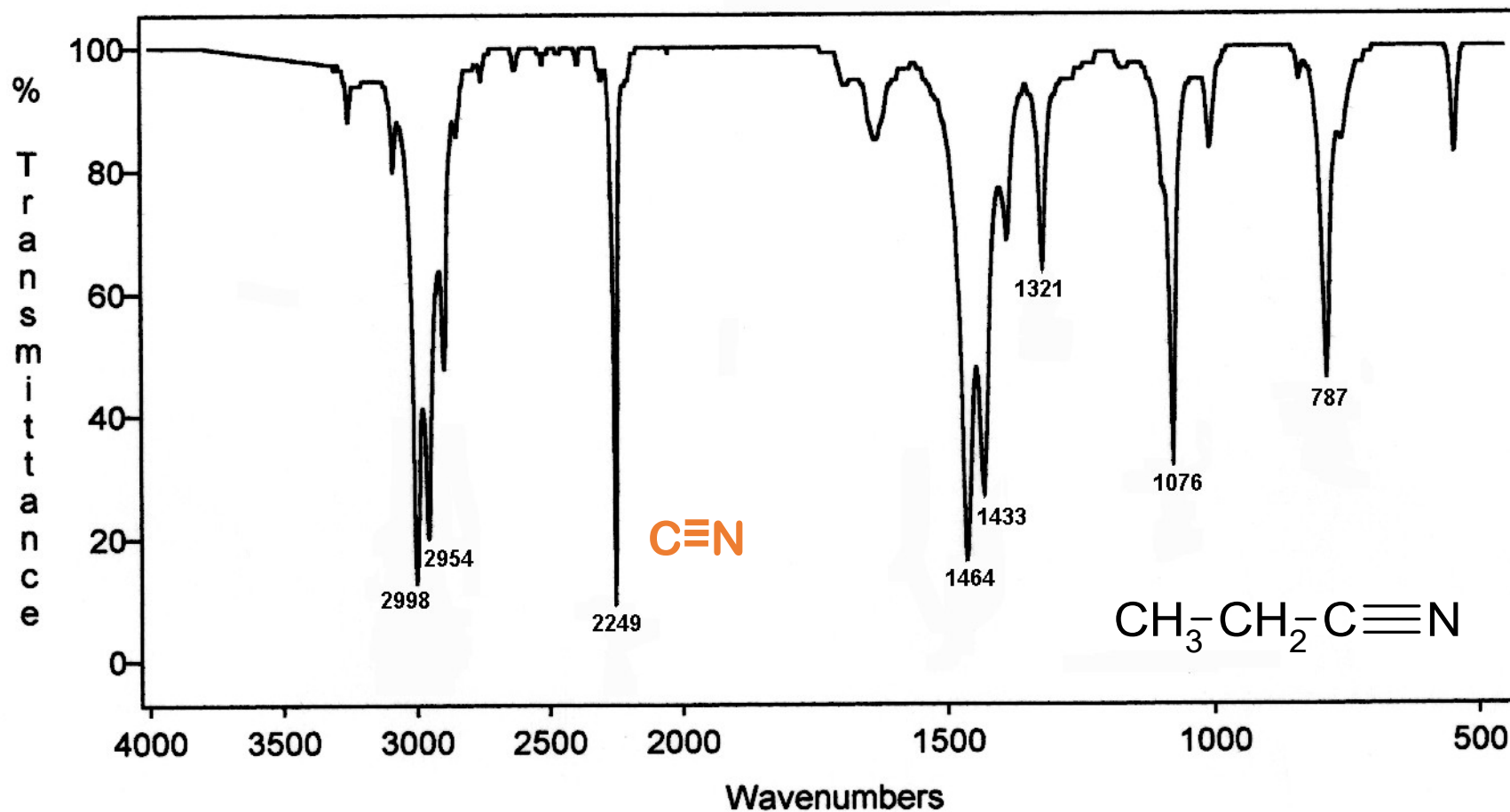
1-Hexyne



Propanenitrile

NITRILE

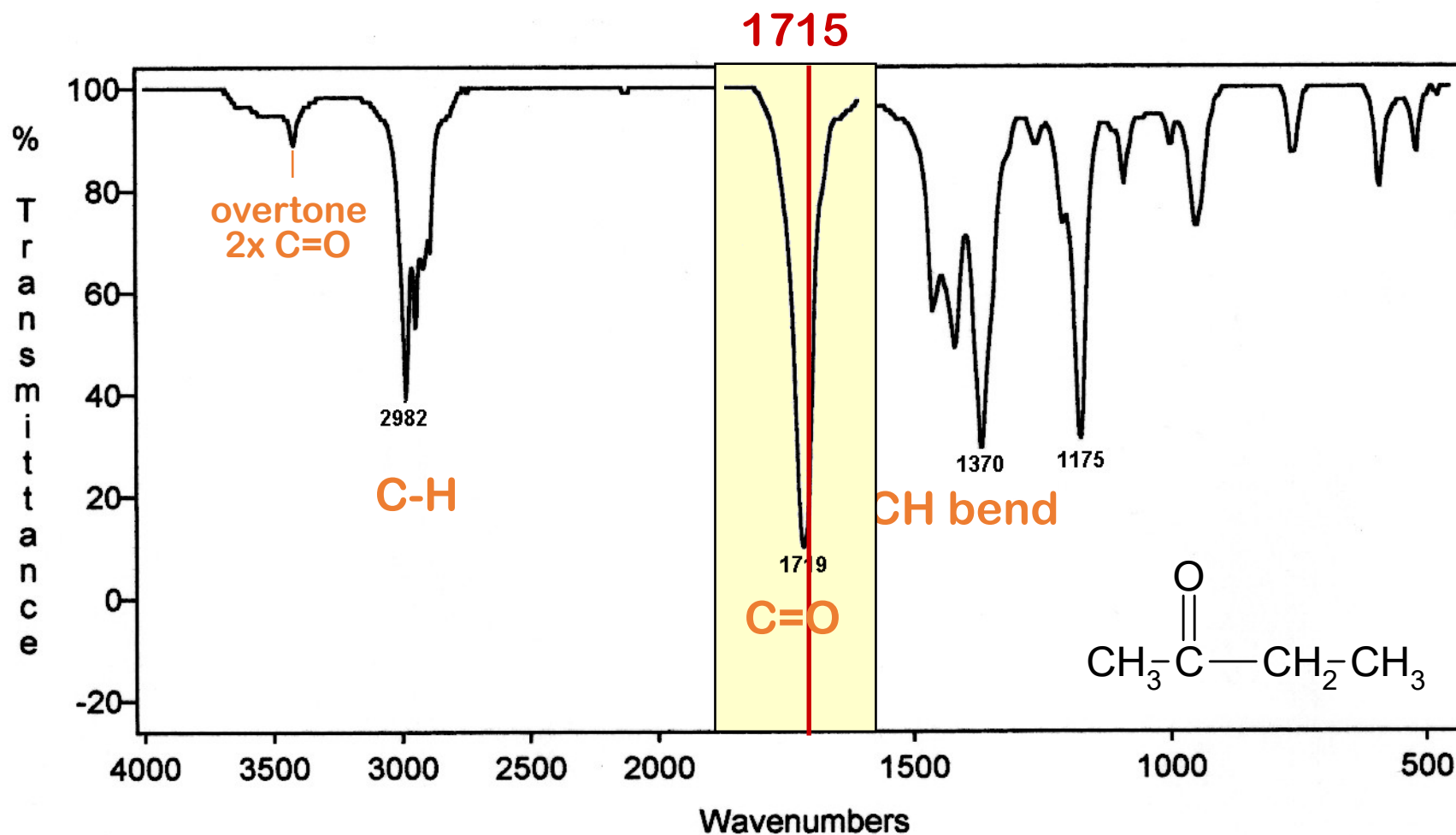
BASE = 2250



KETONE

BASE = 1715

2-Butanone



KETONE

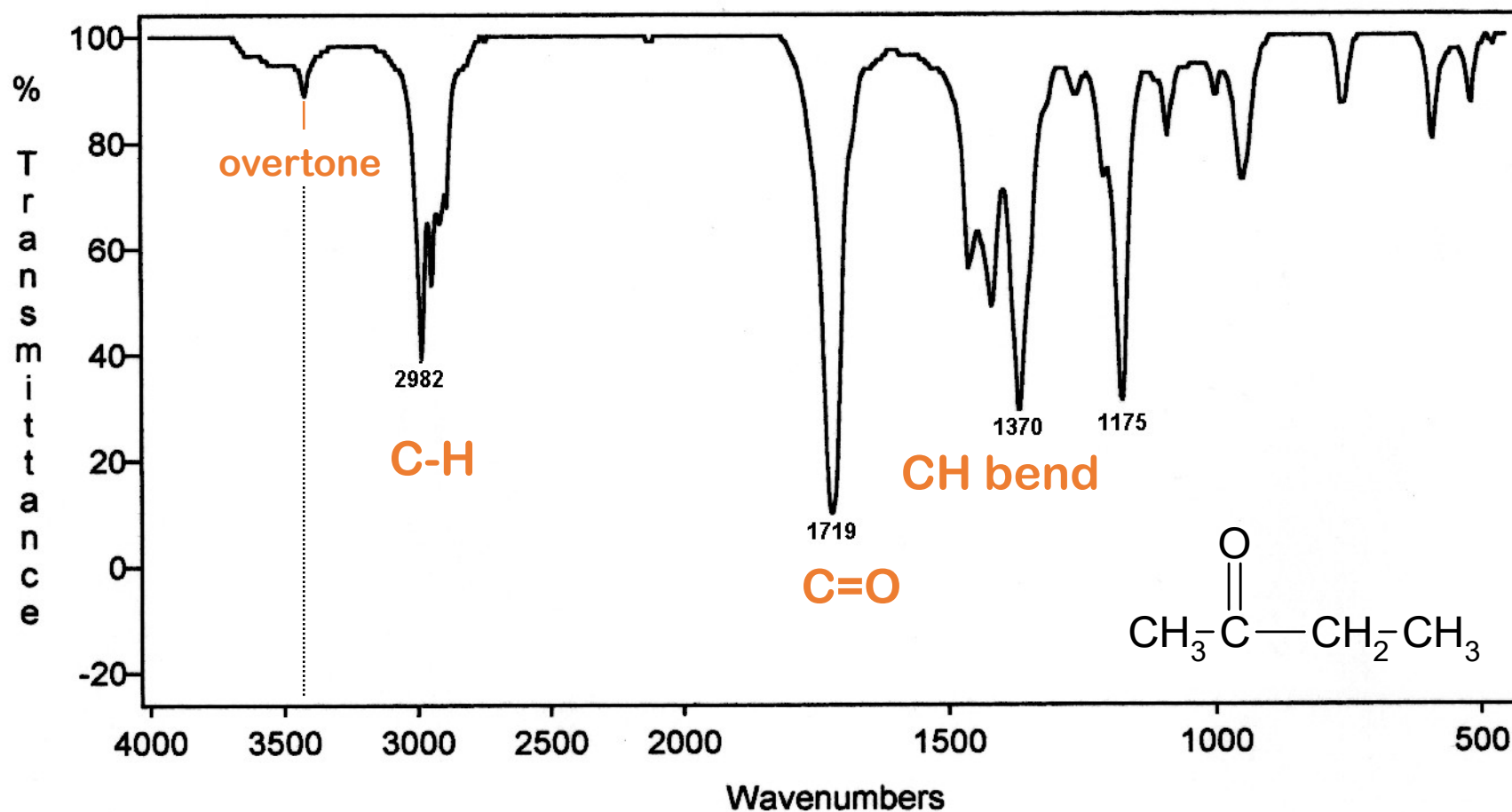
overtone of strong C=O peak

$$1719 \times 2 = 3438$$



2-Butanone

BASE = 1715

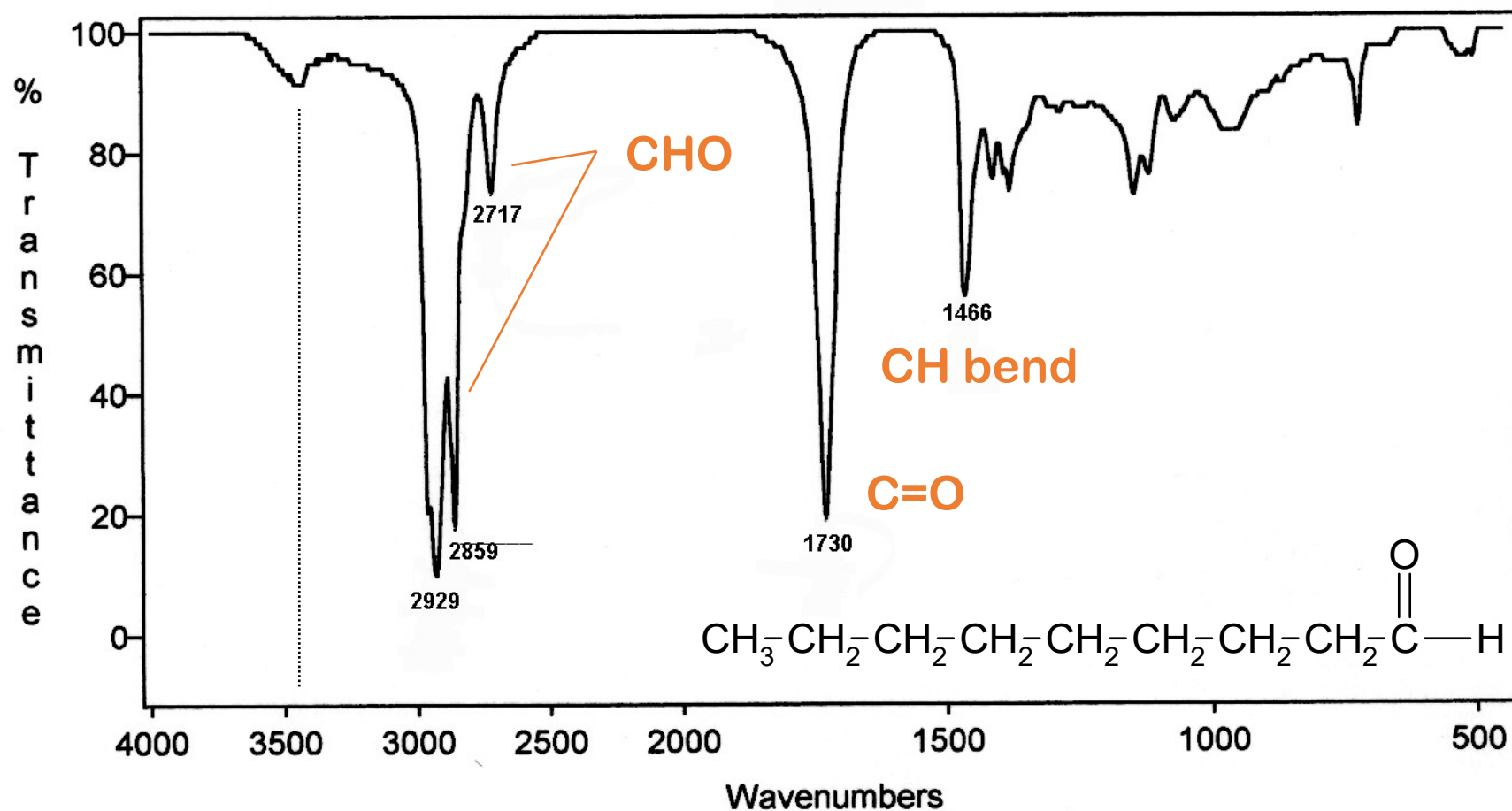


3438

ALDEHYDE

BASE = 1725

Nonanal

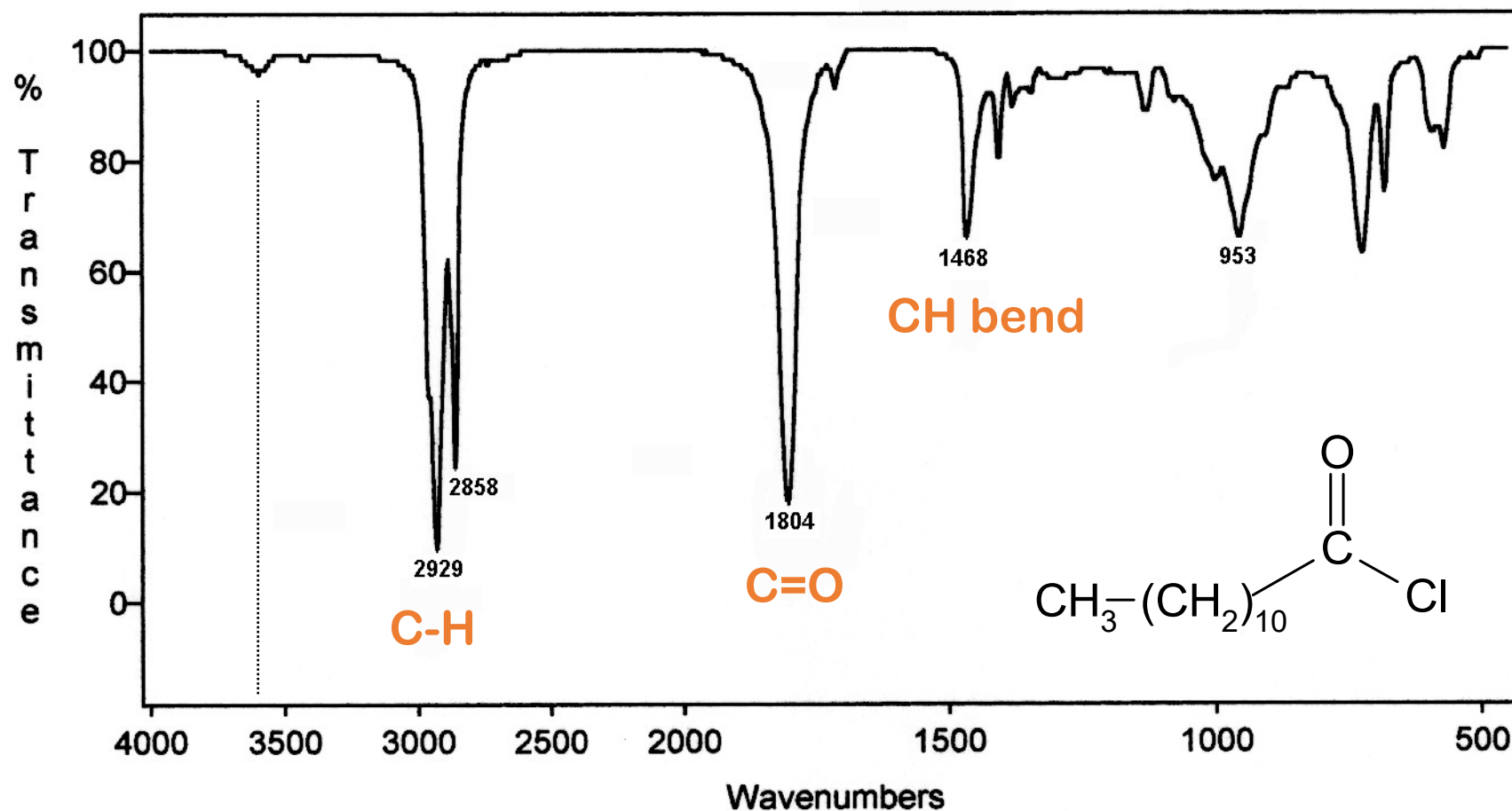


3460

ACID CHLORIDE

Dodecanoyl Chloride

BASE = 1800

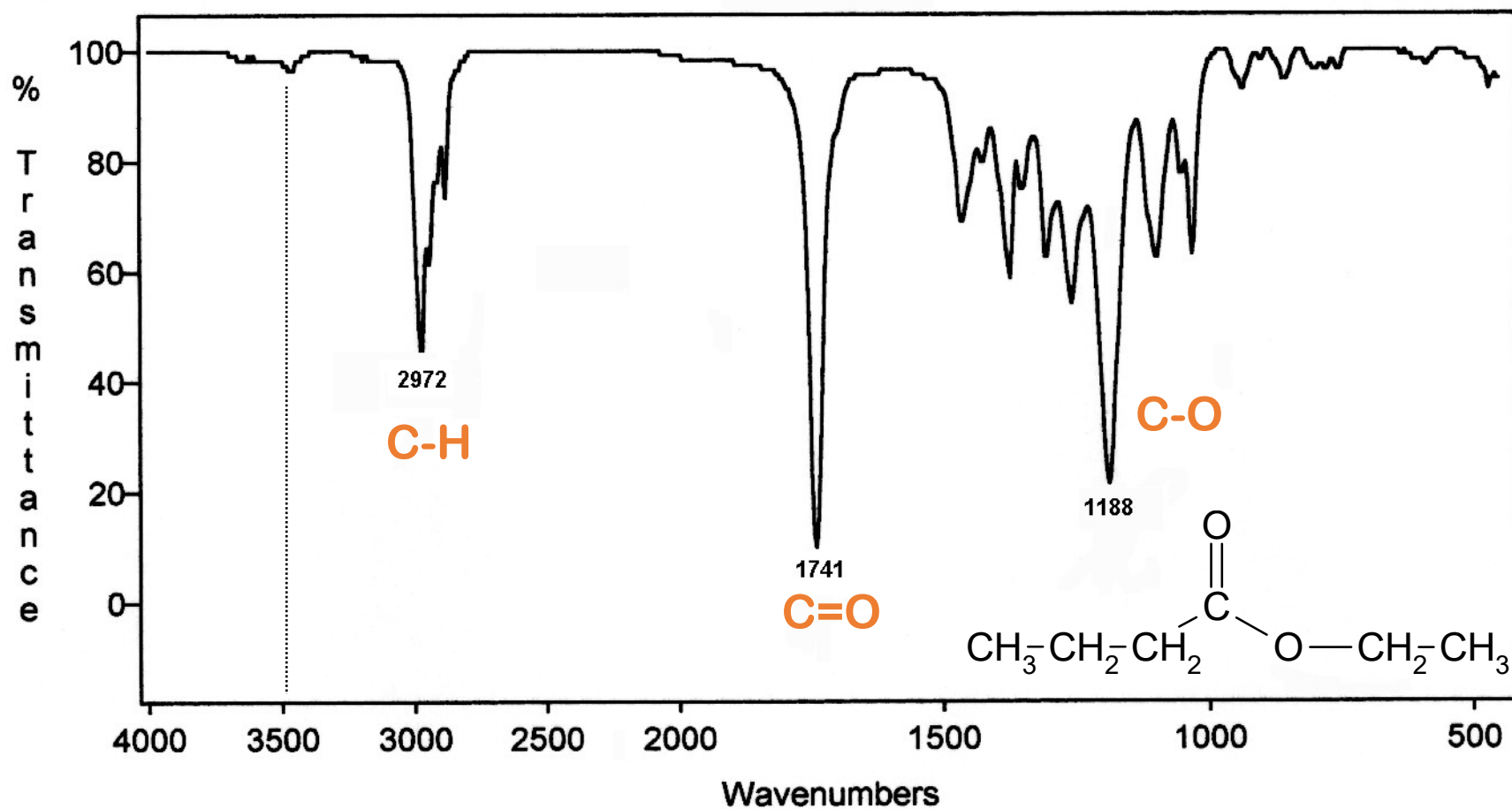


3608

ESTER

BASE = 1735

Ethyl Butanoate

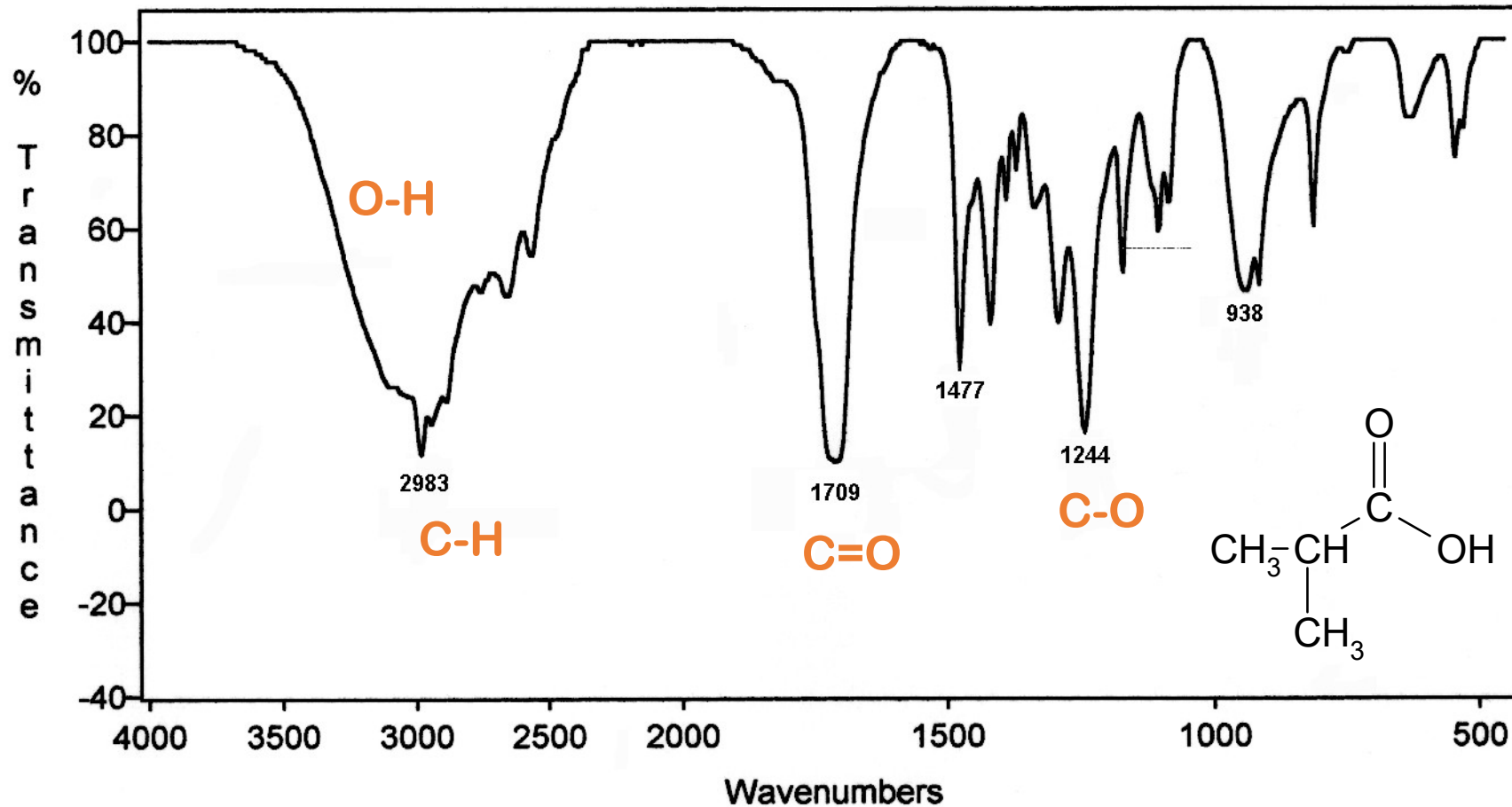


3482

CARBOXYLIC ACID

BASE = 1710

2-Methylpropanoic Acid

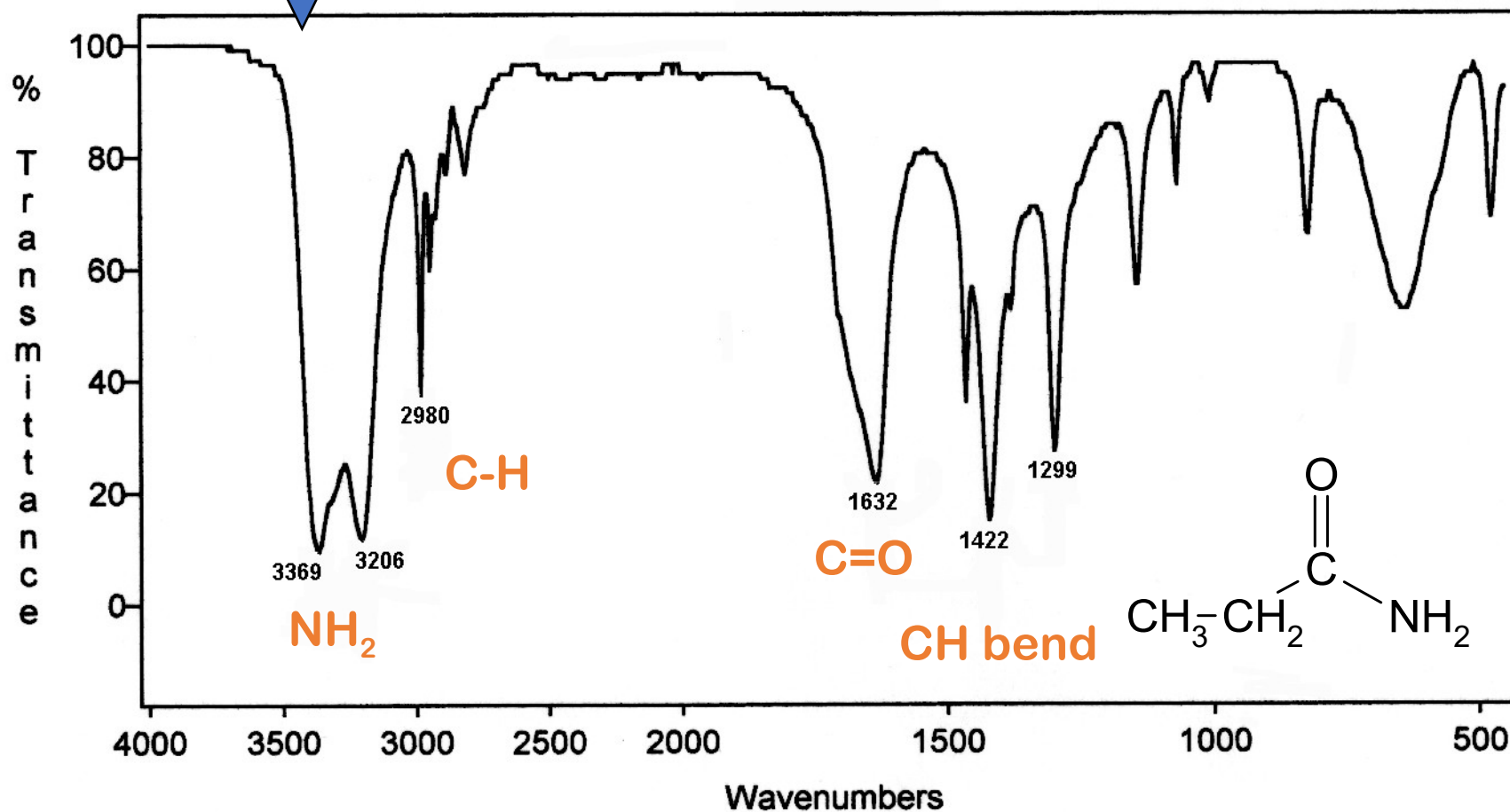


AMIDE

BASE = 1690

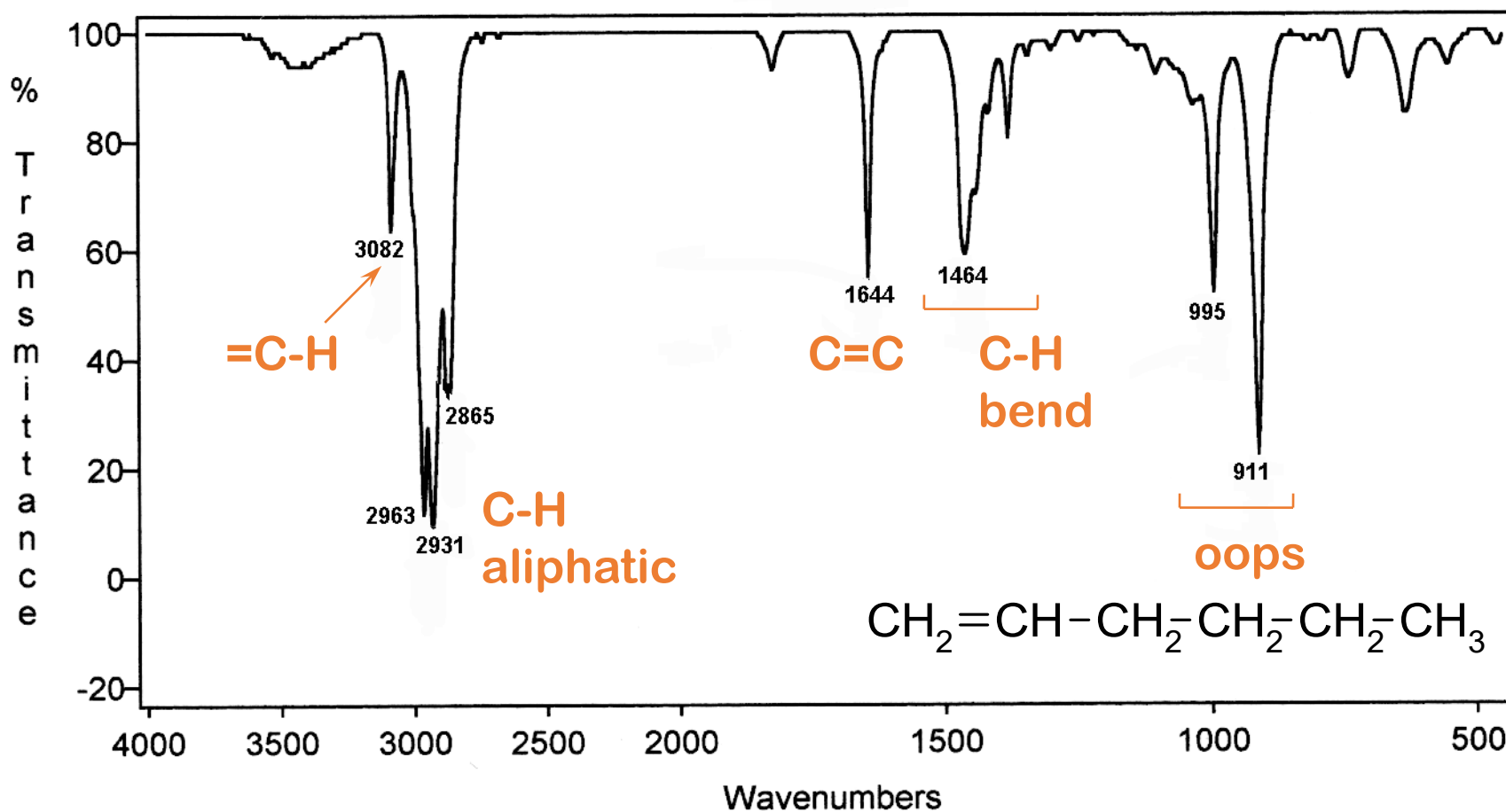
two peaks
sym / asym

Propanamide



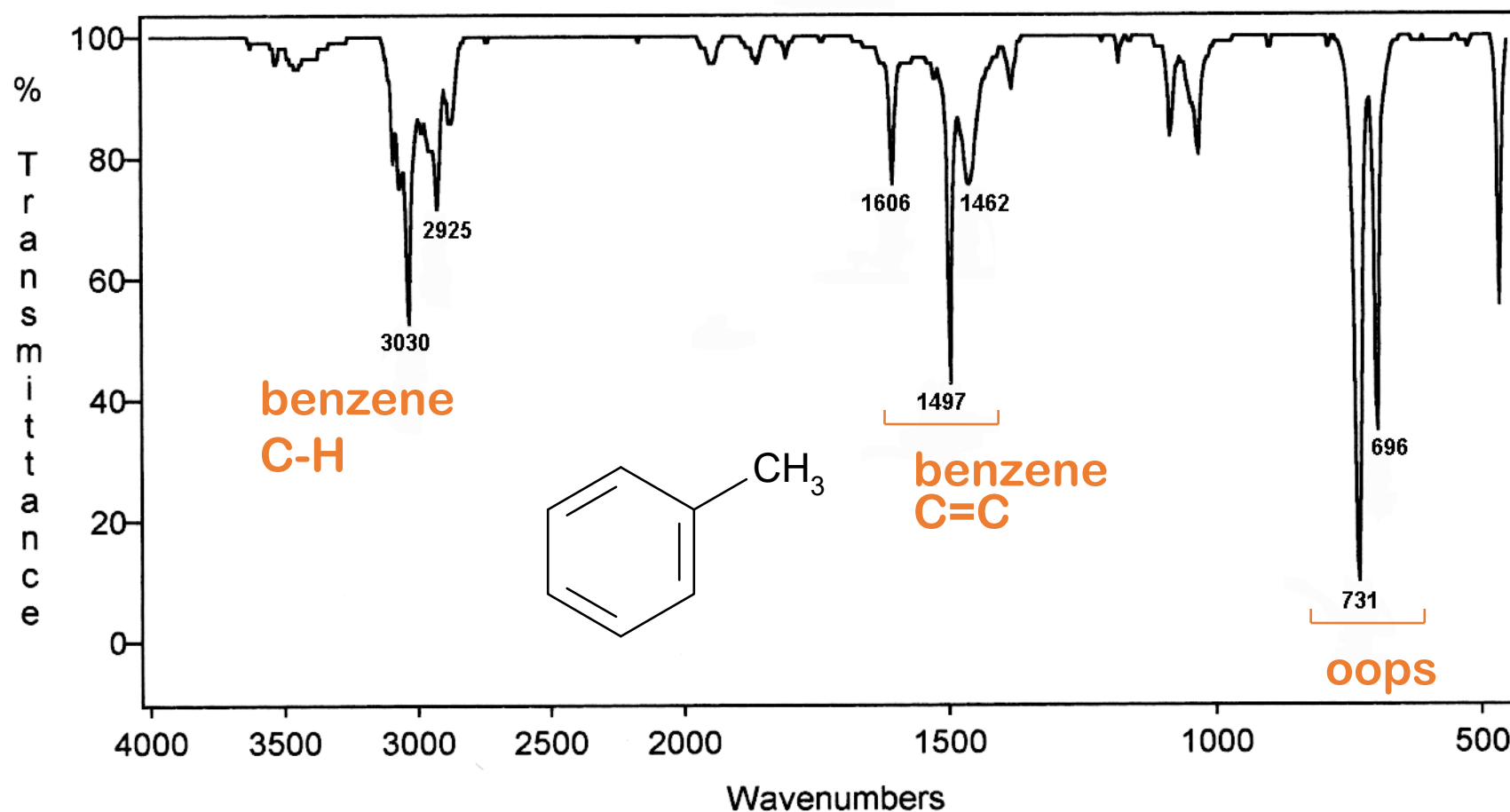
ALKENE

1-Hexene



Toluene

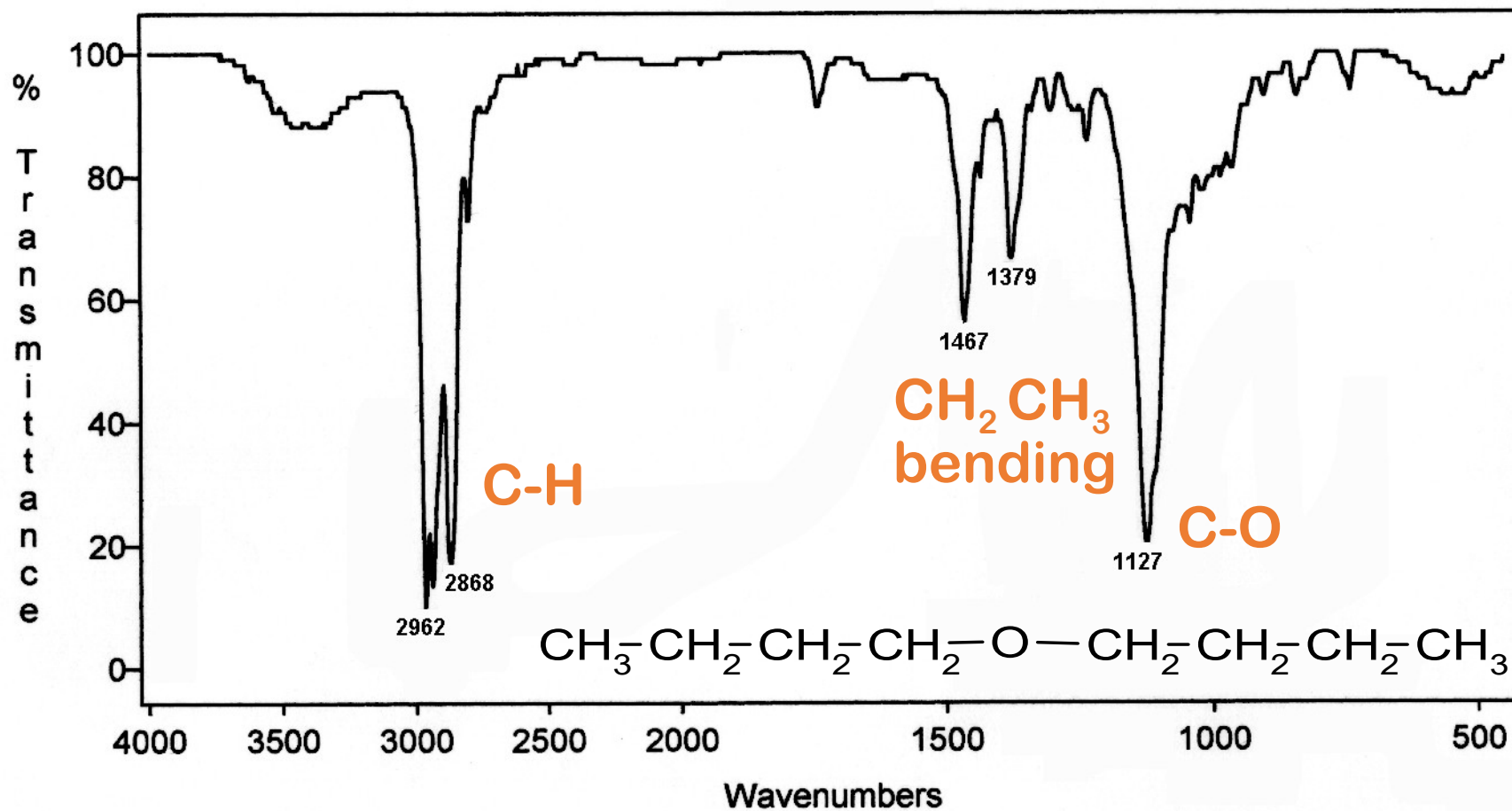
AROMATIC



ETHER

BASE = 1100

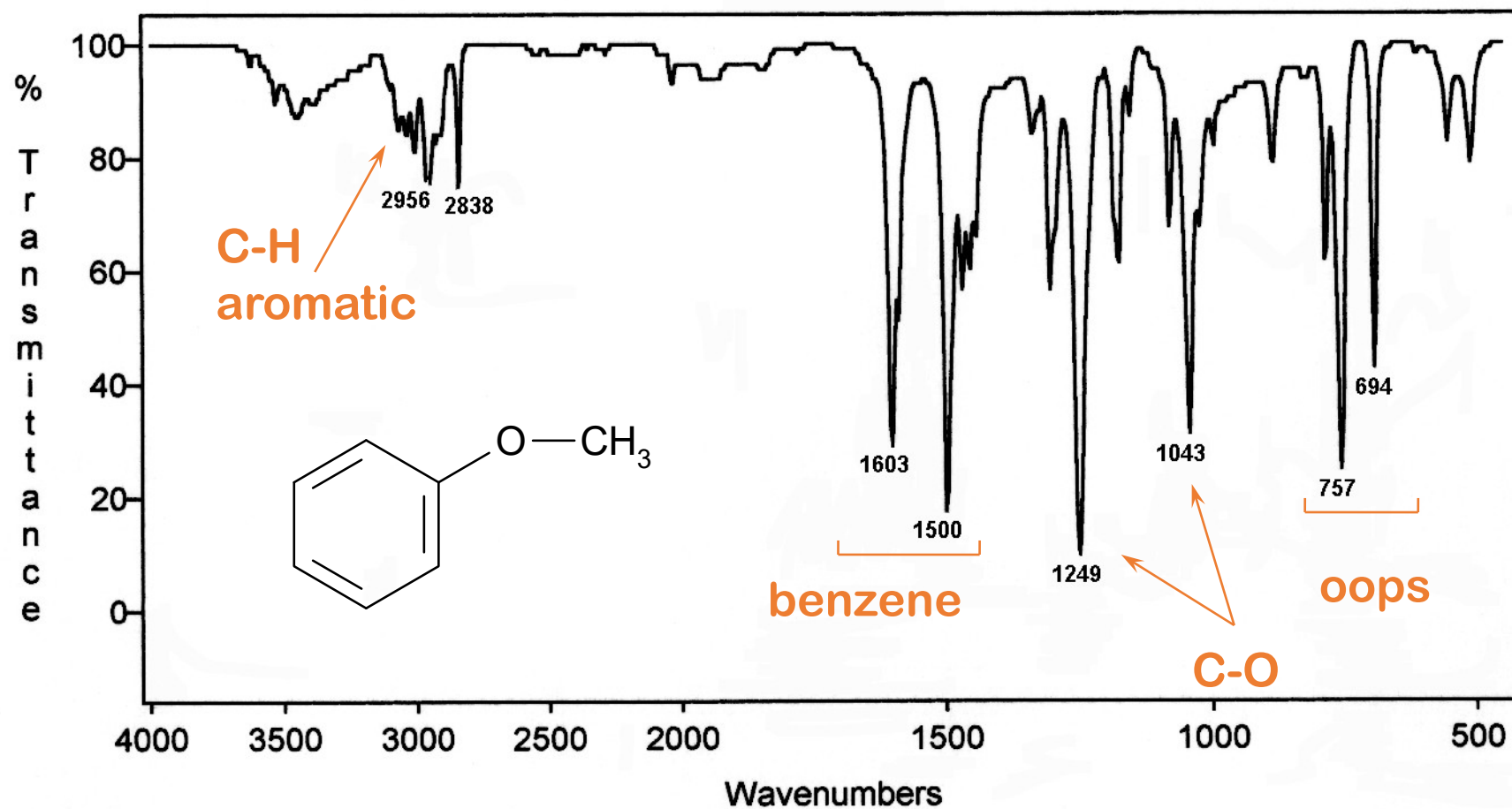
Dibutyl Ether



AROMATIC ETHER

BASE = 1100

Anisole

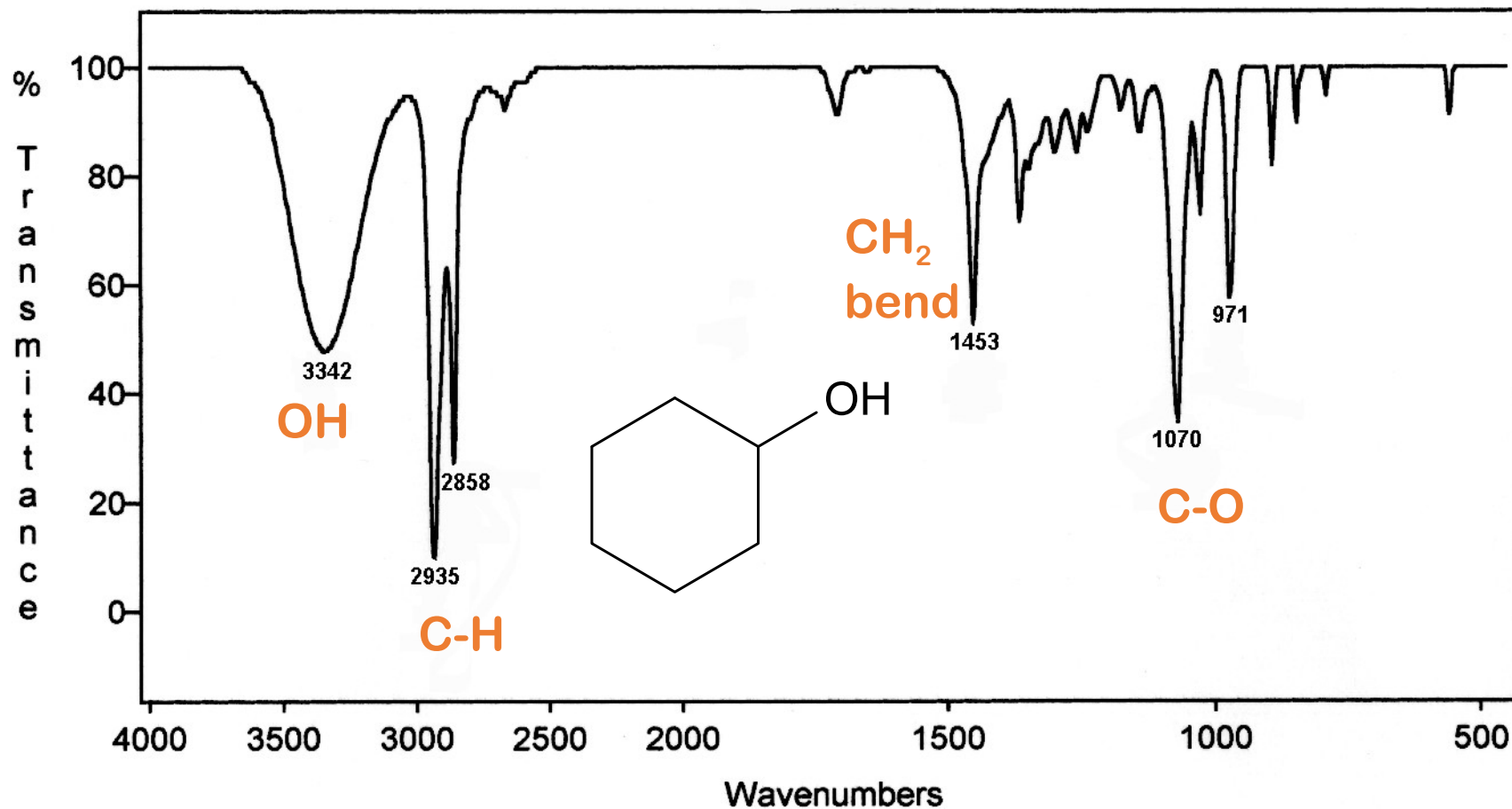


ALCOHOL

BASE = 3600

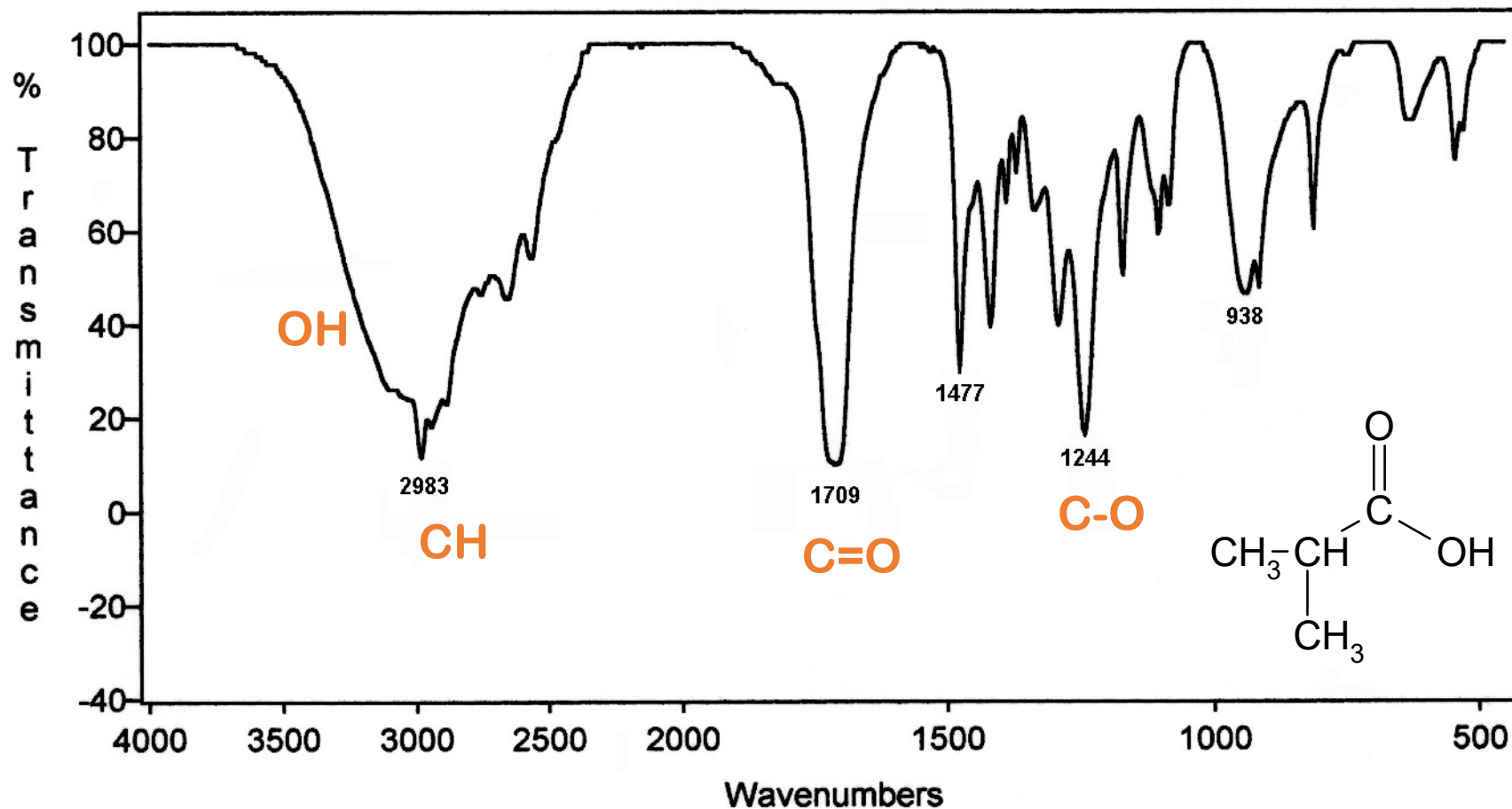
BASE = 1100

Cyclohexanol



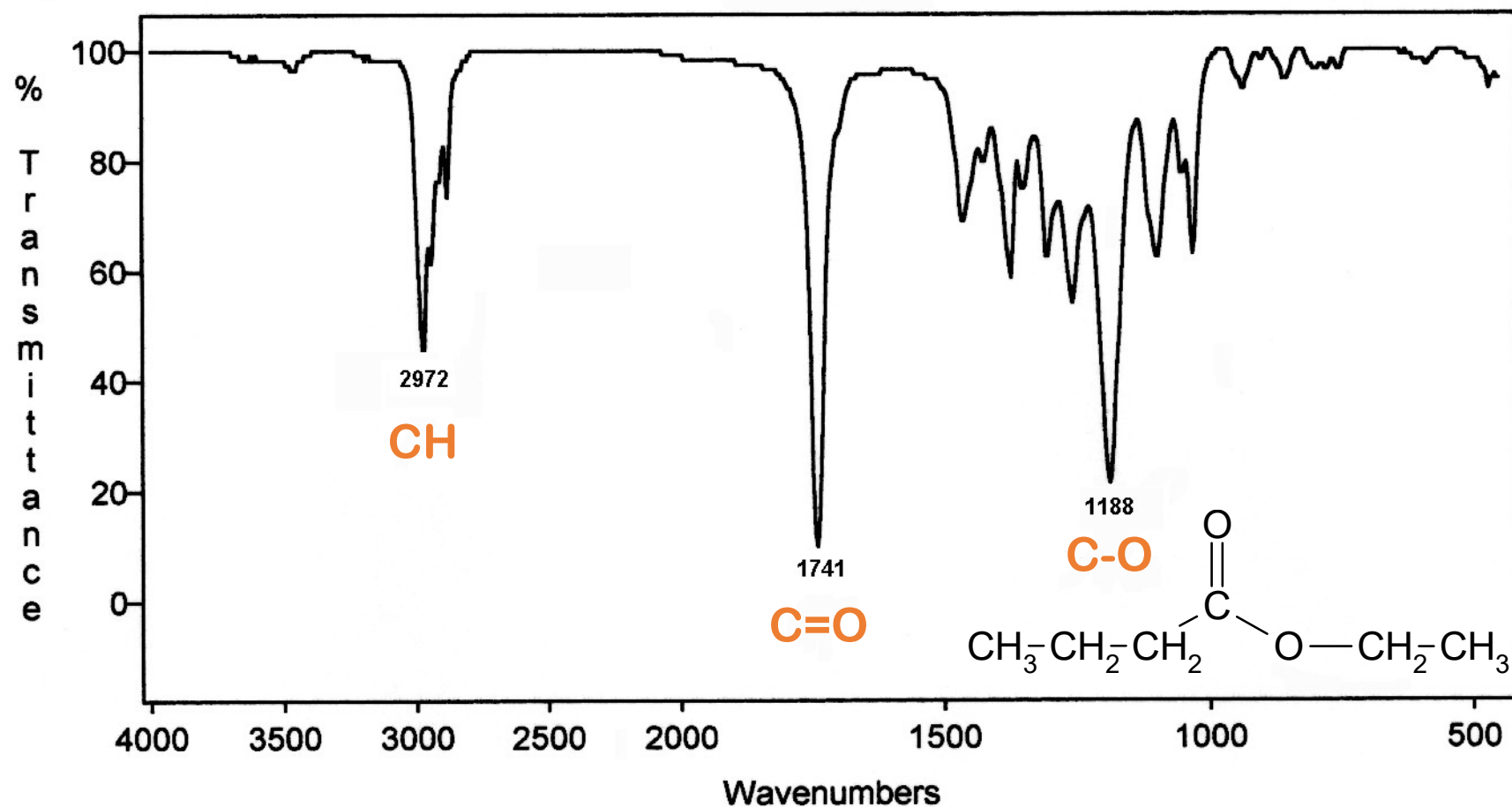
CARBOXYLIC ACID

2-Methylpropanoic Acid



ESTER

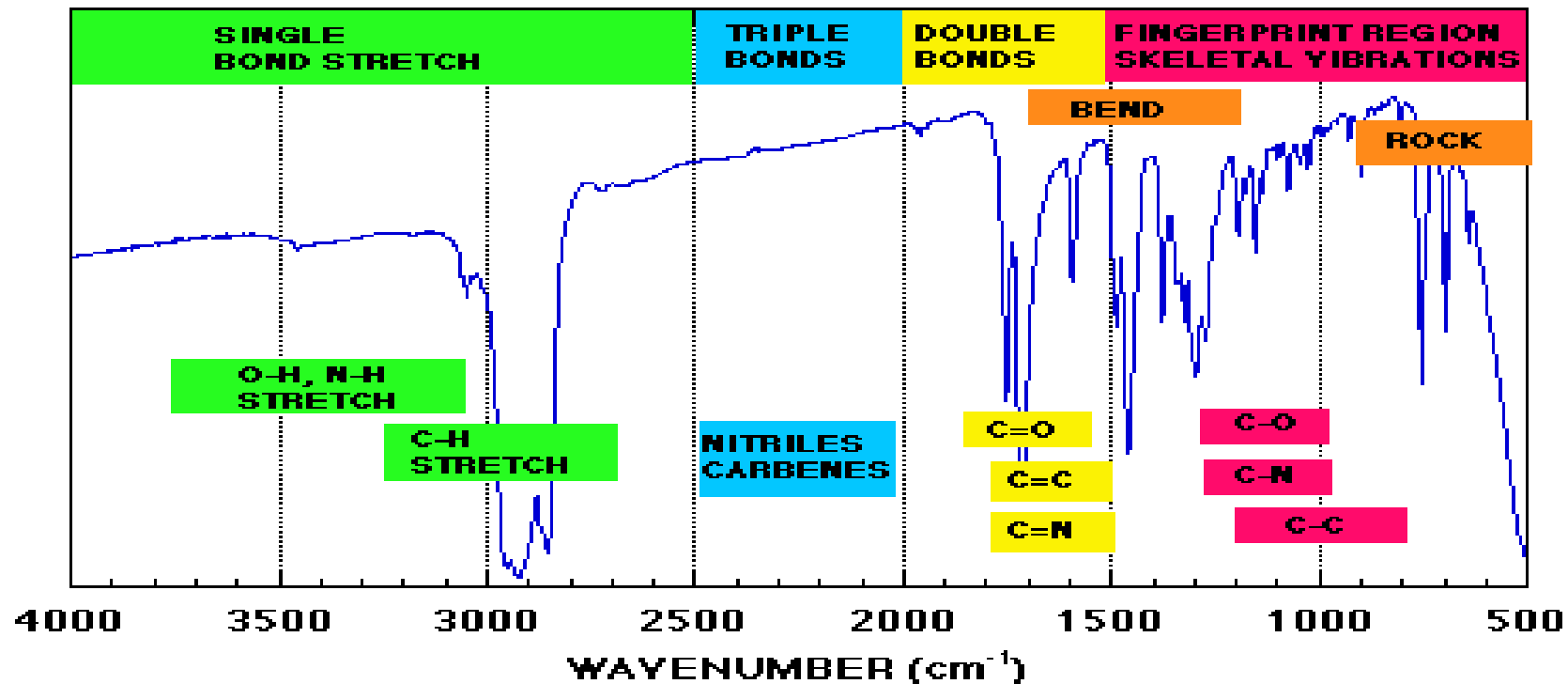
Ethyl Butanoate



Application of IR

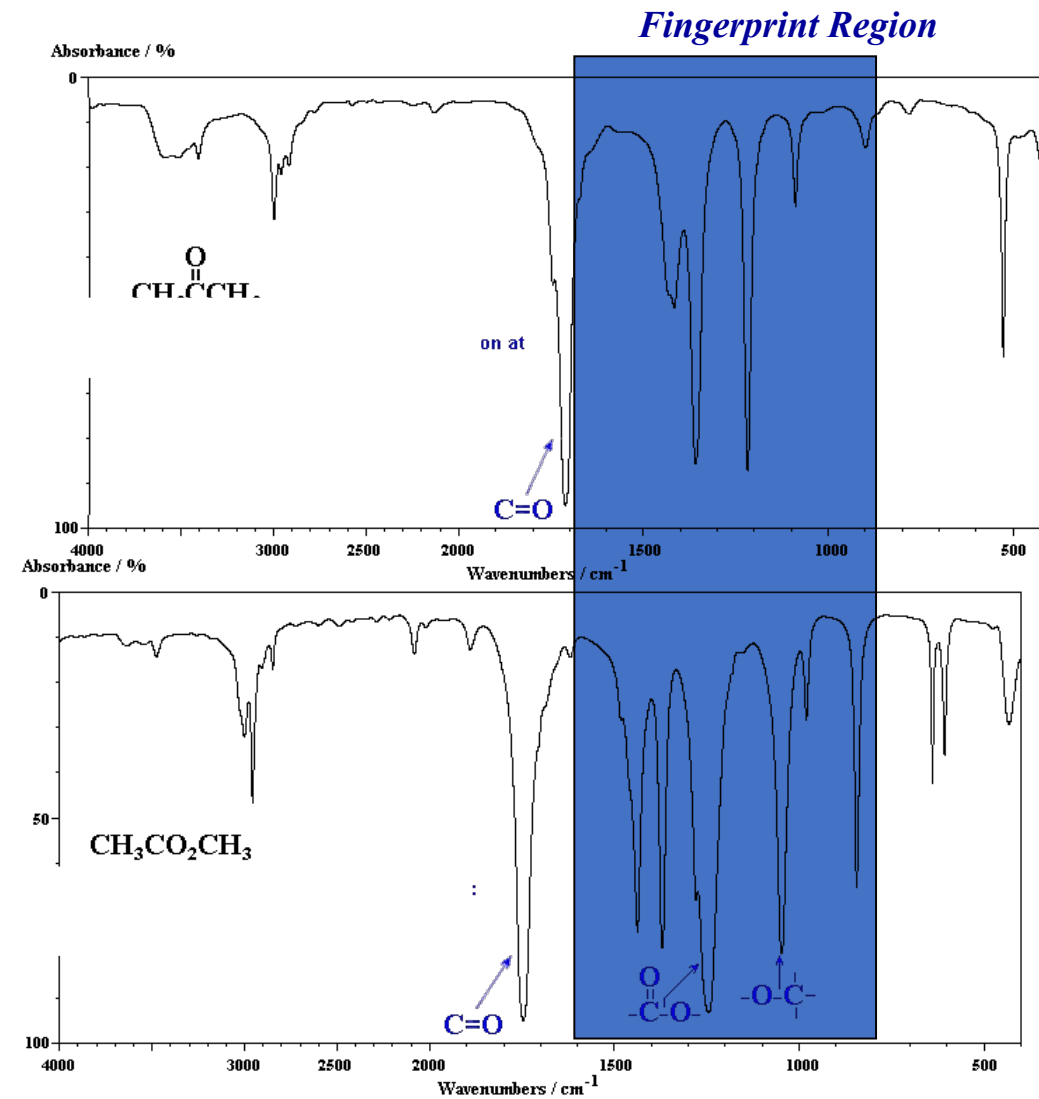
1. Qualitative Analysis (Cpd Identification)

- main application
 - i. Examine what functional groups are present by looking at group frequency region



ii. Fingerprint Region

- Region of most single bond signals
- Many have similar frequencies, so affect each other & give pattern characteristics of overall skeletal structure of a cpd.
- Exact interpretation of this region of spectra seldom possible because of complexity
- **Complexity → uniqueness**



2. Quantitative Analysis

- not as good as UV/Vis in terms of accuracy and precision

- ▶ more complex spectra
- ▶ narrower bands (Beer's Law deviation)
- ▶ limitations of IR instruments (lower light throughput, weaker detectors)
- ▶ high background IR
- ▶ difficult to match reference and sample cells
- ▶ changes in ϵ ($A=\epsilon bc$) common

- potential advantage is **good selectivity**, since so many compounds have different IR spectra

- ▶ one common application is determination of air contaminants such as Carbon Monoxide, Methylene ketone, Methyl alcohol, Ethylene oxide, Chloroform